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UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

# The study of Higgs properties in the four-lepton final state at CMS

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# *Abstract*

Institute of High Energy Physics
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Doctor of Science

**The study of Higgs properties in the four-lepton final state at CMS**

by Tahir JAVAID

This dissertation presents the measurements of the integrated fiducial cross sections of the Higgs boson production via $H \to 4\ell$ decays ($\ell = \mathrm{e}$, $\mu$) in pp collisions at $\sqrt{s} = 13$ TeV. The Higgs fiducial cross section measurements for $m_H = 125.09$ GeV are made with the data collected with CMS detector at Large Hadron Collider corresponds to integrated luminosities of 137.1 $\mathrm{fb}^{-1}$at 13 TeV. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale. Any deviation of these measurements from Standard Model expectations would provide a direct hint to Beyond Standard Model contributions.

Differential fiducial cross sections are presented using the 13 TeV data as a function of several kinematic observables which are sensitive to the Higgs-boson production mechanisms: rapidity and transverse momentum of the four-lepton system, associated jet multiplicity and transverse momentum of the leading jet.

All the cross sections measurements are extracted within a fiducial phase space defined by the selections on lepton kinematics and event topology. The integrated fiducial cross sections are measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\mathrm{stat.})^{+0.24}_{-0.19}(\mathrm{syst.})$ fb at 13 TeV. All the integrated and differential measurements are compared with the Standard Model based theoretical predictions and no significant deviation has been observed.

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# Chapter 1

# Introduction

The Standard Model (SM) of the particle physics gives successful explanation about the constituents of the physical matter and the fundamental forces through which they interact with each other and with composite particles. The electromagnetic, strong and weak forces are described by dedicated theoretical descriptions Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Quantum Flavourdynamics (QFD) which are encoded in the SM. Later on, electromagnetic and weak forces among these, were unified as electroweak (EWK) force at the order of 100 GeV unification energy scale. Sheldon Glashow, Abdus Salam, and Steven Weinberg had significant contribution in this milestone. The experimental verifications of the existence of the electroweak interaction were established by the In 1973, existence of neutral currents announced by the Gargamelle collaboration and then in 1981, the discovery of two massive gauge bosons (W and Z) by UA1 and UA2 collaborations verified the existence of EWK force. These boson should be massless (like photon in QED) according to Goldstone's theorem. To overcome this issue, spontaneous symmetry breaking was introduced that generates the masses of the particles by the Higgs mechanism and separates weak interaction from electromagnetic (EM) interaction (**Reference7**; **Reference8**; **Reference9**; **Reference10**; **Reference11**; **Reference12**). Higgs boson, a new scalar particle, was introduced by the theory being last missing patch of the SM. This triggered the search for this boson of the SM at experimental facilities which lasted for several years. One of the main design and construction motivations of the Large Hadron Collider (LHC) at the CERN was also to look for this new particle given unitary and above the Large Electron-Positron (LEP) constraints, i.e from 114 GeV to 1 TeV. After a long search, the CMS and the ATLAS collaborations of LHC, announced the existence of the heavy scalar boson (the Higgs) on 4 July 2012 (**Reference20**; **Reference21**). Immediately after the discovery, further tests to measure its properties were begun. Mass of the newly discovered particle, the only free parameter in Higgs sector, has been measured to be $m_H = 125.09 \pm 0.21(stat.) \pm 0.11(syst.)$ GeV (**Reference22**). Other properties such as spin, parity (**Reference21a**) and couplings of this boson to different particles (**Reference21b**; **Reference21c**) have also been measured and observed to consistent with the predictions of SM. Therefore, the new particle very probable appears as the SM Higgs boson, affiliated with EWK spontaneous symmetry breaking. SM has passed enormous experimental tests in last 50 years and no significant deviation has been seen yet. Experimental evidence of the Higgs bosonis the biggest achievement of the SM. Other success includes the experimental verification of its prior predictions about the W and Z gauge bosons, gluon, top and charm quarks.

Although SM has passed numerous experimental tests with great precision, it is not considered to be theory of everything. There are many phenomena which are not explained by SM i.e Hierarchy problem, neutrino massess and existence of dark matter and dark energy etc. There are several theoretical extension posed in SM such as the Minimal Super Symmetrical Model(MSSM) and the Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical descriptions, such as string theory and extra dimensions. More precise Higgs boson properties measurements are needed to capture any possible deviation from SM predictions which can be a doorway to new physics.

Model independent production cross section for a particular process is an important property of H boson, where the cross section gives the possibility of the occurrence of that process. In this dissertation, measurements of the integrated or inclusive and differential fiducial cross sections for H boson production in $H \to 4\ell$ decay channel ($\ell = e, \mu$) in proton-proton collisions at $\sqrt{s} = 13$ TeV are presented. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of SM at the TeV energy scale. Any deviation of the integrated or the differential cross sections from SM expectations would provide a direct hint to BSM contributions. Nearly similar measurements have already been reported by the ATLAS experiment in $H \to \gamma\gamma$ (**ATLAS:Hgg_8TeV**; **ATLAS:Hgg_8TeV_13TeV**), $H \to 4\ell$ (**ATLAS:H4l_8TeV**; **ATLAS:H4l_8TeV_13TeV**) decay channels, combination of both channels (**ATLAS:H4l_Hgg_8TeV**; **ATLAS:H4l_Hgg_8TeV_13TeV**) and by the CMS experiment in $H \to \gamma\gamma$ decay mode (**CMS:Hgg_8TeV**).

The Higgs cross section measurement helps to probe the structure, nature of the interaction of H boson with decaying particles. Differential fiducial cross sections are measured using the full Run 2 CMS data recorded at 13 TeVcorresponding to several kinematic observables which are sensitive to the H boson production mechanisms: rapidity and transverse momentum of the H boson, associated jet multiplicity and transverse momentum of leading jet. These differential distributions are sensitive to production mechanism of gluon fusion, colliding protons' PDFs, theoretical modeling of hard quarks radiations and relative contributions of different production mechanisms of H boson.

The fragmentation of the dissertation is systematized as follows. An overview of the SMof particle physics, production and decays of H boson are described in Chapter 2. Overview of the LHC and its CMS experiment are presented in Chapter 3. The calibrations and reconstructions of the physics objects within the CMS detector are described in Chapter 4. Chapter 5 presents the procedure of measurements of inclusive and differential fiducial cross sections of H boson production at $\sqrt{s} = 13$ TeV. Finally, the conclusions and outlook of the dissertation are summarized in Chapter 6.

# Chapter 2

# Theoretical Background

## 2.1 The Standard Model

The main goal of particle physics is to fully understand the nature of the constituents of the matter and the fundamental forces through which they interact. There are many theories purposed to understand the physics laws at particle level but the Standard Model(SM) is the one which stood firm against high precision tests (**Reference1**). It is a collaborative theory of fundamental world, developed in 20th century which can express the fundamental particles (bosons and fermions) and the underlying interactions with the concept of mediating gauge bosons. The last unverified part of the SM, the Higgs boson, was discovered in 2012 at LHC of European Organization for Nuclear Research(CERN) (**Reference20**; **Reference21**). Despite of incredibly successful theory of particle world, the SM does fail to fully describe the known universe.

## 2.2 Fundamental Particles and underlying Interactions

### 2.2.1 Fundamental Particles

Particle with unknown structure is called fundamental particle. Fundamental particles can be categorized into fermions (characterized by Fermi-Dirac statistics) and bosons (Bose-Einstein statistics). The spin-statistics theorem classifies the particles with integral spin as bosons, while particles with half-integral spin as fermions. Fermions includes all leptons, quarks and baryons (composite particles three quarks). Ordinary matter is composed of leptons and quarks which are categorized in three "generations" as illustrated in Table 2.1. First generation includes an "up" type quark which carries electric charge $+\frac{2}{3}$ and a "down" type quark carrying electric charge $-\frac{1}{3}$. Quarks also carry color charge (red, green, or blue) which is responsible for the strong interaction. The types of the quarks in order of increasing mass are "up", "down", "strange", "charm", "bottom" and "top". The whole visible world is only made of light type of quarks,"up" and "down". The heavy quarks can be produced only in high energy collisions (cosmic rays, particle accelerators) and quickly decays into "up" and "down" quarks.

Quarks interact through all three Standard Model forces, weak, strong, and electromagnetic. According to the Quantum Chromodynamics (QCD), two quarks (quark antiquark pair) or three quarks can combine as mesons ($q_1\bar{q}_2$) or baryons ($q_1q_2q_3$) respectively to make color-neutral hadrons. Out of hadrons, protons and neutrons are most widely recognized making nuclei of matter which are formed of $uud$ and $udd$ quarks respectively.

| **Fermions** | **1st Gen.** | **2nd Gen.** | **3rd Gen.** | **Charge** | **Interactions** |
|---|---|---|---|---|---|
| **Quarks** | $up$<br>$down$ | $charm$<br>$strange$ | $top$<br>$bottom$ | $+\frac{2}{3}$<br>$-\frac{1}{3}$ | all |
| **Leptons** | $e$<br>$\nu_e$ | $\mu$<br>$\nu_\mu$ | $\tau$<br>$\nu_\tau$ | $-1$<br>$0$ | weak, electromagnetic<br>weak |

TABLE 2.1: Generations, charge and interaction types of quarks and leptons.

Leptons also have three generations like quarks. Leptons only interact via electromagnetic and weak interaction, since they have no color charge. In fact, quarks and gluons are only particles carrying the color charge. Lepton generations each a charged lepton and a neutrino associated. Neutrinos are electrically neutral and carry no color charges, so interact through weak force only. Charged leptons masses are listed in order of increasing mass $m_e = 0.511$ MeV, $m_\mu = 105.7$ MeV and $\mathrm{m}_\tau = 1.78$ GeV. Charged leptons carry $-1$ and their antiparticle carry $+1$ electric charge. In addition, all leptons lepton flavor quantum number: all leptons have assigned $+1$ while antileptons have assigned $-1$.

Carriers of fundamental forces are bosons, more details would be described in Section 2.2.2. Electromagnetic force is mediated by photon which is massless, neutral and of spin-1. On the other hand, weak force is carried by $W^\pm$ (80.4 GeV) and Z (91.2 GeV) bosons having electric charges $\pm1$ and $0$ respectively. At last, strong forces is carried by massless, neutral spin-1 gluons. Gluon is the only gauge boson which carries color and anti-color charge, thus it can interact with other gluons.

TABLE 2.2: Gauge bosons and their masses, coupling constants and interaction ranges for the three forces of the SM

| Force | Weak | Electromagnetic | Strong |
|---|---|---|---|
| Mediator boson | $W^\pm$ and $Z$ | photon ($\gamma$) | Gluon (g) |
| Mass(GeV) | 80.4, 91.2 GeV | 0 | 0 |
| Coupling Constant | $G_F \approx 1.2 \times 10^{-5}$ GeV$^{-2}$ | $\alpha = \frac{1}{137}$ | $\alpha_s \approx 0.1$ |
| Interaction Range | $10^{-16}$ cm | $\infty$ | $10^{-13}$ cm |

### 2.2.2 Fundamental Interactions

The three fundamental interactions (electromagnetic, strong and weak) are explained three dedicated theoretical descriptions Quantum Electrodynamics (QED),

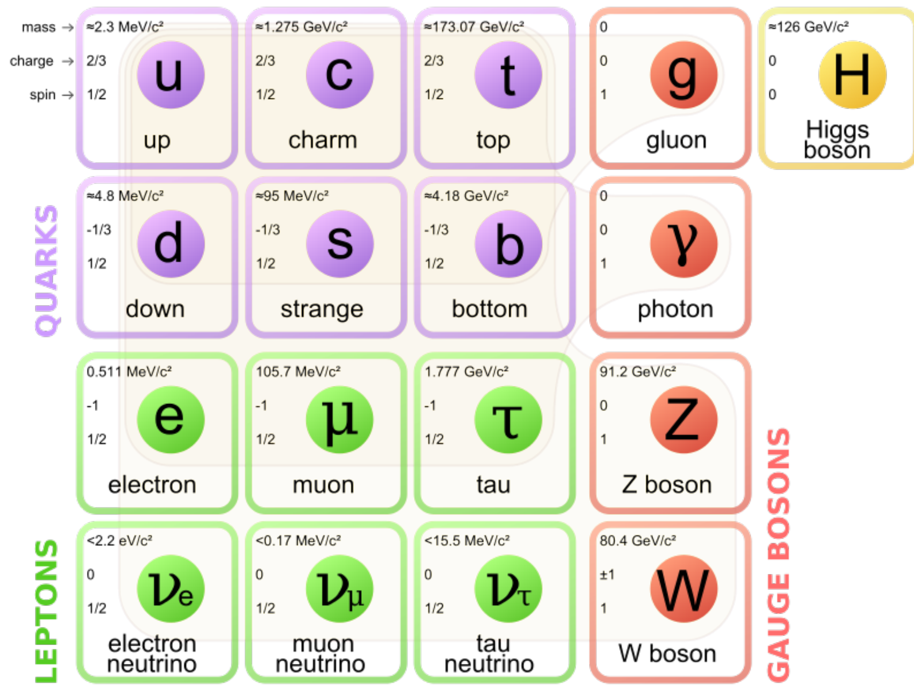



FIGURE 2.1: The elementary particles given in three generations of matter, the gauge bosons in fourth column and the Higgs boson in the fifth.

Quantum Chromodynamics (QCD) and Quantum Flavourdynamics (QFD) which are encoded in SM. Electromagnetic and weak interactions were then unified into electroweak (EWK) interaction with significant contributions by Abdus Salam, Sheldon Glashow and Steven Weinberg. The mathematical description of interaction of charged particles are described by QED. Photon of QED is the force carrier of electromagnetic force. QED is the first theory which explains the interaction between light and matter that patches up special relativity and quantum mechanics. Figure 2.2 shows an example of QED, electron-positron annihilation. Annihilation is the process of destruction of particle and anti particle pair to any set of particles obeying conservation of quantum numbers, energy and momentum. QED is perturbative theory based on fundamentals of quantum mechanics established by three distinguished physicists W. Heisenberg, W. Pauli and P. A. M. Dirac during 1920s. Later on, Freeman J. Dyson, Richard P. Feynman and Julian S. Schwinger showed that charged particle can possibly interact through a sequence of processes with growing complexity, and all of these processes could be described by Feynman developed graphical technique. QED is abelian gauge theory with $U(1)$ being its symmetry group.

Weak interaction, one of the four fundamental forces, was first revealed by the understanding of beta decay in 1933 by Enrico Fermi called Fermi's interaction (**Referencea**). Weak force is mediated by $W^{\pm}$ and $Z$ bosons. In terms of a field theory, it is represented by gauge theory with symmetry group $SU(2)$, having three gauge bosons $(W^{+}, W^{-}, Z)$. Weak force, rarely known from quantum

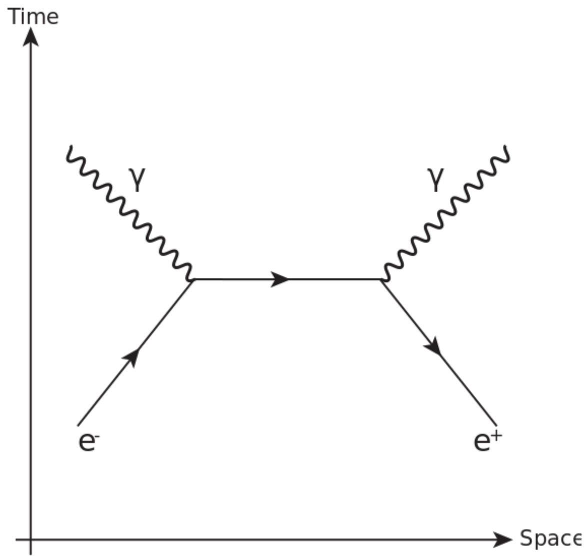



FIGURE 2.2: A Feynman diagram of annihilation of electron positron.

flavordynamics (QFD), is the only force which is proficient to change the flavor of quarks. This is very short range force because exchange bosons ($W^{\pm}$, $Z$) are massive. All fermions and the Higgs boson can interact via weak force. In 1970s, electromagnetic and weak forces were unified as electroweak force above unification energy (on the order of 100 GeV). The Physics Nobel Prize was awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their role in electroweak unification. In the field theory representation, Electroweak unification is settled under the gauge group $\mathrm{SU(2)_L \otimes U(1)_\gamma}$ . Experimental observation of existence of electroweak interaction was accomplished in two phases, the first was disclosure of existence of neutral current in scattering of neutrino in 1973 by Gargamelle collaboration and then the discovery of $W$ and Z gauge bosons in proton-antiproton collisions by both the UA1 and the UA2 collaborations in 1983 (**Reference2**; **Reference3**; **Reference4**; **Reference5**).

The representation of SM is made by $\mathrm{SU(3)_C \otimes SU(2)_L \otimes U(1)_\gamma}$ gauge symmetry. Where $\mathrm{U(1)_\gamma}$ is representation of QED, $\mathrm{SU(2)_L}$ represents weak theory and $\mathrm{SU(3)_C}$ represents QCD. By the unification of electromagnetic and weak force represented by $\mathrm{SU(2)_L \otimes U(1)_\gamma}$
gauge symmetry (**Reference6**), Glashow, Salam and Weinberg was able to formulate the fields to account the $W^{\pm}$, $Z$ and $\gamma$ (A) boson in footings of the original basis fields $A'$ and $B$ as expressed in equations 2.1, 2.2 and 2.3. The spontaneous breaking of $\mathrm{SU(2)_L \otimes U(1)_\gamma}$ gauge symmetry is primary reason for the generation of masses of the particles and separate electroweak (EW) interaction from electromagnetic (EM) and weak interactions. At higher energies($> 100\ GeV$), both weak and EM forces acts as similar style. Further details would be described in 2.3 section.

$$W^{\pm}_{\mu} = \sqrt{\frac{1}{2}}(A^1_{\mu} \pm iA^2_{\mu}) \qquad M_W = \frac{1}{2}g\mathrm{v} \tag{2.1}$$

$$Z_\mu^\pm = \sqrt{\frac{1}{g^2 + g'^2}}(gA_\mu^3 - g'B_\mu) \qquad M_Z = \frac{1}{2}\mathrm{v}\sqrt{g^2 + g'^2} \tag{2.2}$$

$$A_\gamma = \sqrt{\frac{1}{g^2 + g'^2}}(gA_\mu^3 - g'B_\mu) \qquad M_A = 0. \tag{2.3}$$

One of the the four fundamental forces is the strong force which is responsible for strong nuclear force which bounds the nucleus of an atom. As the name speaks for itself, strong force is strongest among four fundamental forces, although it act at a distance of $10^{-15}$ meter (femtometer) or less. Strong force is described by QCD and only exist between quarks and gluons which carry non-vanishing color charges. In the field theory representation of the SM, the strong force is represented by the symmetry group of $\mathrm{SU}(3)_\mathrm{C}$ . Here, C represents QCD color, which must be conserved in all such interactions. The strong interaction is mediated by gluons and contrary to the other fundamental forces, strong force does not diminish with gain in distance between quarks know as color confinement. As an implication of this property, no signature of elementary quark or gluon particle are observed in High Energy Physics (HEP) colliders, instead they bound with each other to form streams of hadrons by phenomenon known as hadronization called jets.

## 2.3 The Brout-Englert-Higgs Mechanism (BEH)

Gauge fields must acquire mass in order to correctly describe weak interaction and to be consistent with experimental findings. Before electroweak symmetry breaking (EWSB), all particles were supposed not to have masses, however mass term in electroweak Lagrangian for a gauge boson must be introduced to guarantee the renormalization of theory that would violate gauge invariance. Which hints that some basic thing is missing in the theory apart from fermions and gauge boson fundamental fields, which can maintain the local gauge invariance. The Higgs mechanism was postulated as a part of SM, which introduces the existence of a new scalar Higgs field. This field transformation ensures the renormalizability of the theory and gauge invariance. By this formulation the concept of breaking of symmetry and origin of masses of the fermions and bosons ($W^\pm, Z$ boson) are introduced (**Reference7**; **Reference8**; **Reference9**; **Reference10**; **Reference11**; **Reference12**). The Higgs mechanism introduces a doublet (under isospin) of complex scalar field

$$\phi = \begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^1 + i\phi^2 \\ \phi^3 + i\phi^4 \end{pmatrix} \tag{2.4}$$

which can be substituted in electroweak lagrangian given below

$$\mathscr{L}_{EWSB} = (D_\mu\phi)^\dagger(D_\mu\phi) + \mathrm{V}(\phi^\dagger\phi) \tag{2.5}$$

here $D_\mu = \delta_\mu - igt_aW_\mu^a + \frac{i}{2}g'YB_\mu$ represents covariant derivative and the lagrangian described in Equation 2.5 is gauge invariant under the $\mathrm{SU}(2)_\mathrm{L} \otimes \mathrm{U}(1)_\gamma$ transformation. The kinetic energy term in the lagrangian is written in terms of the covariant

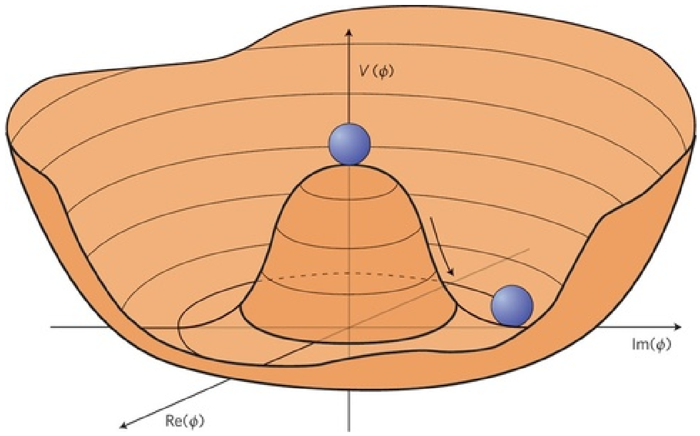



FIGURE 2.3: The Higgs potential as in Eq.2.6

derivatives whereas the potential only depends upon the product $\phi^\dagger\phi$. $\phi$ is the scalar field and can be codified in terms of quantum numbers which are isospin ($I_3$), hypercharge (Y) and electric charge (Q):

| | $I_3$ | Y | Q |
|---|---|---|---|
| $\begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix}$ | $\begin{pmatrix} \frac{1}{2} \\ \frac{1}{-2} \end{pmatrix}$ | $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ | $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ |

while the potential term in Equation 2.5 can also be expressed as

$$V(\phi^\dagger\phi) = -\mu^2\phi^\dagger\phi - \lambda(\phi^\dagger\phi)^2 \tag{2.6}$$

here $\mu$ is mass term and the parameter $\lambda$ represents the strength of the interaction. In the scenario, $\mu > 0$ and $\lambda > 0$ and the potential shapes becomes very interesting, which is essential for Higgs mechanism shown in Figure 2.3. There occurs a minima for

$$\phi^\dagger\phi = \frac{1}{2}(\phi_1^\dagger + \phi_2^\dagger + \phi_3^\dagger + \phi_4^\dagger) = \frac{-\mu^2}{2\lambda} \equiv \frac{V^2}{2} \tag{2.7}$$

but on the circle of points where the magnitude of z is $\phi$. This minimum does not exist at a single value of $\phi$, but at circular non-zero values.The ($\phi^\dagger\phi^0$) selection that affiliated with the ground state is arbitrary and in ($\phi^\dagger\phi^0$) plane the selected point is not invariant under rotations which is called spontaneous symmetry breaking. To settle the ground state on the $\phi^0$ axis, the vacuum expectation value (VEV) of field $\phi$ is

$$\langle\phi\rangle = \frac{1}{\sqrt{2}} = \begin{pmatrix} 0 \\ V \end{pmatrix}, \qquad V^2 = -\frac{\mu^2}{\lambda}. \tag{2.8}$$

The field $\phi$ given below, is derived by rewriting it in some generic gauge, on footings of its VEV:

$$\phi = \frac{1}{\sqrt{2}} e^{\frac{i}{\nu}\phi^a t_a \phi} = \begin{pmatrix} 0 \\ H + \nu \end{pmatrix}, a = 1, 2, 3 \tag{2.9}$$

here $\phi_a$ and $\phi_4 = H + \nu$ represents massless fields called Goldstone fields. New degrees of freedom is introduced along with six degrees because of polarizations vector bosons $W^{\pm}$ and $Z$, as they are massless and scalar. A unitary gauge transformation in the field is fixed by

$$\phi' = e^{-\frac{i}{\nu}\phi^a t_a}\phi = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ H+\nu \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ \phi^4 \end{pmatrix}$$

so here we have zero expectation value for the Higgs field.
Now the lagrangian in Eq. 2.5 can be rewritten and disintegrated into three components

$$\mathscr{L}_{EWSB} = \mathscr{L}_H + \mathscr{L}_{HW} + \mathscr{L}_{HZ} \tag{2.10}$$

where three terms is given, assuming approximation V $= \mu^2 H^2 + \cos t$ and ignoring higher order terms:

$$\begin{aligned}
\mathscr{L}_H &= \frac{1}{2}\delta_a H \delta^a H + \mu^2 H^2 \\
\mathscr{L}_{HW} &= \frac{1}{4}\nu^2 g^2 W_a W^{\dagger a} + \frac{1}{2} H W_a W^{\dagger a} \qquad (2.11) \\
&= m_W^2 W_a W^{\dagger a} + g_{HW} W_a W^{\dagger a} \\
\mathscr{L}_{HZ} &= \frac{1}{8}\nu^2 (g^2 + g'^2) Z_a Z^a + \frac{1}{4}\nu (g^2 + g'^2) H Z_a Z^a \qquad (2.12) \\
&= \frac{1}{2} m_Z^2 Z_a Z^a + \frac{1}{2} g_{HZ} H Z_a Z^a.
\end{aligned}$$

For $W^{\pm}$ and $Z$ bosons, mass terms are included in Eqs. 2.11 and 2.12. This adds up three additional degree of freedom; however, three of the total four Goldstone bosons have also been vanished from the Lagrangian $\mathscr{L}_{EWSB}$ not affecting overall number of degrees of freedom. A new scalar field has been introduced referred as Higgs field, introduced in the theory. The boson associated of Higgs field.
Briefly, the BEH mechanism ensures the renormalizability of theory and explain existence of the massive weak gauge bosons, despite of the fact that the fundamental gauge boson in theory still stays to be massless. In fact, the symmetry still exist in equation so using "hidden" is preferred over broken. According to BEH mechanism, fermions also acquire masses via their interaction with Higgs field. The invariance of the Higgs field minimum for $\mathrm{U}(1)_\gamma$ group shows that the electromagnetic symmetry is not broken spontaneously that keeps photon massless.

### 2.3.1 Masses and Couplings of Vector Bosons

Equations. 2.11 and 2.12 show that masses of $W^{\pm}$ and $Z$ bosons are associated with coupling constants of electroweak and a parameter $\nu$ which is VEV that defines a mass scale property of EWSB:

$$\begin{cases} m_W = \frac{1}{2}\nu g \\ m_Z = \frac{1}{2}\sqrt{g^2 + g'^2} \end{cases} \rightarrow \frac{m_W}{m_Z} = \frac{g}{\sqrt{g^2 + g'^2}} = \cos(\theta_W). \tag{2.13}$$

Couplings of Higgs boson to the bosons can also be obtained by Eqs. 2.11 and 2.12

$$g_{HW} = \frac{1}{2}\nu g^2 = \frac{2}{\nu}m_W^2 \tag{2.14}$$

$$g_{HZ} = \frac{1}{2}\nu(g^2 + g'^2) = \frac{2}{\nu}m_Z^2. \tag{2.15}$$

Using above Eqs. 2.14 and 2.14 helps to get the ratio of Higgs decay to $W^{\pm}$ and $Z$ bosons

$$\frac{\mathcal{B}(H \to W^+W^-)}{\mathcal{B}(H \to ZZ)} = \left(\frac{g_{HW}}{\frac{1}{2}g_{HZ}}^2\right) = 4\left(\frac{m_W^2}{m_Z^2}\right)^4 \approx 2.4. \tag{2.16}$$

At last, the characteristic mass scale for EWSB may be obtained by the relation between the Fermi constant $G_F$ and the $\nu$ parameter:

$$\nu = \left(\frac{1}{\sqrt{2}G_F}\right) \approx 246GeV. \tag{2.17}$$

### 2.3.2 Fermions Masses and Couplings

Fermions couples to Higgs field to acquire mass by Higgs mechanism. This can be achieved by introducing an invariant term $\mathrm{SU}(2)_\mathrm{L} \times \mathrm{U}(1)_\gamma$ (which is called yukawa term) in SM lagrangian which can describe interactions between Higgs and the fermion fields. $\phi$ is an iso-doublet and the fermions can be split into right-handed and left-handed singlet. So, the yukawa term for each fermion will have expression given below for leptons:

$$\mathscr{L}_\ell = -G_\ell.\bar{l}_\ell\phi\ell R + \bar{\ell}_R\phi^\dagger l_\ell. \tag{2.18}$$

A mass term will only evolve for second component of $l_\ell$ because first part of $\phi$ is zero. Actually, this also depicts that the neutrinos are massless.

$$\mathscr{L}_\ell = -\frac{G_{H\ell}}{\sqrt{2}}\nu\bar{\ell}\ell - \frac{G_{H\ell}}{\sqrt{2}}H\ell\bar{\ell}. \tag{2.19}$$

For quarks, down type quarks(i.e. down, strange and bottom) behave similar to the leptons while up type quarks (i.e. up , charm, top) should couple to the charge conjugate of $\phi$

$$\phi_c = -i\tau_2\phi^* = \frac{1}{\sqrt{2}}\begin{pmatrix} \phi^3 + i\phi^4 \\ -\phi^1 + i\phi^4 \end{pmatrix} \tag{2.20}$$

in unitary gauge that becomes

$$\phi_c = \frac{1}{\sqrt{2}} \begin{pmatrix} \eta + \nu \\ 0 \end{pmatrix}. \tag{2.21}$$

Therefore, the yukawa Lagrangian can be expressed as

$$\mathscr{L} = -G_{H\ell}\bar{L_L}\phi\ell_R - G_{Hd}\bar{Q_L}\phi d_R - G_{Hu}\bar{Q_L}\phi^c u_R + h.c.. \tag{2.22}$$

The couplings constant and mass of a fermion can be derived from Eq. 2.19, written as

$$m_f = \frac{G_{Hf}}{\sqrt{2}}\nu \tag{2.23}$$

$$g_{Hf} = \sqrt{G_{Hf}}\sqrt{2} = \frac{m_f}{\nu}. \tag{2.24}$$

The fermions mass can not be predicted by the theory as $G_{Hf}$ is free parameters. An fermion object acquires mass when Higgs field interacts with it to trans the right-handed chirality into the left-handed chirality and vice versa. The strength of the interaction of Higgs fields with particles is reflected by its mass.

This evaluation is rather interpretative, as it not accumulating the fact of three families of leptons and quarks in reality. The difference between eigenstates of weak mass and the eigenstates of physical mass makes real analysis complicated. Transformation of the weak mass basis to physical basis in essential to deal with the physical particles. For quark sector, it is achieved by Cabibbo-Kobayashi-Maskawa (CKM) matrices and in case of lepton sector, it is done by Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrices.

## 2.4 Higgs boson Production and its Decay

After electroweak unification, spontaneous symmetry breaking and addition of Higgs mechanism to SM solved all the puzzles with an assumption that there exists the Higgs boson. This boson was considered as the last missing subject of the SM and has been searched at LHC within full mass range allowed i.e from 114 GeV to 1 TeV (given unitary and above LEP constraints). Discussing the production and decay modes of Higgs would allow us to point out the most suitable decay channels Higgs boson discovery point of view and its properties measurements. Theory does not predict the Higgs mass; however, its couplings with bosons are proportional to the square of their masses as shown in Eqs. 2.14 and 2.15. In case of fermions, there is direct proportionality between their couplings with the Higgs and their masses as in Eq. 2.24. For this reason, the most promising channels for Higgs searches are its decays to $W^+W^-$, $ZZ$ and third generation of leptons. Also, photon being massless particle does not couple to Higgs directly but through loop contributions (most probable are top quark and $W^{\pm}$ boson loops).

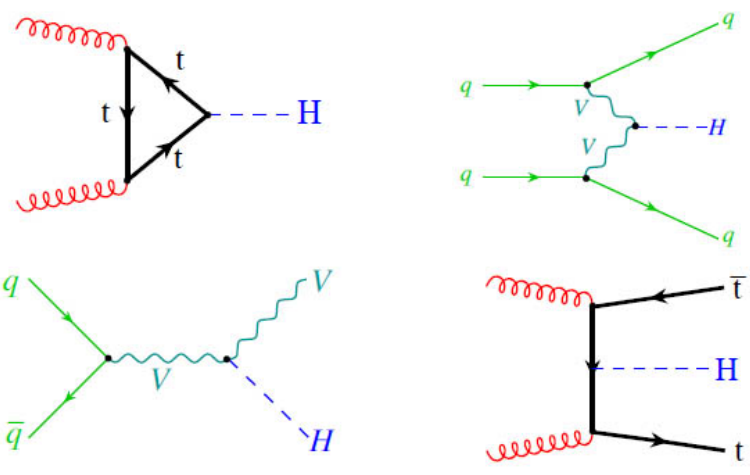



FIGURE 2.4: Feynman diagrams of Higgs production at tree level: gluon-gluon fusion (upper left);VV fusion (upper right); associated production with W and Z bosons (bottom left); associated production with $t\bar{t}$ (bottom right).

### 2.4.1 Higgs boson Production

The production modes of Higgs which are dominant in proton proton collisions at LHC are illustrated in Figure 2.4, corresponding production cross sections as a function of its mass at center-of-mass energies i.e. $\sqrt{s} = 7$, 8 and 13 TeV are shown in Figure 2.5. A considerable increase in production cross section of Higgs is found as we move towards higher center-of-mass energies (Figure 2.6 (**Referenceb**)).

#### Gluon-gluon Fusion

The most dominant production mode of Higgs at LHC is gluon-gluon fusion as illustrated in Figure 2.4 (top left) (**Referenceb**). As LHC and tevatron are hadron colliders, it highly probable that two gluons responsible for binding of hadron collide. The gluons couples to Higgs through the loop of virtual quarks, as coupling is directly proportional to fermion mass, top quark loop is most dominant. For SMHiggs ($m_H = 125$ GeV) the ggH cross section at 7 TeV, 8 TeV and 13 TeV are approximately 15, 19, 44 pb respectively. As we move towards higher energies the production cross section increases significantly, i.e 27% from 7 TeV to 8 TeV and 227% from 8 TeV to 13 TeV at $m_H = 125$ GeV. The cross section numbers in Table 2.3 are evaluated at NNLO (QCD) and NLO (EW) accuracies.

#### Vector Boson Fusion (VBF)

VBF is second most important production mode of Higgs at LHC and LEP as shown in Figure 2.4 (upper right). Its cross section is roughly 10 times lower than ggH at $m_H = 125$ GeV, and becomes comparable at $m_H = 1$ TeV. For SM Higgs at $m_H = 125$ GeV its cross section at 7 TeV, 8 TeV and 13 TeV are approximately 1.2,

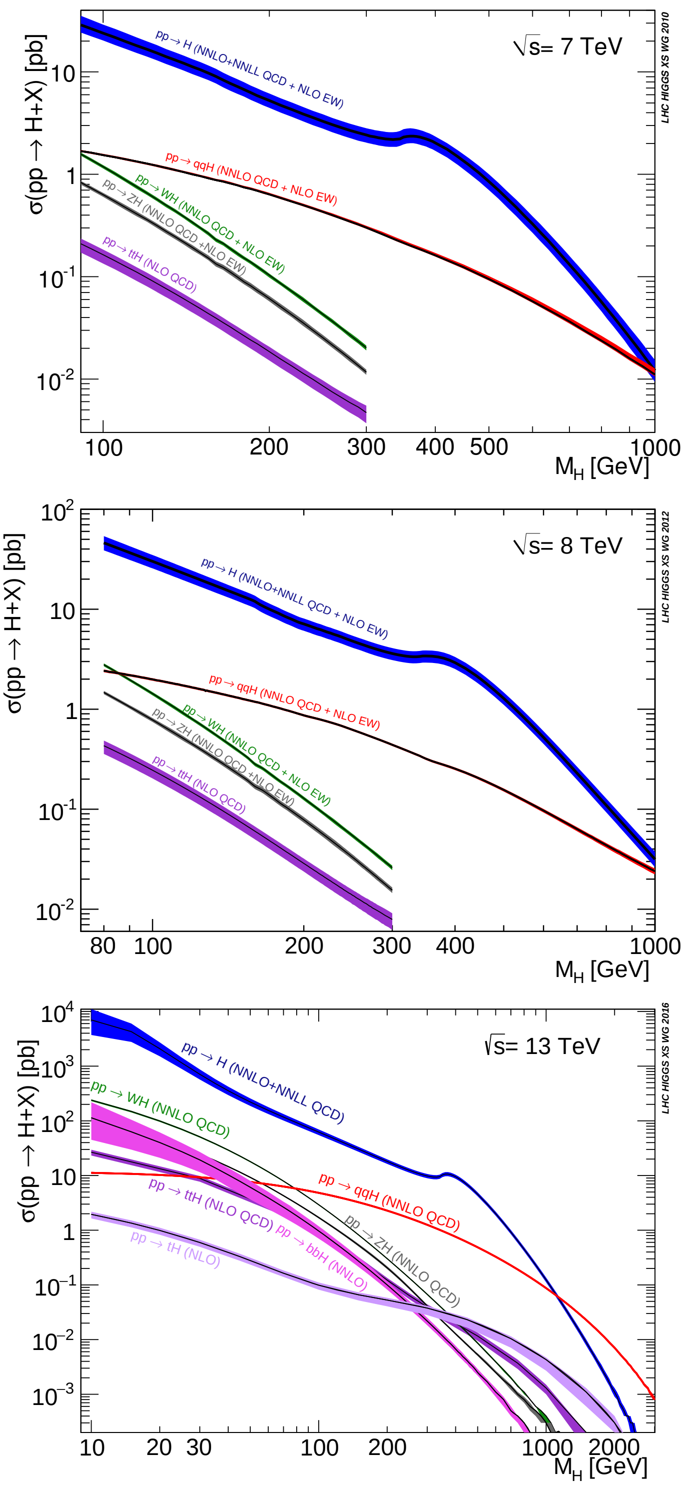



FIGURE 2.5: Production cross sections of SM Higgs boson at 7, 8 and 13 TeV as function of its mass.

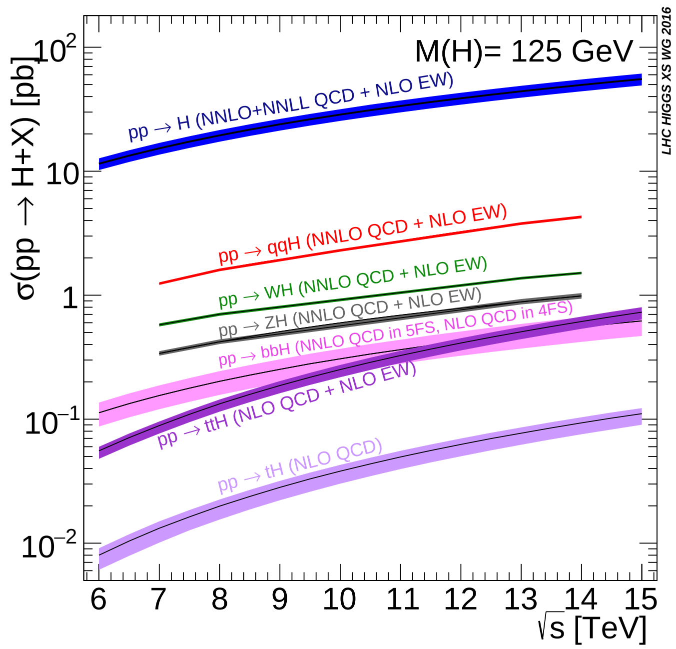



FIGURE 2.6: Production cross sections of Higgs boson ($m_H = 125$ GeV) as a function of center of mass energies

TABLE 2.3: Higgs boson production cross sections in pb for different Higgs masses and center-of-mass energies by ggH and VBF productions mechanism at LHC.

| $m_H(GeV)$ | ggH | | | VBF | | |
|---|---|---|---|---|---|---|
| | 7 TeV | 8 TeV | 13 TeV | 7 TeV | 8 TeV | 13 TeV |
| 120 | 16.61 | 21.07 | 47.38 | 1.30 | 1.68 | 3.9 |
| 125 | 15.31 | 19.47 | 44.14 | 1.24 | 1.60 | 3.78 |
| 130 | 14.15 | 18.03 | 41.23 | 1.19 | 1.53 | 3.64 |

1.6, 3.8 pb respectively. The cross section numbers in the Table 2.3 are computed at NNLL QCD and NLO EW precision (**Referenceb**). In VBF production mode, two $W^{\pm}$ or $Z$ boson fusion create Higgs that have been emitted by a (anti)fermion pair collision. The VBF experimental signal is very clean and can be well distinguished from ggH with two high energy jets in detected in forward region. These jets signature helps to increase signal to background ratio.

### Production of Higgs in association with W and Z bosons

Such production of Higgs is known as Higgs Strahlung as illustrated in Figure 2.4 (bottom left). In this process, Higgs is radiated by high energetic $W^{\pm}$ or $Z$ bosons that is produced by fusion of two fermions. This production mode was leading mechanism at LEP and sub-leading mechanism at Tevatron. At LEP it becomes very

TABLE 2.4: Higgs boson production cross section in pb for different Higgs masses and center-of-mass energies by associative productions mechanism at LHC.

| $m_H(GeV)$ | WH | | | ZH | | | ttH | | |
|---|---|---|---|---|---|---|---|---|---|
| | 7 TeV | 8 TeV | 13 TeV | 7 TeV | 8 TeV | 13 TeV | 7 TeV | 8 TeV | 13 TeV |
| 120 | 0.66 | 0.81 | 1.57 | 0.39 | 0.48 | 0.99 | 0.11 | 0.15 | 0.57 |
| 125 | 0.58 | 0.70 | 1.37 | 0.34 | 0.42 | 0.88 | 0.09 | 0.13 | 0.51 |
| 130 | 0.51 | 0.62 | 1.21 | 0.30 | 0.37 | 0.79 | 0.08 | 0.12 | 0.45 |

probable to produce $W^{\pm}$ or $Z$ boson with electron positron collisions. In case of hadron collider, tevatron makes quark antiquark collisions more likely as compared to LHC. AT LHC this production mechanism is at third number. The WH and ZH cross section for Standard Model Higgs at center-of-mass energies of 7 TeV, 8 TeV and 13 TeV are listed in Table 2.4 which are computed at NNLL QCD and NLO EW precision (**Referenceb**).

#### Top Fusion

The least dominant process of Higgs production is illustrated in Figure 2.4 (bottom right). In this process two gluons collide producing pair of top antitop which produces Higgs by fusion in association $t\bar{t}$. The ttH cross section for Standard Model Higgs at center of mass energies of 7 TeV, 8 TeV and 13 TeV are listed in Table 2.4 which are computed at NLO QCD and NLO EW precision (**Referenceb**).

### 2.4.2 Higgs Decay

As Higgs boson is electroweak particle, so it wants to decay quickly. Likelihood of its decay depends on many aspects, i.e. its mass, interaction strength and etc. Most of the factors are controlled by SM except the Higgs boson mass itself. The life time of SM 126 GeV Higgs is $1.6 \times 10^{-22}$ s. It can decay to all SM particles with mass but it likes to decay most heavier particles available. Higgs boson decay branching ratio to a variety of particles are illustrated in Figure 2.7 (**Referenceb**).

$H \to b\bar{b}$ (at $m_H \approx 125$ GeV) is the most dominant decay mode of Higgs. But the QCD background in this channel is of many order higher than signal which makes it less sensitive channel. However, CMS and ATLAS collaborations and other collaborations (**Reference13**), (**Reference14**) and (**Reference15**)(CDF and D0) have exploited this channel, in association with a vector boson which decays to leptons, in boosted regime. B quarks are also difficult to reconstruct in a detector. When they decay in the detector, they fragment into jets of many hadrons. These jets are difficult to reconstruct perfectly. Because of this, it is necessary to exploit other channels in this region. $H \to \gamma\gamma$ is one most favorable decay channel due to signature with two high energy photons with excellent mass resolution.

The main backgrounds in this channel are $q\bar{q} \to \gamma\gamma$, gg $\to \gamma\gamma$ and instrumental background where one or two jets are misidentified to be photons. As the Higgs mass at 125 GeV, Higgs boson decay to diboson($WW$, $ZZ$) boson are also available,

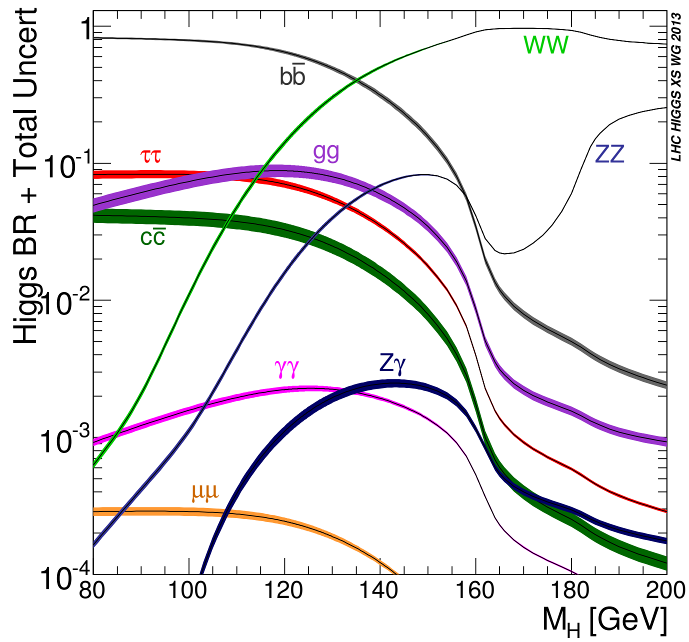



FIGURE 2.7: Branching ratios for different Higgs boson decay channels as a function of the Higgs mass.

where one or both boson can be off-shell. In case of $WW$ decay, its impossible to completely reconstruct $WW$ boson in detector as it decays $H \rightarrow WW \rightarrow \ell\nu\ell\nu$ have two neutrinos and $H \rightarrow WW \rightarrow \ell\nu jj$ have one neutrino and two jets. These objects pulls mass distribution to be wide and signal to background ratio to be low, which makes $WW$ decay channel less sensitive. While $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$ decay is labeled as golden channel in spite of low branching ratio. The clean signature of at least four leptons makes this channel almost background free. These leptons are high energetic leptons which can reconstructed with excellence in the detector. The branching ratio of the Higgs in intermediate range is illustrated in Figure 2.8 (**Referenceb**).

## 2.5 Experimental Observations

The electroweak unification theory lead to series of impressive experimental discoveries. In 1973, the evidence of neutral current through inelastic scattering was found by Gargemelle buddle chamber collaboration at CERN (**Reference16**). This evidence provide starting platform to the Standard Model of electroweak interaction.

Super Proton Synchrotron proton-atniproton collider at CERN played its role to next breakthrough that came a decade later. The existence of $W$, $Z$ boson announced by UA1 and UA2 collaboration through the channels $Z \rightarrow \mu^+\mu^-$, $Z \rightarrow e^+e^-$ and $W \rightarrow e\nu$ (**Reference2**; **Reference3**; **Reference4**; **Reference5**). This was a

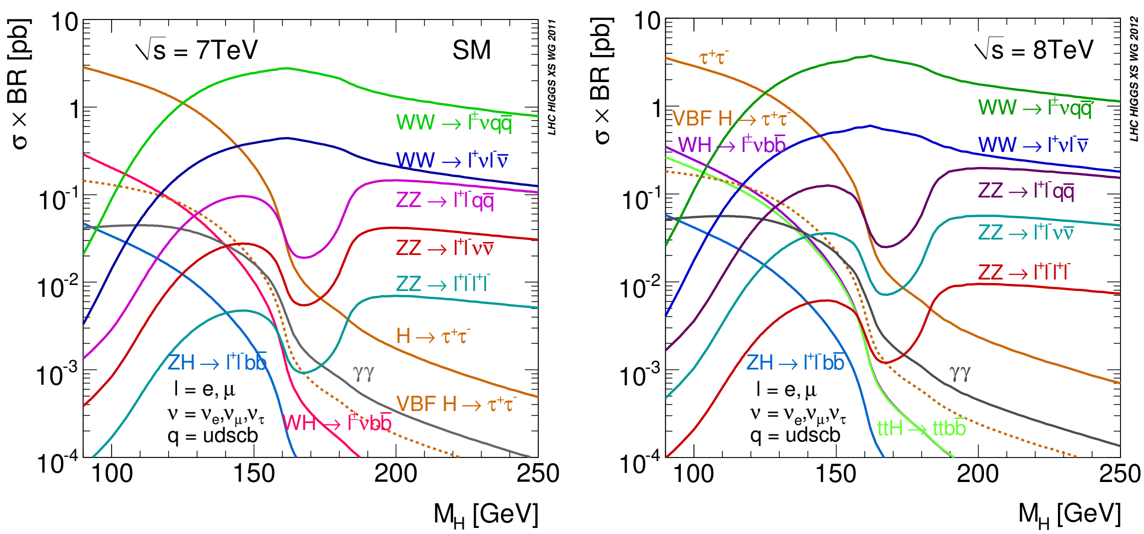



FIGURE 2.8: Cross sections of Higgs times branching ratios at 7 TeV and 8 TeV (left and right diagrams respectively).

first observation of $W, Z$ boson that mediates weak force. For their contributions, Simon van dar Meer and Carlo Rubbia were awarded with Nobel Prize for their contribution in 1984.

These important experimental discoveries and growing knowledge on weak interaction convinced to construct the LEP collider. at CERN, for deeper look into $W, Z$ further decays, their precise mass measurement (**Reference17**; **Reference18**). LEP was equipped with four detectors around the ring; OPAL (Omni-Purpose Apparatus for LEP), DELPHI (Detector with Lepton and Hadron Identification), ALEPH(Apparatus for LEP Physics) and L3 (Letter of Intent 3). These experiments reported many precise results of the SM, specially masses of W and Z bosons and confirming three neutrino species consistent with Standard Model (**Reference18**).

Around 2000, construction for the Large Hadron Collider (LHC) (**Reference19**) at CERN was started to look for last missing patch of SM, the Higgs boson. LHC is equipped with four major detectors; ATLAS (A Toroidal LHC Apparatus), CMS (Compact Muon Solenoid), ALICE (A Large Ion Collider Experiment), LHCb (Large Hadron Collider beauty). LHC had shown brilliant performance since being operational. It has already recorded data of proton-proton collisions at center-of-mass energies 7,8 and 13 TeV in from 2011 to 2018. After a long search , the CMS and ATLAS collaboration of LHC, certified the presence of the Higgs, a heavy scalar boson, on 4 July 2012 (**Reference20**; **Reference21**). Further test to measure of the properties of this scalar boson immediately begins, in March 2013, CERN announced the tentative confirmation of consistency of its properties with SM (**Reference22**; **Reference23**; **Reference24**; **Reference25**).

## 2.6 Beyond Standard Model

Although SM has passed numerous experimental test with great precision, it is not considered to be theory of everything. There are many phenomena which are not explained by SM i.e Hierarchy problem, lack of direct detection of dark matter and dark energy, and masses of neutrinos etc. There are several theoretical extension posed in Standard Model such as Minimal Super Symmetrical Model(MSSM) and Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical description, such as string theory, extra dimension. The simplest broadening of Higgs sector of SM is the the electroweak singlet (EWS) model (**Reference26**). In this model, a heavy singlet is defined along with SM scalar Higgs field and then two CP-even Higgs bosons comes out as result of spontaneous symmetry breaking. Another generic extension to SM Higgs sector is 2 Higgs Doublet Model (2HDM), which introduces two scalar, complex doublet $\phi_1$ and $\phi_2$ (**Reference27**). Spontaneous symmetry breaking results five Higgs boson, a neutral CP-odd one (A), two charged ($H^{\pm}$) and two CP-even particles (h and H). Experiments are looking for any direct or indirect hints of new physics. Some of main aspects where SM is unable to explain are discussed below.

### 2.6.1 Hierarchy Problem

Larger discrepancy between features of the gravitational and the weak forces is known as "Hierarchy Problem". There does exist a convincing explanation of the problem why weak force is $10^{32}$ times stronger than gravitational force. The field theory breaks down because of existence of quantum gravity at plank scale ($M_p \approx 10^{19}$ GeV). Writing the Standard Model Lagrangian for the Higgs mass will lead to following expression below

$$m_H^2 = m_{oH}^2 + \delta m_H^2 \tag{2.25}$$

here $m_{oH}^2$ represents mass parameter exist in the Standard Model Lagrangian, and $\Delta m_H^2$ represents the quantum radiative corrections to the mass that are predicted to be heavier. These corrections can be found as

$$\Delta m_H^2 \approx \frac{\lambda^2}{16\pi^2}\Lambda^2 \tag{2.26}$$

here $\lambda$ represents the Yukawa coupling constant, and $\Lambda$ represents the energy at which the Standard Model is invalid, or about 1 TeV. This was also one of the reason to expect Higgs boson mass at order of few hundred GeV. The Super symmetry (SUSY) provide the solution of Hierarchy problem by predicting the existence of heavier super partners of all SM particles (bosons and fermions). Several direct and indirect searches has been conducted at LHC to hunt for SUSY, but no evidence has been found so far.

### 2.6.2 Dark Matter and Dark Energy

Latest observation from cosmological experiments shows that universe is composed of ordinary matter, dark matter and dark energy with percentage as 4.9%, 26% and 70% respectively. SM does not provide any clue about the presence of dark matter and dark energy.

### 2.6.3 Matter Antimatter Asymmetry

Today, if we look around in the universe, we see matter dominance over antimatter. Matter and anti-matter are supposed to be produced in equal quantities at the time of Big-Bang. This gives rise to one of the greatest mysteries, where all antimatter has gone. Standard Model has unable to interpret this imbalance between matter and the antimatter.

### 2.6.4 Grand Unification and Beyond

One of the main goals of the particle physics is to unify all fundamental force at some energy scale to describe the theory of everything as Standard Model unify electromagnetic and weak force in to electroweak. But Standard Model does not provide any clue about unification of gravity along with other fundamental force generally referred as Grand Unification Theory (GUT). This implies that Standard Model itself is a patch of Grand Unification. Search for any new physics (beyond SM) is important and justified that could improve our understanding about these unsolved mysteries.

## 2.7 Higgs Boson as a Probe to BSM

After the discovery of the Higgs boson, it became very natural and important to measure Higgs boson properties as precise as possible. Measurements of its properties provide a direct probe into SM Higs sector of perturbative QCD calculations at TeV energy scale.

Couplings of Higgs boson with fermions are predicted by SM through Yukawa interactions. Through Yukawa interactions with Higgs, quarks and leptons get masses. While in SM the coupling strength of Higgs particle with other particles is directly proportional to mass of that particle. The masses of leptons and quarks are well know by experiments, so it possible to estimate the decay rates of Higgs to fermions precisely. Experimental measurements can test these prediction which will provide a direct hint to BSM if any deviation is observed.

The cross section measurements of the Higgs boson in a process illustrate the possibility of occurrence of that process. Any deviation from SM prediction in integrated or differential cross sections will guide to BSM effects. The observation and studies of the Higgs boson production like gluon gluon fusion ($\mathrm{gg} \rightarrow \mathrm{H}$), vector boson fusion (VBF), production in association with a vector boson (WH, ZH) and with a pair of top quarks ($t\bar{t}$H) also shed light on its coupling with vector boson and top quark etc.

# Chapter 3

# Experiment

## 3.1 The Large Hadron Collider (LHC)

LHC is a particle accelerator of the world that has tendency to smash two protons beams at the energy up to 7.0 TeV each beam. It is the most advanced, complex and largest experiment in the history of mankind, designed and constructed by a collaborative team of more than 10,000 technician, engineers and scientists from over 100 countries. The LHC was constructed at CERN from 1998 to 2008, in a tunnel of 27 Kilometers (17 mi) circumference below the France-Switzerland border close to Geneva, Switzerland. The LHC is accommodated in a 3.8 meters wide circular tunnel at a depth varies from 75-150 meters underground. Previously, this tunnel also bore Large Electron-Positron collider which was constructed from 1983 to 1988.

The first beam of protons was injected into LHC on September 10, 2008. The LHC was able to fire the protons around the tunnel in the segments of three kilometers each time. The initial test to circulate the beams in clockwise and counterclockwise throughout the collider was also successfully completed by the same day. On 19 September 2008, an electric defect triggered the magnet quench in about a hundred bending magnets. About six tonnes of liquid helium was lost due to which 50 superconducting magnets and their mounts were damaged. Three months later, CERN unfolded the details of the incident and published technical report. Particle accelerator was repaired after this incident that took about an year. Finally, on November 20, 2009, LHC became operational again and a low energy beam was successfully rotated within tunnel. On November 30, shortly after, LHC became world's strongest particle accelerator after having achieved 1.18 TeV energy each beam thus defeating the Tevatron's highest achieved energy of 0.98 TeV. In 2010, LHC kept boasting the beam energies and reached center of mass energy up to 7 TeV (3.5 TeV each beam). In 2011 and then in 2012, LHC consistently delivered data at 7 and 8 TeV energies respectively. During 2013-14 , LHC machine was shutdown to do major parts upgrade on the LHC machine and detectors installed on it. In 2015, upgraded LHC started again at 13 TeV close to full stipulated operating energies (14 TeV). After successfully delivering proton-proton collisions data during Run 2 till the end of 2018, LHC machine is shutdown for Long Shutdown 2 (LS2) upgrades and is expected be operational again for Run3 in 2021 after several upgrades.

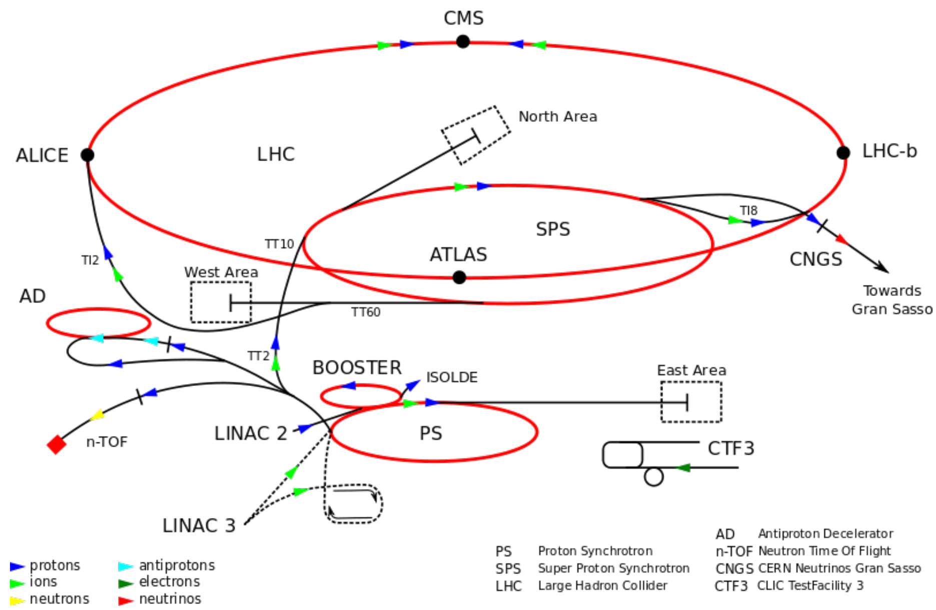



FIGURE 3.1: LHC experiment at CERN

### 3.1.1 Machine Layout and Performance

#### Machine Layout

Basic layout of LHC and LEP machines are similar, having eight linear and eight arcs sections as elaborated in Figure 3.2 (**Reference28**). The LHC is designed to have two separate rings containing clockwise and counterclockwise proton beams, equipped with separate magnetic fields and vacuum chambers. These two rings only intersect each other at the points where detectors are installed. Point 1 and point 5, opposite to each other, where ATLAS and CMS, two general purpose detectors of LHC, are installed respectively. The other two important detectors, ALICE and LHCb, are installed at point 2 and point 8 which are designed to study heavy ion Physics and B Physics respectively. Two protons beams are injected in the vertical plane in both rings close to point 2 and point 8. Among 8 sections, there are four with no beam crossing at all. Point 3 and point 7 are equipped with two independent collimation systems for both rings. At point 4, two RF system are installed one for each beam. While at point 6, kicker magnets are installed to dump the beams.

#### LHC Performance Goals

LHC design goal is to test SM Physics to ultimate precision and hunt for any kind of new physics up to 14 TeV. Number of events (N) for a certain process produced at LHC are given as

$$N = \mathscr{L}\sigma_{process} \tag{3.1}$$

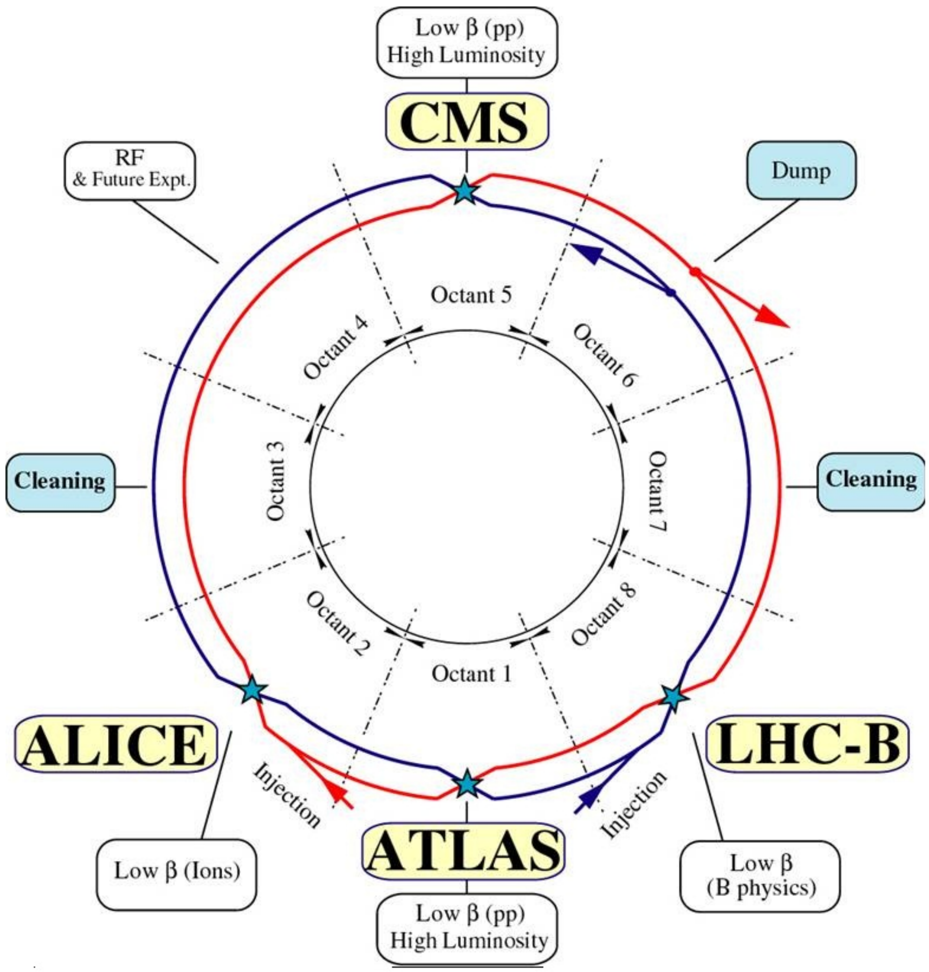



FIGURE 3.2: The Schematic layout of the LHC

Here $\mathscr{L}$ represents machine luminosity and $\sigma_{process}$ is the production cross section of a certain process. All possible processes' cross sections produced during proton-(anti) proton collisions at LHC machine luminosity $\mathscr{L} = 10^{33}\ cm^2\ s^{-1}$ are shown in Figure 3.3. Higgs production at LHC facility is rare as compared to other QCD and weak boson $(W, Z)$ productions which act as backgrounds in the Higgs boson searches. In order to reconstruct Higgs boson at LHC, one need to exploit specific final states providing enough signal to background discrimination. From Figure 3.3 its clear that Higgs boson production rises with increase in center-of-mass energies. So increase in machine luminosity and center-of-mass energies enables us to get more statistics of Higgs boson. The Nominal luminosity for CMS and ATLAS experiment is $\mathscr{L}$ = $10^{34}$cm$^2$s$^{-1}$, and nominal center of mass energy of LHC is $\sqrt{s} = 14$ TeV. The cross section in Figure 3.3 (**Reference29**) shows that LHC has possibility to produce a single Higgs boson after every 100 s at mass 500 GeV. To get ultimate number of Higgs bosons produced at certain luminosity, one need to multiply this with specific branching ratio, selection and reconstruction efficiencies.

### Nominal Luminosity and Beam Parameters

The machine luminosity can be expressed in terms of beam parameters as:

$$\mathscr{L} = \frac{\gamma_r N_b^2 f_{rev} n_b}{4\pi\epsilon_n\beta^*} F \tag{3.2}$$

here $\gamma_r$ represents relativistic factor; however, $N_b$ and $n_b$ are the number of particles in a bunch and number of bunches in a beam respectively. While $\beta^*$ represents the beta function at the collisions point and F is the factor to account for geometrical depletion in luminosity due to cross angle at interaction point (IP). This factor can be written as below

$$F = \left(1 + (\frac{\theta_c \theta_X}{2\sigma^*})^2\right) \tag{3.3}$$

where $\theta_c$, $\theta_z$ and $\sigma^*$ are crossing angle at point of interaction, root mean square (RMS) length of bunch and $\sigma^*$ the transverse RMS size of beam at IP. The expression above is valid for two circular beams with same parameters.

The two major experiment of LHC, the CMS (**Reference30**) and the ATLAS (**reference31**), are designed to operate at luminosity of $\mathscr{L} = 10^{34} cm^2 s^{-1}$. The summary of the parameters to obtain this nominal luminosity is summarized in Table 3.1. Such high luminosity can not be attained by proton-anti-proton collisions, leaving only possibility of proton proton collisions for LHC machine. To align two opposite moving proton beams in both rings for collisions needs dipole magnetic with opposite fields direction. The LHCb (**Reference32**) detector is specially designed to study B Physics with peak luminosity $\mathscr{L} = 10^{32} cm^2 s^{-1}$. And the ALICE (**Reference33**) is the detector to study heavy-ion collisions targeting highest luminosity $\mathscr{L} = 10^{27} cm^2 s^{-1}$.

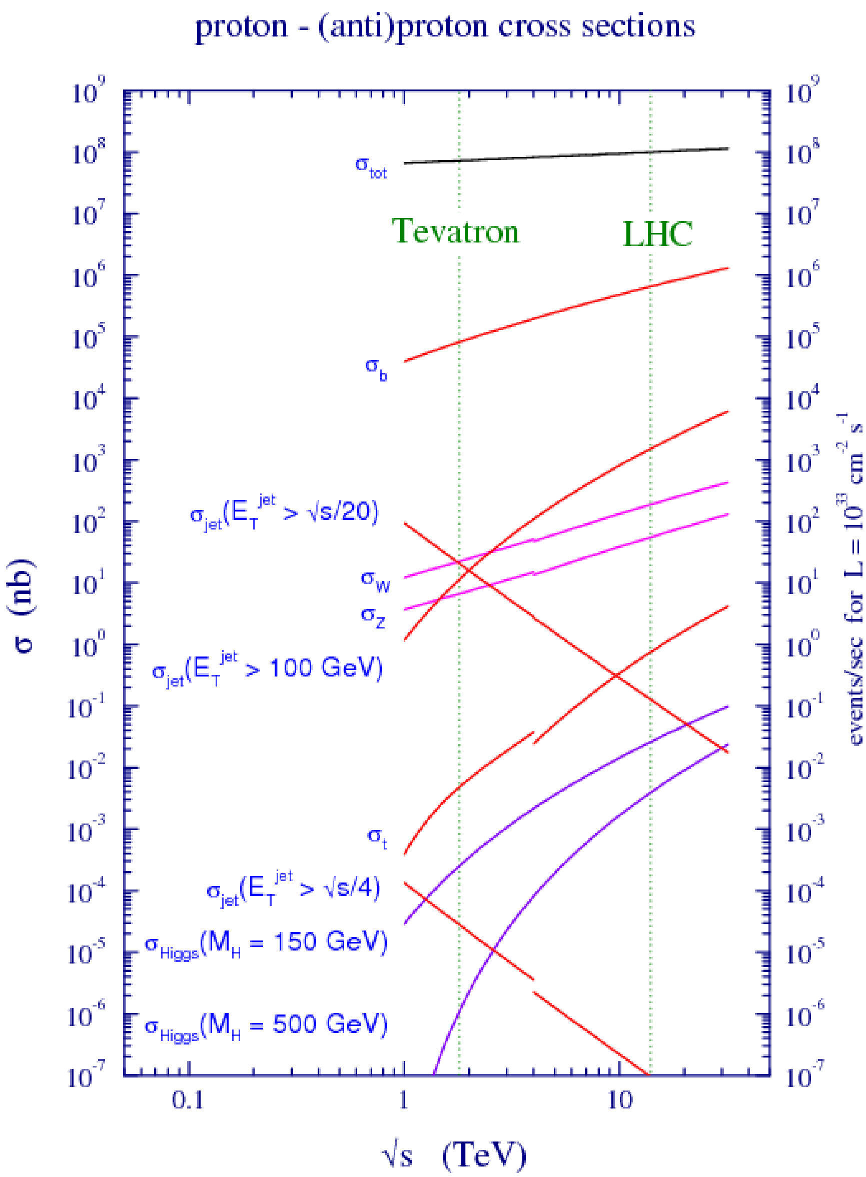



FIGURE 3.3: Cross section values for different process corresponding to machine luminosity $\mathscr{L} = 10^{33}\, cm^2\, s^{-1}$.

TABLE 3.1: Parameters values to obtain LHC nominal luminosity for detectors.

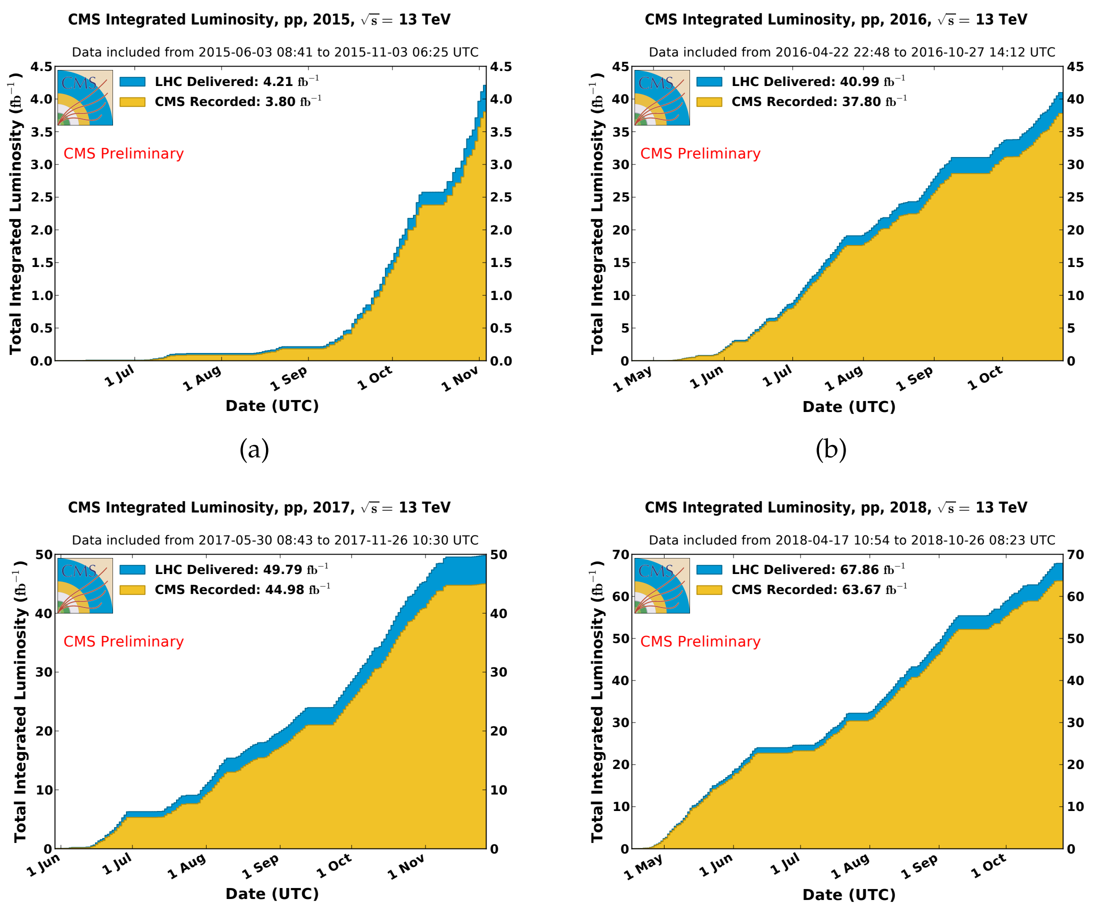
| Parameter | Value |
|---|---|
| circumference | 26.7 km |
| center of mass energy ($\sqrt{s}$) | 14 TeV |
| Luminosity (nominal) | $10^{34} cm^2 s^{-1}$ |
| Bunch spacing | 25 ns |
| distance between bunches | 7.5 m |
| No. of bunches in ring | 2808 |
| No. of protons in a bunch | $1.15 \times 10^{11}$ |
| beta function at the collision point($\beta^*$) | 0.55 m |
| Transverse RMS size of beam at impact point($\sigma^*$) | 16.3 $\mu$m |
| Magnetic field strength at 7 TeV(B) | 8.33 T |
| Temperature of magnet (T) | 1.9 K |

### 3.1.2 Data Taking from 2015 to 2018 (Full Run 2)

After successful data taking during Run 1, LHC machine was shutdown for two years in 2013 and 2014 for major upgrades in LHC machine and detectors. LHC started again in 2015 at center-of-mass energies to 13 TeV(close to its design value i.e 14 TeV). LHC delivered 4.2 $\mathrm{fb}^{-1}$during 2015 at 13 TeV center of mass energy and subsequently, it delivered 41.0 $\mathrm{fb}^{-1}$, 49.8 $\mathrm{fb}^{-1}$and 67.9 $\mathrm{fb}^{-1}$during the years 2016, 2017 and 2018 as shown in the Figure 3.4(a), 3.4(b), 3.4(c) and 3.4(d) respectively.

## 3.2 The Compact Muon Solenoid (CMS) Experiment

The CMS detector is 28 m in length, 15 m in height and 14,000 tonnes heavy general purpose particle detector installed on LHC ring. It is capable to study large areas from SM Physics (including Higgs Physics) to search for extra dimensions and the dark matter. CMS is efficient enough to cover a large range of physics with energy ranges from a few 100 GeV to TeV scale (**Reference34**). CMS and ATLAS experiment shares the same physics goal though both independent experiment uses different technical solution and detectors design. It is apparent from Figure 3.3 that the total cross section proton-proton collisionis at $\sqrt{s} = 14$ TeV is approximately 100 mb. While designed LHC luminosity leads to have $10^9$ inelastic events/s approximately. It is huge experimental challenge to deal with such high rate collisions. CMS trigger system helps to select online events and reducing the huge rate to approximately 100 events/s to be stored and used for physics analysis later. The read-out and trigger system designed to meet the short bunch crossing time, 25 ns.
Being CMS as general purpose detector, followings are the main requirements to fulfill physics goal of LHC:




(a) (b)

(c) (d)

FIGURE 3.4: Integrated luminosities of LHC delivered (blue) and CMS recorded (yellow) data for year 2015(a), year 2016(b), year 2017(c) and year 2018(d).

- Good resolution for muon identification and momentum over wide range of dimensions and determination of the charge on muons with ultimate precision with momentum up to 1 TeV.
- Healthy inner tacker reconstruction efficiency and resolution of charged particle momentum. Efficient trigger system and better discrimination of $\tau$'s and b-jets.
- Good dielectron and diphoton mass and energy resolutions.
- Good dijet-mass and missing-transverse-energy (MET) resolutions .

Choice of magnetic field configuration is very important feature of a particle detector. Stronger magnetic field is needed to measure high energy charged particle momentum with precision. In CMS, this is achieved by selecting the superconducting magnets.

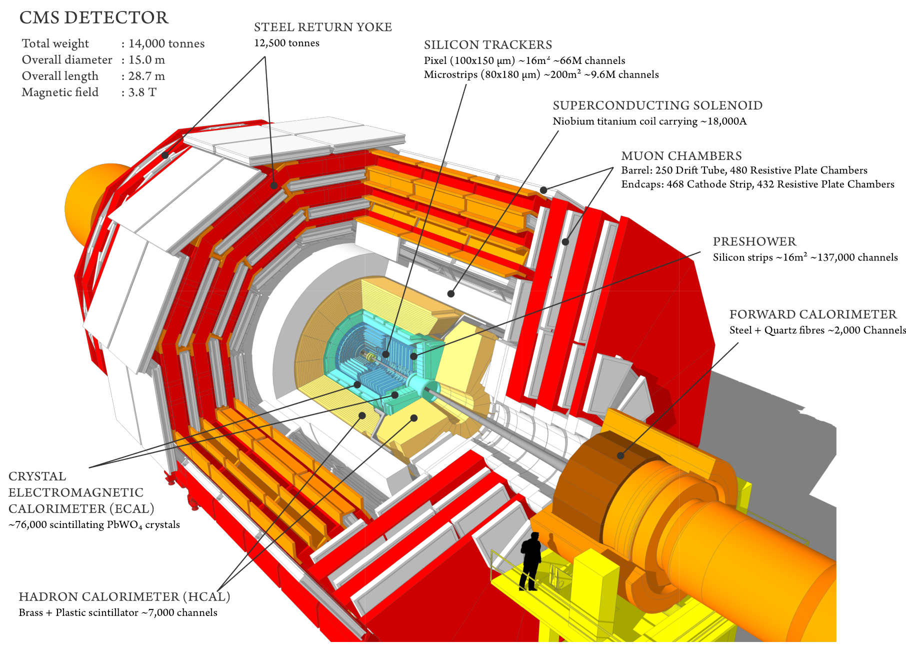



FIGURE 3.5: The CMS detector sectional view.

CMS overall design is shown in Figure 3.5. CMS magnet is superconducting solenoid which is 13m length with inner diameter of 6m. It is placed at the heart of CMS and is capable to produce a magnetic field of 3.8 T. The inner tracker system (ITS), electromagnetic calorimeter (ECAL) and hadronic calorimeter(HCAL) are the sub-detectors of CMS which are fitted inside the large solenoid magnet. Inner tracker of CMS have large track multiplicity as being closest to beam pipe where collisions happens. ECAL is build with crystals of lead tungstate which covers pseudorapidity up to $|\eta| < 3.0$. After ECAL, hadronic calorimeter is installed which is made of brass. Outermost layer of the CMS detector is the Muon System. Details of sub-detectors, trigger system and data acquisition systems will be described in next Sections.

### 3.2.1 CMS Detector Geometry

The coordinate system would be described in this Section and will be used throughout. The CMS detector is like a closed cylinder around beam axis (Z axis) as illustrated in Figure 3.5. The center of the experiment lies inside beam pipe where two beams collides; the y axis is directed vertically upwards and the x axis being horizontal towards the LHC ring center. The $r$ represents the radial component in transverse (xy) plane while $\phi$ is the the azimuthal angle measured from the x axis. $\theta$ is the polar angle which is measured from the z axis as shown in Figure 3.6. Direction of any particle within CMS is described by $(\eta, \phi)$, where $\eta$ being the pseudo-rapidity defined as $\eta = -\ln\tan(\theta/2)$. LHC is colliding facility of protons

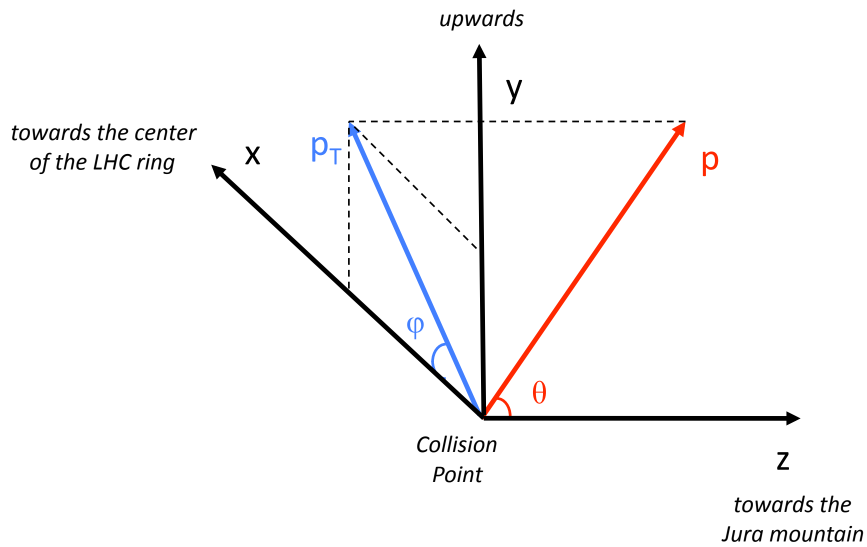



FIGURE 3.6: The CMS Experiment coordinate system.

mainly, which contains partons (i.e. quarks and gluons) carrying different fractions of proton's total momentum. As a results, events in CMS detector will have boosted center-of-mass as compared to lab frame along z axis. New defined quantity pseudorapidity is invariant under these boosts. In short, pseudorapidity is approximation of rapidity in case of high energetic particles which are assumed to be massless and is defined below

$$y = \ln(\frac{E + p_z}{E - p_z}) \tag{3.4}$$

$$\approx \ln(\frac{1 + \cos\theta}{1 - \cos\theta}) \tag{3.5}$$

$$= \ln(\cot^2\theta). \tag{3.6}$$

Pseudorapidity also helps to differentiate the barrel regions from the endcap regions of sub detectors. In terms of pseudorapidity, the barrel region are defined by $|\eta| < 1.5$ whereas the endcap region corresponds to $1.5 < |\eta| < 3.0$. Formally, electromagnetic calorimeter and tracking system provides a coverage a region up to $|\eta| < 2.5$, while muon station covers up to $|\eta| < 2.4$. So, in CMS muons and electrons are selected with $|\eta| < 2.4$ and $|\eta| < 2.5$ respectively. As there is no transverse momentum (along y axis) before collisions which leads to have transverse momentum approximately zero. The transverse energy of any particle is orthogonal to z axis, defined as $E_T = E\sin\theta$ and similarly the transverse momentum $p_T = p\sin\theta$. For a massless particle approximation, electrons and muons, transverse momentum is equal to transverse energy i.e. $E_T = p_T$.
As the CMS is hermetic, the energy of those particle which escaped from detector

TABLE 3.2: Types of detectors installed in CMS tracking system.

| Section | Radial distance (cm) | Detector type | Cell size |
|---|---|---|---|
| Pixel | $r < 10$ | pixel | $100\mu m \times 150\mu m$ |
| TIB+TID | $20 < r < 55$ | microstrips | $10cm \times 150\mu m$ |
| TOB+TEC | $55 < r < 110$ | strips | up to $25cm \times 180\mu m$ |

is understood as missing transverse energy. These particles can be new physics particles, neutrinos and neutralinos which have no or very little interaction with detector material.

### 3.2.2 Inner Tracking System (ITS)

The CMS ITS is most inner sub-detector surrounding the beam pipe and region of interaction as shown in Figure 3.5. Its is 2.5 m in diameter and 5 m in length feeding $|\eta|$ coverage up to 2.5. The purpose of inner tracking system is to determine the trajectories of the charged particles needed to calculate their momentum and to reconstruct the secondary vertices. The whole ITS is placed in CMS solenoid under influence of a uniform magnetic field of strength 3.8 T. At LHC nominal luminosity of $10^{34}cm^2s^{-1}$, there is an average of more than 20 proton-proton interactions (for 2018 it was above 30) leading to average about 1000 particles emerging after every bunch crossing, 25 ns. To distinguish the trajectories and assign correct bunch crossing, fast response and high granularity detector technology is needed. To protect against radiation damage caused by intense particle flux and priority to minimize the material to avoid from multiple scattering, nuclear reaction, bremsstrahlung and photon conversion, a compromise was found to choose silicon detector technology. CMS inner tracker is made of about 200 $m^2$ of silicon hence being the largest tracker of its kind ever made.

CMS tracker detector should be robust which can reconstruct tracks of charged particle in whole pseudorapidity range with momentum above 1 GeV to achieve LHC physics goals. Tracker also helps to identify other particles like electron, muons along with electromagnetic calorimeter. CMS inner operates under intense particle flux with hit density varies from 1 MHz/mm$^2$ at radial distance of 4cm decreasing to 60 KHz/mm$^2$ and to 3 KHz/mm$^2$ at distance of 115cm. Keeping in view of hit densities, different types of detectors are installed at various radial distances of inner tracker listed in Table 3.2. Silicon detectors are chosen due its unique properties of fast response and good spatial resolution which enables it to work efficiently even in high radiation environment. To limit the damage mainly by radiation the whole tracker system is kept at about -20° C.

The CMS inner tracker schematic drawing (as it stood before phase 1 upgrades) is shown in Figure 3.7 (**Reference34**). Tracker most inner region is the pixel detector that is nearest to the interaction point consists of disk modules on each side and barrel layers. It provides accurate traces in $z$ and $r - \phi$ with a pixel cell of size $100 \times 150\mu m^2$ covering the $|\eta|$ range up to 2.5. Pixel detector also plays its role for impact parameter resolution that helps to reconstruct secondary vertex (**Reference34**). Pixel detector delivers three spatial point with excellent precision throughout the coverage of pixel.

During data taking of Run 1 operations of LHC, some dynamic inefficiencies (about 5% at an event pile-up of 40 ) were reported in the first layer of barrel of pixel sub-detector, due to the limited size of the readout bandwidth. Subsequently, a new pixel detector was installed in March 2017 Ref. [see comment], equipped with one additional barrel layer and one additional forward disk. In the new pixel detector, the innermost layer is closer to the interaction point and the outermost one is further away from it as it is shown in Fig. 3.8 It also uses new Readout Chips (ROCs) benefitted from reduced data loss at higher collision rates with respect to Phase 1 operations (at 160 Mbit/s). Bandwidth of the readout electronics was increased approximately by a factor of eight (320 Mbit/s). In addition, DC-DC power converters were installed to allow reusing the existing fibers and cables. Hence, the upgraded pixel detector has a higher rate capability and number of channels, providing increased tracking efficiency, reduced fake rate, and improved impact parameter resolution. In the figure 3.9 clear improvments in the dynamic efficiency of 2017 (with pixel upgrades) with respect to the 2016 (without pixel upgrades) can be seen.

Silicon Strip is the second layer detector after pixel detector which is installed at radial distance from 20cm to 116cm. This detector surrounds both sides of pixel with its tacker inner barrel (TIB) and inner tracker disk (TID) regions. There are 3 disks on each endcap and 4 barrel layers where silicon microstrips of thickness $320\mu$m are installed. Microstrips are aligned along with beam axis in the barrel region and in radial direction on disks. The strip pitch in barrel region varies from 80$\mu$m to 120$\mu$m from inner to outer barrel layer. In the endcap region strip pitch changes from 100$\mu$m to 141$\mu$m. Finally, the most outer part of CMS tracker system is tracker outer barrel (TOB) installed at radial distance from 55cm to 116cm containing 500$\mu$m thick 6 barrel layers with strip pitch of 183$\mu$m and 122$\mu$m. TOB provides z coverage up to 118 cm, beyond this point 9 disks of tracker endcaps (TEC+, TEC-) are located that covers 124 cm$<|z|<$282 cm and 22.5 cm$<|r|<113.5$ cm.

To measure the momentum in CMS tracker, we simply exploit the property of the charged particles to bend in the magnetic field. High momentum particles will bend less in detector and vice versa, making resolution worse due to less curvature.

The upgraded pixel detector improved $\mathrm{H} \to 4\ell$ analysis as there was significant increase in lepton number in an event thus improving the reconstruction of Z candidates and hence an increase in the selection efficiency upto much extent. [Ref. CMS-TDR-011]

### 3.2.3 Electromagnetic Calorimeter (ECAL)

ECAL is the cylindrical shaped, hermetic and homogeneous calorimeter build with crystals of lead tungstate ($PbWO_4$). Its barrel ergion (EB) provides pseudorapidity coverage of $|\eta| < 1.479$ and contains 61200 $PbWO_4$ crystals. Endcap region of the ECAL (EE) provides pseudorapidity coverage up to $\eta < 3$ and is build with

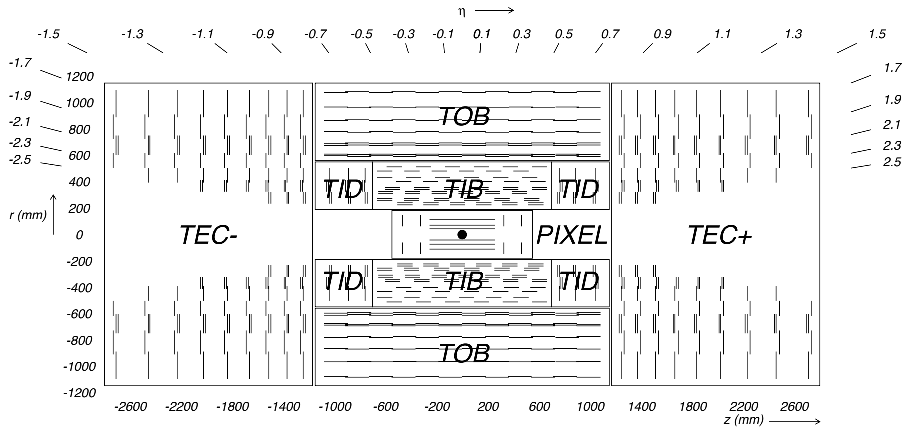



FIGURE 3.7: Schematic diagram of CMS tracker system. Lines represent detector modules (before phase 1 pixel upgrades ).

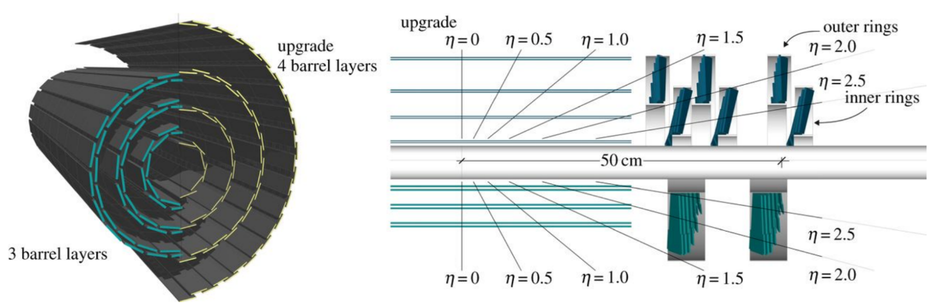



FIGURE 3.8: Schematic diagram of pixel detector's upgrade. Old pixel detector is shown by the lower half detector, and the new pixel detector is represented by the upper half.

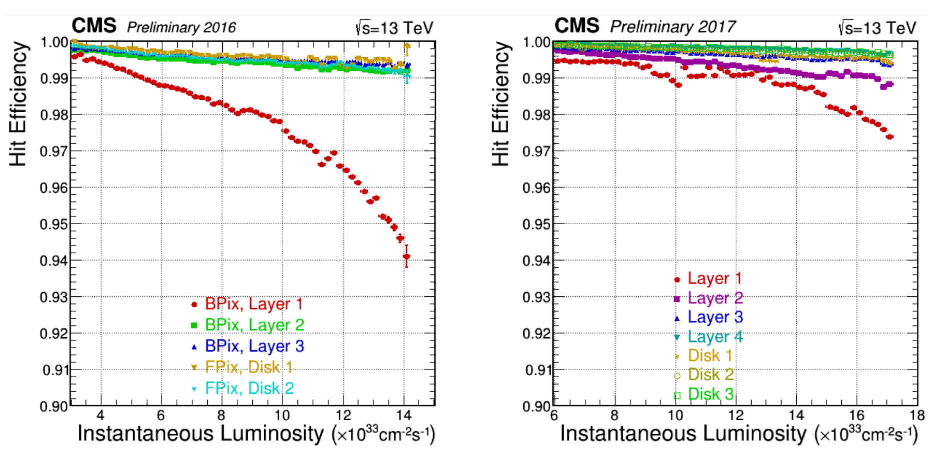



FIGURE 3.9: Pixel detector's hit efficiency as function of instantaneous luminosity corresponding to 2016 data (left) and to 2017 data (right) Ref.[in comments]

7324 crystals on either side in endcap regions. The region of $1.4442 < |\eta| < 1.56$ is regarded as crack region between barrel and endcaps as shown in Figure 3.10 (**Reference34**) is excluded for physics analysis. A preshower detector is installed in endcap regions to distinguish between single and diphoton close to each other e.g neutral pion decay ($\pi^\circ \to \gamma\gamma$). Specially designed Avalanche photodiodes (APDs) and the Vacuum phototriodes (VPTs) are employed in barrel and endcaps region respectively ensure efficient working under high magnetic field. Higgs discovery (both in $H \to \gamma\gamma$ and $H \to ZZ \to 4\ell$ channels) is based on one of the ECAL key feature in the design is to reconstruct electrons and photons with high resolution.

### ECAL Crystals and Geometry

To make a compact and high granularity calorimeter which could fit inside the CMS solenoid, lead tungstate was the best option which is denser than steel (8.28 g/cm$^3$) with short radiation length (X$\circ$) of 0.89 cm (**cms:ECALTDR**). The scintillator ($PbWO_4$) emits light which is directly proportional to incident particle's energy through ionization. Fast response (same order of LHC bunch cross crossing i.e. 25ns) of $PbWO_4$ makes it ideal for ECAL. In barrel region, there are 8.14 m$^3$ active crystals installed weighting up to 67.4 tonne. The each crystals cross section is measured to be 22 × 22 mm$^2$ in barrel and 28.62 × 28.62 mm$^2$ at endcaps. Crystal length in barrel and endcaps is 230mm and 220 mm which is 25.8 and 24.7 times of the radiation length.

The amount of emitted light from scintillator and amplification of photodiodes both are temperature dependent. The nominal temperature at which ECAL operates is around 18°C. For this reason, cooling system has been installed in ECAL to maintain the temperature at nominal value. The cooling system rotates water at 18°C through parallel grid of aluminium pipes a to maintain the temperature in the detector. In the barrel region, each supermodule has its independent supply of water. The cooling system of the ECAL work efficiently and maintain the temperature of $18 \pm 0.05°C$.

TABLE 3.3: ECAL crystal details at endcaps and barrel regions.

| | Barrel | Endcaps |
|---|---|---|
| Number of crystals | 61200 | 14648 |
| Crystal cross section in $(\eta, \phi)$ | $0.0174 \times 0.0174$ | variable |
| Cross section of front face of crystal | $22 \times 22mm^2$ | $28.62 \times 28.62mm^2$ |
| Cross section of rear face of crystal | $26 \times 26mm^2$ | $30 \times 30mm^2$ |
| Crystal length | 230 mm | 220 mm |
| Crystal length(in terms of $X_\circ$) | $25.8X_\circ$ | $24.7X_\circ$ |

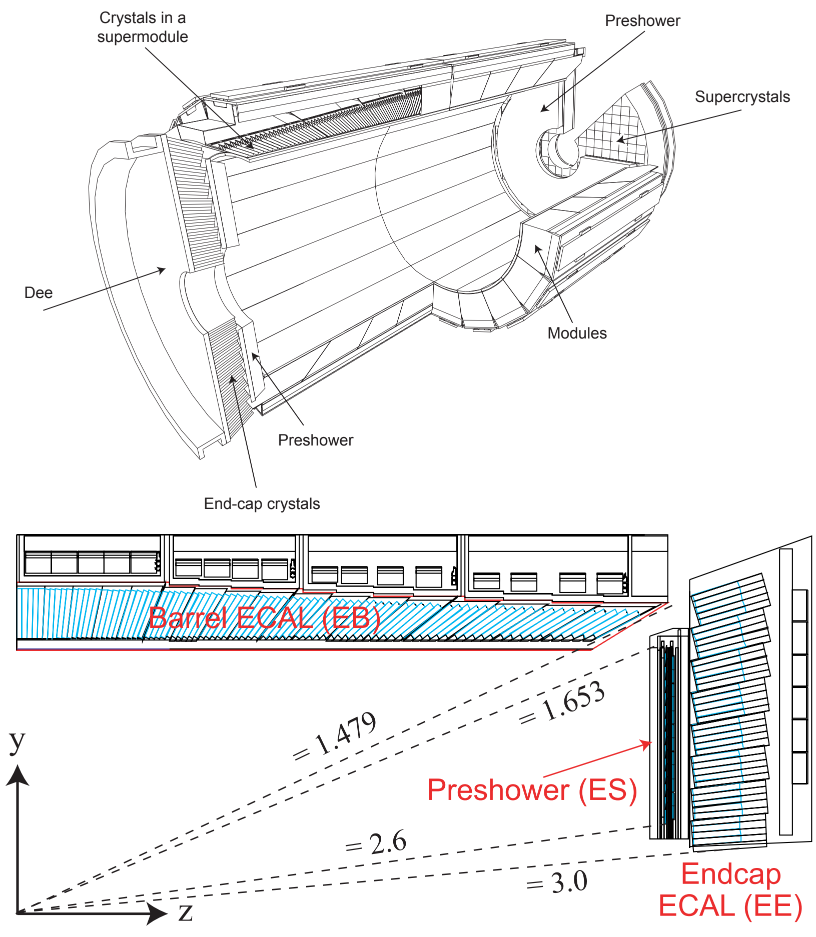



FIGURE 3.10: CMS ECAL layout and pseudorapidity coverage.

#### Photodetectors

The requirements on photodetectors are to be fast, robust and able to work efficiently in a region that contains 3.8 T magnetic field. Considering these requirements, different types of photodetectors are installed both in the barrel and the endcap regions. In the barrel region, each crystal a pair of avalanche photodiodes are mounted with of $5 \times 5$ mm$^2$ active area. Gain of the working APD is 50 and in parallel read out. In endcap region, a specially designed vacuum phototriodes at National Research Institute Electron(St. Petersburg) for CMS ECAL was used. These are single gain photomultipliers with mean gain being 10.2 in absence of field. The diameter of each VPT is 22mm. The total active area is almost 280mm$^2$ where one VPT is attached at the back of every crystal. Their fine coper anode makes them capable to work under intense magnetic field. As compared to APDs, VPTs have lower quantum efficiency and the internal gain which is balanced out with the larger back face crystal coverage. The summary of number of the crystals with their dimensions are summarized in Table 3.3.

#### Preshower

Purpose of the installation of preshower detector is the identification of neutral pions detected in the endcaps regions. Due to its high granularity, the preshower helps to differentiate between electron and minimum ionizing particles (MIPs) and to determine the position of electrons and photon. It provides the pseudorapidity coverage of $1.653 < |\eta| < 2.6$.

#### Energy Resolution

It can defined as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{S}{\sqrt{E}}\right)^2 + \left(\frac{N}{E}\right)^2 + C^2 \tag{3.7}$$

where C is the constant of proportionality and is extracted by fitting the measured points to the function, N being noise term and S is stochastic term. Recently relatively energy reslution study of ECAL using 2017 data and MC has been performed using Z events decaying to a pair of electrons. The resolution plots for bremsstrahlung electrons ($R_9$>0.94 and $R_9$<0.94) is separately shown for data and MC in the Figure 3.11 (**comments**). For $\mathrm{H} \to 4\ell$ analyses, excellent energy resolution for leptons is important for precise measurements for which ECAL of CMS plays an important role.

### 3.2.4 Hadron Calorimeter

CMS being a general purpose detector is capable to study several final state signatures involved in a large range of high energy processes. The hadron calorimeter (HCAL) provides the measurement of energies and positions of hadronic jets (for example neutrons, protons, kaons and pions )(**cms:HCALTDR**). HCAL being hermetic is also helpful to calculate MET that can be assigned to neutrinos or other

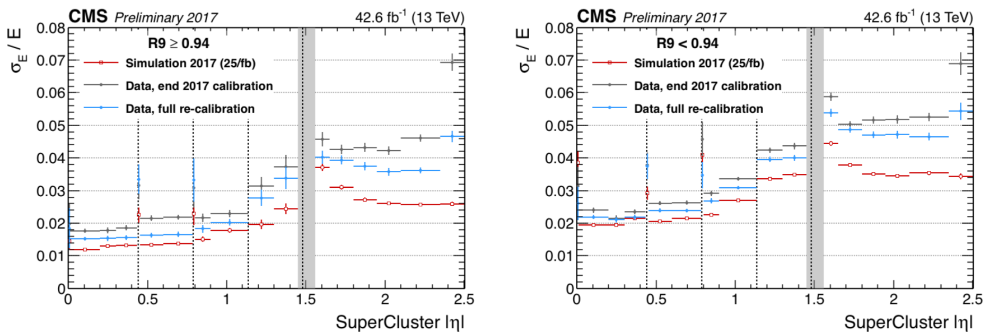



FIGURE 3.11: CMS ECAL energy resolution.

invisible particles. There are no gaps or cracks in HCAL to make sure to capture all known particles, leaving MET with only possibility of escaped particles. It is fitted between ECAL and the CMS solenoid at radial distance from 1.77 m to 2.95 m providing a coverage of pseudorapidity up to $|\eta| < 3.0$, beyond this point a forward hadron calorimeter (HF) is installed at a distance of 11.2 m from the collision point on both ends extending pseudorapidity coverage to $|\eta| < 5.2$. As HCAL is placed inside 3.8 T of magnetic field, so absorber is chosen to be a material which is non-magnetic. Also, due to requirement on maximum number of interaction length, cost and mechanical property brass has been chosen for CMS HCAL.

HCAL is also splitted into barrel and encap regions, HB and HE respectively. HB provides coverage of pseudorapidity of $|\eta| < 1.33$ and HE of $1.3 < |\eta| < 3$. HB is split into two halves ($HB^+$ and $HB^-$) which consist of 36 similar wedges each weighing 26 tonnes illustrated in Figure 3.12. The wedges are built with the absorbant, the plane brass plates kept along beam axis. All plates are made of brass with physical properties described in Table 3.4 except inner and outer most made of steel for structural strength. Table 3.5 shows the thickness of each layer. HCAL is constructed with alternate layers of plastic and brass (the absorber and scientillator respectively) which helps to measure the arrival time, position and energy of a particle. Plates of brass are 79 mm thick each with 99 mm gaps for the scintillators installation. Optical fibers collect all the light emitted when a particle passes through scintillator and passed it to readout box. Total energy is calculated by summation of the light over many layers in depths called tower. The scintillator in HCAL are arranged in the from of 70,000 tiny tiles. individual tiles makes scintillator tray by grouping them in $\phi$ layers to limit the number of readout boxes. These trays works as individual units, so it can be replaced or tested easily. Rather than single-particle energy resolution, design of HCAL endcap was more motiviated from the goal that cracks between to minimize the cracks between barrel and endcaps regions of HCAL rather than energy resolution of single-particle.

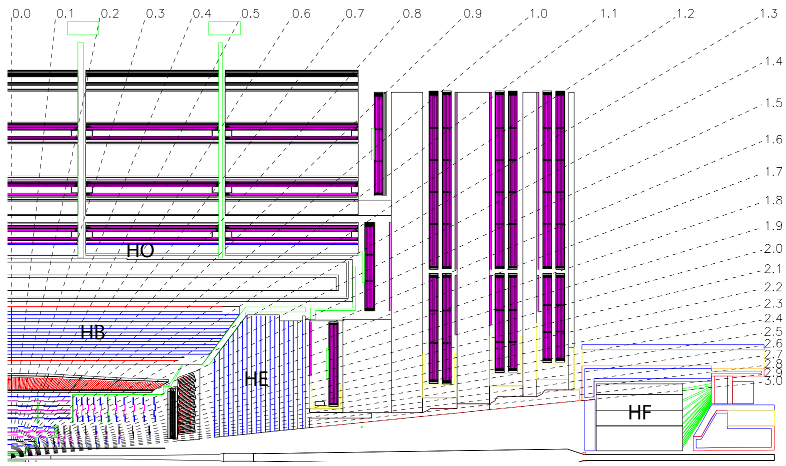



FIGURE 3.12: CMS HCAL layout.

TABLE 3.4: Properties of brass of HCAL.

| | |
|---|---|
| chemical composition | 70% Cu, 30% Zn |
| density (g/cm$^3$) | 8.53 |
| radiation length (cm) | 1.49 |
| interaction length (cm) | 16.42 |

**Forward Calorimeter**

The two HF are placed on both sides of CMS to catch up the particle emerging from proton- proton collisions at very small angle relative to beam axis as shown in Figure 3.12 (**Reference34**). HF is most forward region of the detector where enormous particle fluxes are expected. On average, the energy deposited in HF only is 8 times more as compared to the whole CMS detector. The design of HF calorimeter for such harsh conditions was a considerable challenge. With integrated luminosity of $5 \times 10^5$ pb$^{-1}$at $|\eta| = 5$, HF will absorb a radiation dose of about 10 MGy. The rate of charged hadrons is also extremely high. Because of hard conditions HF material and design is different from rest of the HCAL.

TABLE 3.5: HCAL absorber thickness.

| layer type | material composition | thickness (mm) |
|---|---|---|
| front plate | steel | 40 |
| 1-8 | brass | 50.5 |
| 9-14 | brass | 56.5 |
| back plate | steel | 75 |

The energy resolution of the HCAL is given as

$$\left(\frac{\sigma(E)}{E}\right)^2 = \left(\frac{90\%}{\sqrt{E}}\right)^2 + (4.5\%)^2 \tag{3.8}$$

and for HF it is given as

$$\left(\frac{\sigma(E)}{E}\right)^2 = \left(\frac{170\%}{\sqrt{E}}\right)^2 + (9.0\%)^2. \tag{3.9}$$

There were subsequent upgrades in HCAL during Run 2 era. In 2017, PMTs were replaced with new multi-anode PMTs and additionally electronics replaced for better differentiation of real hit from the muons which hit the PMT window. In 2017, HE of HCAL was upgraded and photo detectors were replaced with latest silicon photo multipliers (SiPMs) for noise elimination and better photo detection efficiency. (reference in comments)

### 3.2.5 Superconducting Magnet

The CMS central is superconducting solenoid (**cms:MFTDR**; **Reference36**; **Reference37**; **Reference38**) designed to have bore of diameter 6 m and length 12.5 m producing magnetic field of 3.8 T which is 100,000 time stronger than earth magnet. The CMS solenoid is the largest ever built magnet weighs 12,000 tonnes, powerful enough in providing a strong magnetic field in its dimensions throughout the detector. It surrounds CMS ITS, ECAL and HCAL. i The task of the magnetic field is to efficiently bend the charged particles when they more throught it. More the momentum of the charged particles less they bend, so high energetic charged particles require a stronger magnetic field for its momentum to be measured accurately. CMS aims to have excellent momentum resolution with such strong magnetic field specially for muons. The CMS magnetic is made of coil of NbTi wire which supplies the uniform magnetic field when current pass through it. The NbTi becomes superconductor at $-268.5°$ C (4 K) which allows a huge amount of current (10,000 A) to pass through it. To maintain this temperature liquid helium is used. A large bending is achieved in tacker by stronger magnetic field providing precise and efficiencies momentum measurement. The magnetic field direction inside (tracker) and outside the solenoid (muon system) allows to make second momentum measurement for muons.

### 3.2.6 CMS Muon System

Muon identification in the CMS detector is provided by its outer most layer of sub-detector called muon system (**cms:muonsystem_TDR**). The detection of the muons is very important for many physics process. In particular, for SM Higgs boson decay to $ZZ/ZZ^*$ which is labeled as a golden channel if each Z boson decays to a pair of opposite charged muons. Best best mass resolution can be obtained with four muons in final state as muons are less effected by detector material as compared to electrons because they interact weakly with detector material. Along with this example, SUSY and other model for new physics searches also encourages the necessity for their reconstruction of muons with excellent resolution.

To have a robust and precise muon system was one of the core motive of CMS as indicated by detector name. Identification, triggering and momentum measurement of the muons are the tasks assigned to the muon system. The flux-return yoke and strong magnetic field are playing crucial role for excellent muons momentum resolution and triggering. The flux-return yoke by absorbing the hadrons helps for muons identification.

Muon system was essentially designed as cylindrical shape to provide full pseudorapidity coverage with detection area of 25000 $m^2$, having barrel and endcaps regions. Muons detection is based on gasses detectors of three types. The strength of magnetic field is uniform in barrel region with less neutron-induced background. Drift tubes are installed providing pseudorapidity coverage $|\eta| < 1.2$, as illustrated in Figure 3.13. The barrel region is arranged to have four muons station, measuring $r - \phi$ bending plane and distance along beam side. The arrangement of the drift cells are offset by the half of the width of cell providing dead spot in efficiency, determination of standalone bunch crossing and excellent time resolution by utilizing mean timer circuits. The direction and number of chambers are optimized to get maximum efficiency of combination of the hits from different stations to get single track which is helpful to reject the background. In endcap region, where non-uniform strength of magnetic field and more background is expected cathode strip chambers (CSCs) are installed that providing coverage $0.9 < |\eta| < 2.4$. Like DT, there 4 CSCs stations on each endcap. Each station is composed of chambers, which are capable to work independently, aligned perpendicularly to z-axis and spread between the flux-return plates. In order to overcome uncertainties of background rates and assignment of correct bunch crossing number, resistive plate chambers (RPCs) are planted both in the barrel and the endcap detector regions. There is a dedicated trigger system for each type of muons gasses detector including RPCs. Good time resolution, fast response and coarser spatial resolution makes RPCs distinct from CSCs and DT. The RPCs also plays vital role for the combination of multiple hits in a chamber to for tracks. The RPCs are installed in between DT and CSCs in region the barrel and the endcap region respectively.

#### Drift Tubes of muon system of CMS

The DTs are planted in the barrel region of CMS muon system that provides pseudorapidity coverage of $|\eta| < 1.2$ as illustrated in Figure 3.13. Whole DT system consists of 5 wheels along z-axis, and 4 stations along radial direction (MB1, MB2, MB3, MB4). Each station is build with 12 chambers except fourth which contains 14 chambers. Each drift chamber of an average size of 2m$\times$2.5m is made of 12 aluminium layers, making 3 groups of four. The central group provides the coordinates measurement along z-axis and other two groups measures $r - \phi$. There are about 60 drift tubes in each chamber, each 4 cm wide made of drift cells each of size 42 mm $\times$ 13 mm. The 50$\mu$m wire made of stainless steel serves as anode while two

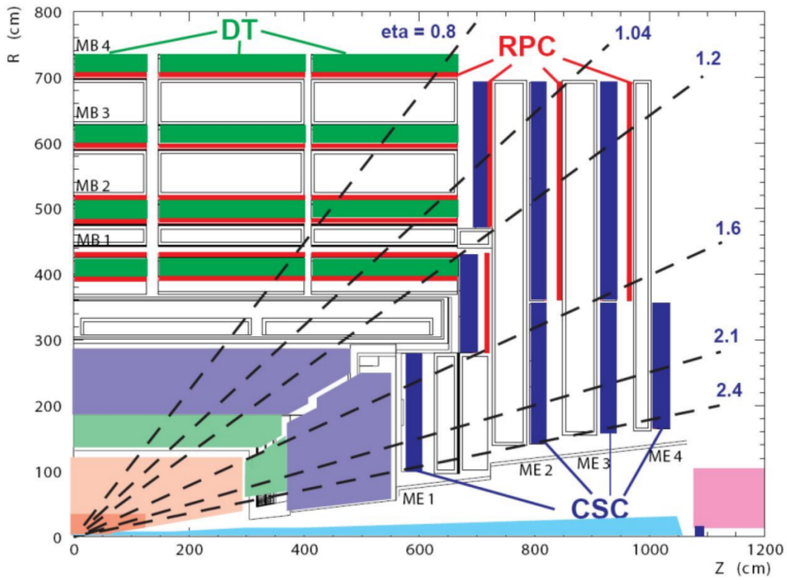



FIGURE 3.13: CMS muon system layout.

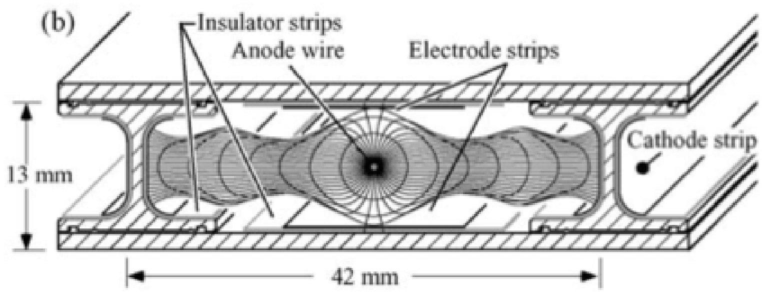



FIGURE 3.14: DT electric field created by cathode, anode and insulating strips .

"I-shaped" aluminum tape each of thickness 50$\mu$m are cathodes of the drift cell as shown in Figure 3.14. As any charges particle pass by the drift cell, it ionizes the gas by removing electrons from its atom. This establish a electric field finishing of at positively charged wire. Distance between the wire and the muon track is estimated by drift time of these electrons. The cell is filled with mixture of $CO_2$ (15%) Ar (85%) gases which is inexpensive and safe. Other qualities of this mixture are good saturation drift velocity (5.4 cm/$\mu$s) and quenching. To keep drift velocity steady, the filled gas is kept at a constant pressure inside the chambers. The pressure in each wheel is continuously monitored by sensors. It is expected to inject 10% new gas everyday.

#### Cathode Strip Chambers of the CMS muon system

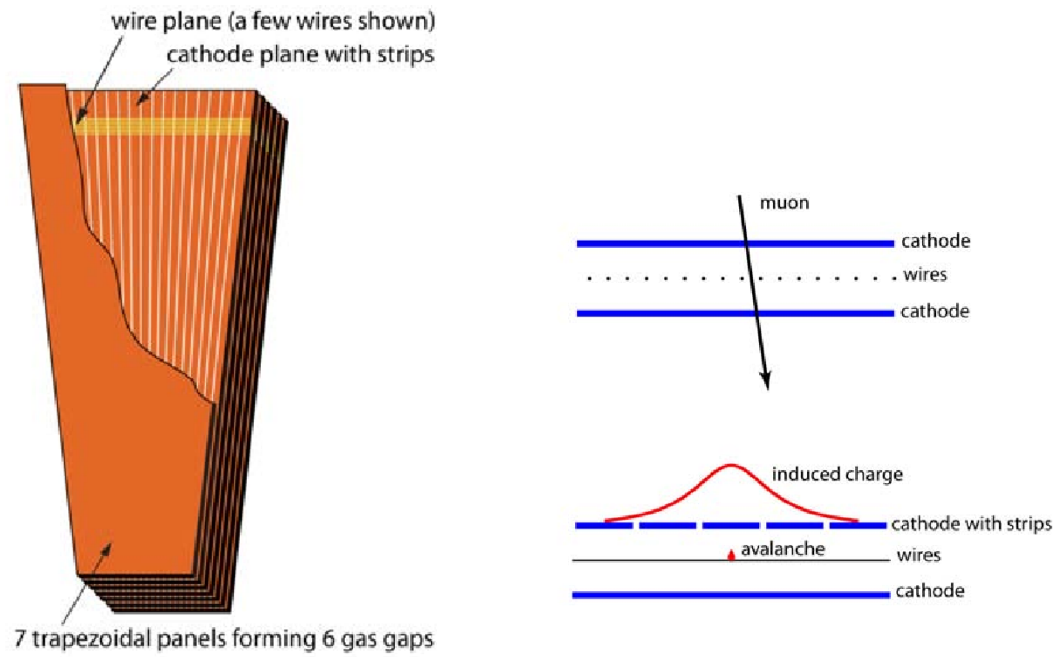
The cathode strip chambers (CSC) are installed at endcap region of CMS muon system providing the coverage of pseudorapidity upto $|0.9| < |\eta| < 2.4$, illustrated in Figure 3.13. In the endcap regions, the strength of magnetic field is non-uniform which forced to use in endcap some different technology from the barrel region. Another challenge for CSCs is higher expected neutron-induced background as compared to the barrel region. The endcaps of the muon systems are made up of 468 CSCs arranged in four station like the barrel region (ME1, ME2, ME3 and ME4). The pseudorapidity range from $|0.9| < |\eta| < 1.2$ is covered by both CSCs and DTs. The chambers are aligned trapezoidal, each chamber is covering $10^{\circ}$ or $20^{\circ}$ in $\phi$. RPCs are also installed between CSCs stations.

A single chamber is the basic unit which capable to work independently which is shown in Figure 3.15(a). The CSC is multiple wire detector made of 6 layers of positively charged anode interleaved with 7 layers of negatively charged cathode wires made of cooper filled within gas volume as shown in Figure 3.15(a). The cathode and anode wires are at right angle to each other and provides two position coordinates of each particle. As the charged particle pass through chamber, it will ionize the gas atoms producing avalanche of electrons to anode wires, illustrated in Figure 3.15(b) . As the wires are nearby, CSCs not only provides excellent momentum resolution but helps to trigger the event as well. It feeds $r - \phi$ resolution of 2 mm to first level trigger, $75\mu$m for ME1/1 and ME1/2 chambers and $150\mu$m for other stations. The CSCs also help to assign bunch crossing with an efficiencies of 92% which can be enhanced up to 99% by simple majority out of all chambers. CSC has tendency to operate at high event rates and in an environment where the strength of magnetic field is non-uniform. This makes CSCs robust which does not require precise gas pressure and temperature. Multilayer CSCs is useful for the combination of hits in CSCs and tracker detectors.

#### Resistive Plate Chambers of muon system of CMS

RPCs are important part of CMS muon system installed in both barrel and endcap region. Time resolution of RPCs is excellent which is their main motivation. Additionally, spatial resolution of RPCs is also good. These very fast responsive detector providing another muon trigger along with CSCs and DTs. RPCs are capable to provide a time resolution of about a nano-second.

RPCs are made of two high resistive plastic positive and negative charged plates, kept parallel to each other, with gas filled inn between them. The gas mixture consists of 95% of $C_2H_2F_4$ and 5% of $C_4H_{10}$. As the muon pass through the RPCs, electrons librated during primary and secondary ionization of the gas create avalanche which is received by outer metallic strips with a short but precise delay in time. Quick momentum measurement is feeded to trigger the event. The RPCs excellent time resolution assign bunch crossing with ultimate precision.




(a) (b)

FIGURE 3.15: CSC layout (left) and induced current (right) when a charged particle pass thorugh it .

**Momentum Resolution**

CMS detector is capable to measure muons transverse momentum independently by using tracker or muon system. The expected transverse momentum resolution of muons system has been studied for the using muons of $p_T = 40$ GeV from Z and W boson decays and found to be 10%(20%) in barrel and endcap regions. Global muons can also be reconstructed at CMS by the combining the information from tracker and muons system. As compared to muon system, tracker gives better resolution for muons $p_T$ less than 100 GeV and vice versa. The transverse momentum resolution is found to be improved sufficiently by using global muons i.e 1%(2%) for $p_T < 100$ GeV in barrel(endcap) regions. For muons with $p_T$ greatar than 100 GeV, transverse momentum resolution of CMS muon system improves upto 8% and 10% in the barrel and endcap regions respectively. Momentum resolution of CMS muon system alone, tracker alone and combination of both are shown in Figure 3.16 (**cms:muonsystem_TDR_2**).

### 3.2.7 Trigger and Data Acquisition System of CMS

The LHC running at its nominal parameters provides particle collisions produced with frequency of 40 MHz. Most of these events are soft collisions and uninteresting for new physics as they are like minimum bias collisions. Though CMS detector is capable to reconstruct all of these events but storing all these events

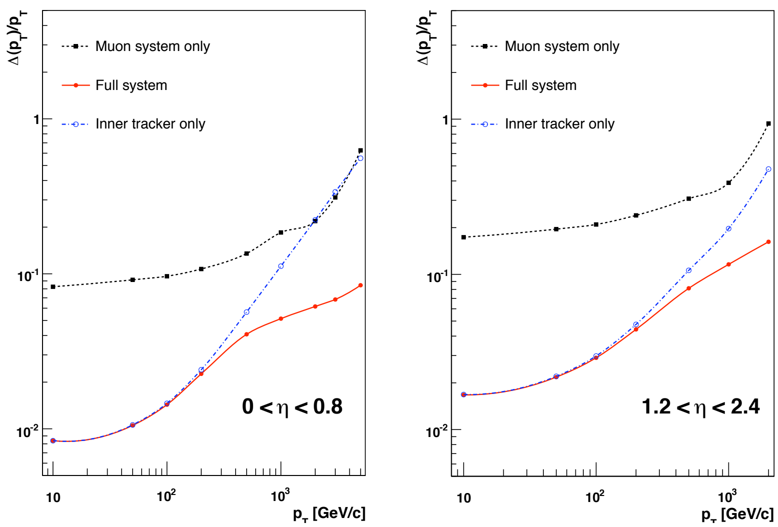



FIGURE 3.16: Momentum resolution of muon as a function of $p_T$ of muons both in barrel and endcap regions.

will fill the CMS storage in days. So trigger system of CMS helps of select interesting events out of all for further steps. The events which does not pass the first step trigger are deleted permanently. Trigger system performs the first step for physics event selection which reduces the event rate from 40 MHz down to 1kHz into two layers Level-1 and the high-level triggers (shortly written as L1 and HLT respectively).

The L1 trigger is made of the programmable electronics, dropping the event rate drastically to 100 kHz. To be fast, L1 trigger events by their energy and momentum calculations estimated by only using coarse in muon system and calorimeter instead of complete high resolution information. The L1 trigger decides to select or reject event in 3.2$\mu$m.

All events with positive L1 decision, with their complete information, are transferred to HLT which makes use of one thousand commercial processors. It uses the growing complex algorithms which exploit high details of the event. Decision time of HLT varies with event number with mean time of about $50$ ms. Since HLT has excess of complete information of an event, it demands a greater bandwidth of the order of 1Tb/s.

For example, to look for Higgs boson decay into 4-leptons final state, the trigger would look for energetic leptons in the detector. For L1 trigger, electron and muons points to a narrow energy deposit in ECAL and hit pattern in tracker and the muon system respectively. While for HLT, momentum and total energy of the leptons are reconstructed by ECAL and combination of muon system and inner tracker.

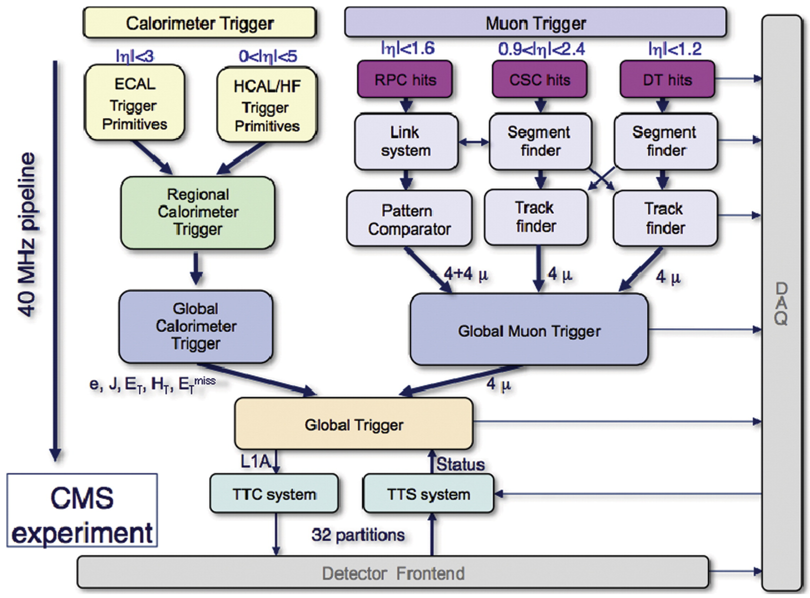



FIGURE 3.17: The configuration of L1 trigger.

### Level-1 Trigger Architecture

Configuration of L1 trigger is elaborated in Figure 3.17. It can be classified into calorimeter and muon triggers. Each ECAL and HCAL has independent regional trigger feeding to global calorimeter trigger. Similarly, muon trigger has local triggers for each sub-detector merging into global muon trigger. Finally, global calorimeter and muon triggers merge into global trigger which makes L1 final decision.

Each local trigger feeds to global/regional trigger by providing trigger primitives groups (TPGs) based on coarse of sub-detector. Regional calorimeter trigger (RCT) builds objects which deposit their energy in ECAL and HCAL such as jets, $\gamma$, electrons, photons and tau's. Similarly, in muon triggering, DT/CSC/RPC track finders feeds muons tracks to muon global trigger (MGT). Global muon and calorimeter trigger feeded four best candidates at most, which are then sent to global trigger (GT). GT compares these candidates with L1 trigger menu to decide weather to keep to reject the event. Some triggers in the menu evolved with increasing LHC center of mass energy by tightening the requirement on candidate or pre-scaled (selecting only the nth event) to control the output rate. Candidates able to fire at least one trigger out of whole trigger menu is kept for further investigation by HLT. For example, the trigger/bit "L1 SingleEG8Iso" in the L1 trigger menu requires one isolated (candidate with zero or little hadronic activity around it) electron or photon at least with $p_T$ greater than 8 GeV.

### High-Level Trigger Architecture

HLT is three level (2, 2.5 and 3) trigger with growing complexity starting at level 2 which usually works on top of the information provided by L1 trigger. Goal

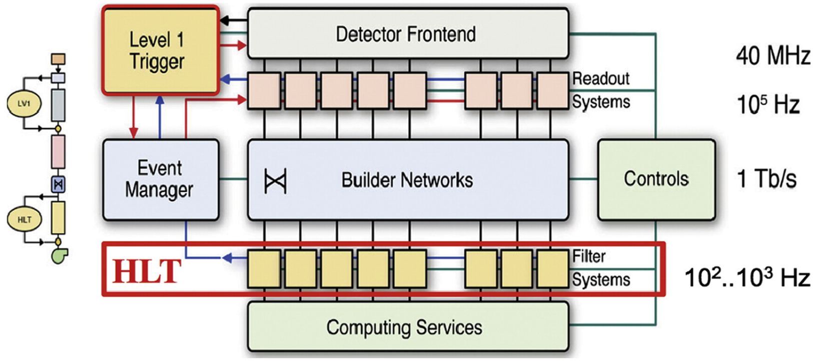



FIGURE 3.18: The CMS DAQ system structure.

of the HLT is to use output event rate of L1-trigger(100kHz) and reduce it upto 1 kHZ. Level 2 HLT is to build fine granularity objects by combining the information from sub-detectors e.g from preshower and ECAL energy clusters are combined for electrons. Bremsstrahlung of the electrons causes its energy to spread into several neighbor cells. To compensate this effect, clusters of ECAL are combined to form superclusters.

At HLT level 2.5, the inner tracker information is combined and matched with calorimeters and muon system. In case of electron, energy supercluster is matched with the hits in the tracker, if matches it form a seed. Tracker hits are matched with the hits in muon system to reconstruct global muon. HLT level 3 reconstructs the complete track, if its found electron seed or matching hits of muon. After these three stages, precise energy and momentum are calculated of the objects that decides to accept or reject the event.

There are more than 200 different HLT paths each derived by at least one L1 bit. The data which pass the HLT trigger is organized into different datasets according to the objects reconstructed in the event and the passed selection e.g an event with two muons reconstructed with transverse momentum greater than 8 and 17 GeV will be placed in "DoubleMu” dataset.

### Data Acquisition (DAQ) System of CMS

The schematic skeleton of CMS DAQ system is presented in Figure 3.18. CMS DAQ provides data from 650 different readout channels to HLT along with enough computing resources to limit the output data rate (**cms:DAQ_TDR**). Frangments of an event with a size of 2KB is provided by each module. In order to read data from detector sensors, Front End Drivers (FEDs) utilizea network whose bandwidth is 100 Gb/s. These FEDs are located underground at the level of the detector called counting room. Events are then transferred to a system where these are filtered at 100 kHz rate maximum. The complete reconstruction of the objects of the event and high machine luminosity is the reason for this high rate.

### 3.2.8 Computing Grid

An important and last step after data collection is to process it with full detailed reconstruction algorithms and made available for physics analysis.Although, trigger system of CMS drastically reduces the amount of data but even then CMS delivers data at the order of petabytes every year. To analyze and store this huge amount of the data, LHC has connected huge number of computing resources from collaborative institutes or universities, computer centers and middleware providers through a grid known as the Worldwide LHC Computing Grid (WLCG).

There are three layers of Tier (Tier-0, Tier-1 and Tier-2) for data flow between different computing sites. The types of dataset is described below briefly

- RAW data that contains all the recorded information from detector along with triggers decisions and other metadata. A copy of RAW data is achieved and placed at permanent storage element. A occupancy of around 1.5 (2.0) MB/event is needed to store data (simulated data) in RAW version.
- Many algorithms and pattern recognition techniques like cluster and track-finding; primary and secondary vertex reconstruction etc. are applied to RAW data to obtain RECO data format. RECO events occupancy is roughly three times less than RAW (0.5 MB/event).
- The most compact format of data AOD (analysis object data) is obtained by dropping unnecessary information and reducing per event size up to 100 KB.

Tier-0 is located at CERN with which copies the recorded data to permanent storage, promptly reconstruct data from RAW version to have first version of prompt-RECO data and copies RAW data to Tier-1 centers. Tier-1 are mainly held in national computing centers and labs from collaborative countries. At each Tier-1, a fraction of detailed the reconstruction of RAW data simulated data has been performed to get AOD. Tier-1 also responsible to store and export AOD format data to Tier-2. The last layer Tier-3 is hosted by CMS collaborative institutes where main analysis activity, alignment, detector related studies and offline calibration is done. Tier-3 are used for Monte Carlo (MC) data production which can be exported to Tier-1 for long term storage.

# Chapter 4

# Physics Objects studies with the CMS detector

## 4.1 Physics Objects

### 4.1.1 Electrons

#### Electron Reconstruction

Preselection of the electron candidates is made through loose cuts on track-cluster matching observables. This done to maintain highest possible selection efficiency of electrons and to reject QCD background significantly. For the analysis, electrons are subjected to have $p_T^e > 7$ GeV within $|\eta^e| < 2.5$, and satisfy a loose primary vertex constraints on longitudinal and transvers components of impact parameter $viz.$ $d_{xy} < 0.5$ cm and $d_z < 1$ cm respectively. Such electrons are called **loose electrons**. Details of electron reconstruction can be explored in Ref. (**ElectronLegacy**).

The data-MC discrepancy is corrected using scale factors as is done for the electron selection with data efficiencies measured through the same tag-and-probe technique outlined later (see Section 4.1.1). These studies for reconstructions are carried out by the EGM POG and the results are only summarised here.

The electron reconstruction scale factors are shown Fig. 4.1 and are applied as a function of electron $p_T$ and super cluster $\eta$.

#### Electron Identification and Isolation

With respect to previous analyses, an important improvement brought in this analysis is usage of a new multivariate discriminant for electron selection in all data taking periods.

Identification and isolation of the reconstructed electrons is now made using Gradient Boosted Decision Tree (GBDT) which is a multivariate classifier algorithm. It exploits inputs from electromagnetic cluster, cluster-track matching, variables which exclusively depend on tracking measurements and as well as particle flow isolation sums. The list of observables used can be seen in the Table 4.1.

For both signal and the background, the classifier is trained on 2017 MC samples of Drell-Yan(DY) plus jets. Fig. 4.2 shows the output of the multiclassifier discriminant for prompt electrons from DY events and fakes from jets in DY events.

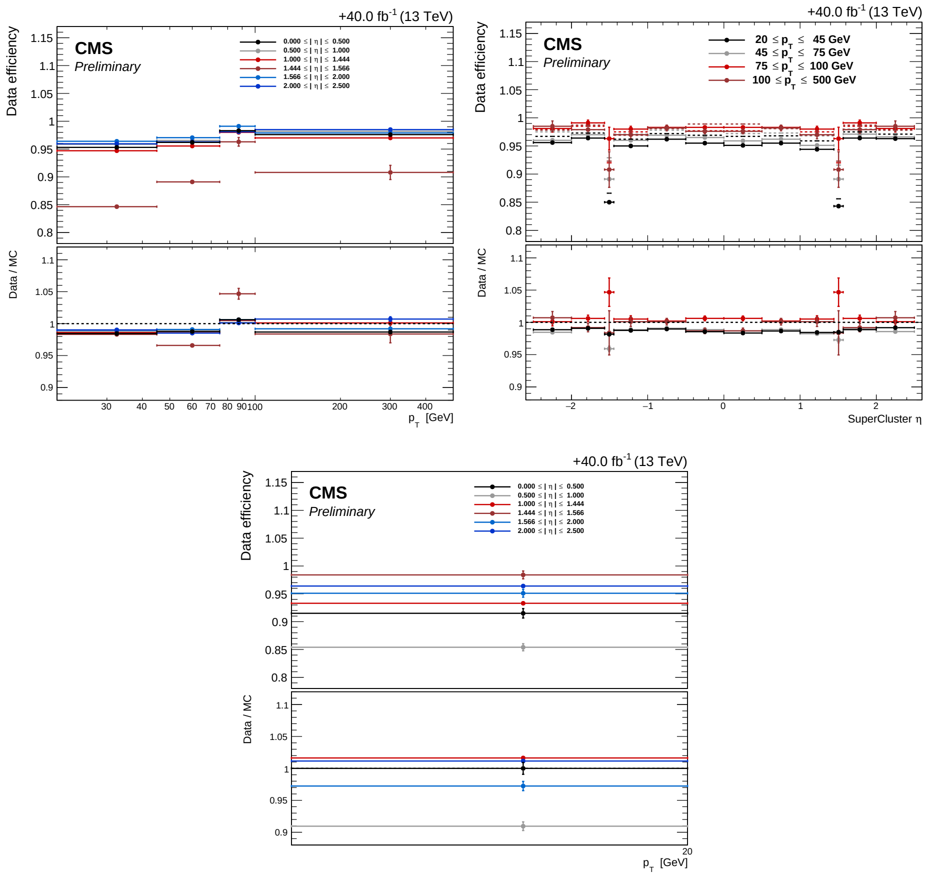



FIGURE 4.1: Reconstruction efficiencies of electron in data versus $p_T$ (left), $\eta$ (right) for electrons having $p_T > 20$ GeV (top) and $p_T < 20$ GeV (bottom) with corresponding data/MC scale factors as‘ăprovided by the EGM POG.

| observable type | observable name |
|---|---|
| 6*cluster shape | RMS value of spectrum of energy-crystal number along $\eta$ and $\varphi$; $\sigma_{i\eta i\eta}$, $\sigma_{i\varphi i\varphi}$ |
| | width of super cluster along $\eta$ and $\phi$ |
| | 'ratio of hadronic energy behind the supercluster of electron‘ăto supercluster energy, $H/E$ |
| | circularity $(E_{5\times5} - E_{5\times1})/E_{5\times5}$ |
| | sum of the seed and adjacent crystal over the super cluster energy $R_9$ |
| | for the endcap training bins: pre-shower energy fraction $E_{PS}/E_{raw}$ |
| 2*track-cluster matching | energy-momentum agreement $E_{tot}/p_{in}$, $E_{ele}/p_{out}$, $1/E_{tot} - 1/p_{in}$ |
| | position matching $\Delta\eta_{in}$, $\Delta\varphi_{in}$, $\Delta\eta_{seed}$ |
| 5*tracking | fractional loss of momentum loss $f_{brem} = 1 - p_{out}/p_{in}$ |
| | number of hits of KF and the GSF track $N_{KF}$, $N_{GSF}$ $(\cdot)$ |
| | reduced $\chi^2$ of KF and the GSF track $\chi^2_{KF}$, $\chi^2_{\text{GSF}}$ |
| | number of expected innre hits (missing) $(\cdot)$ |
| | probability transformation in conversion vertex fit $\chi^2$ $(\cdot)$ |
| 3*isolation | Particle Flow photon isolation sum $(\cdot)$ |
| | Particle Flow charged hadrons isolation sum $(\cdot)$ |
| | Particle Flow neutral hadrons isolation sum $(\cdot)$ |
| 1*For PU-resilience | Mean energy density in the event: $\rho$ $(\cdot)$ |

TABLE 4.1: Overview of variables used at inputs to identification classifier. The Variables not used in the run I MVA are specified with $(\cdot)$.

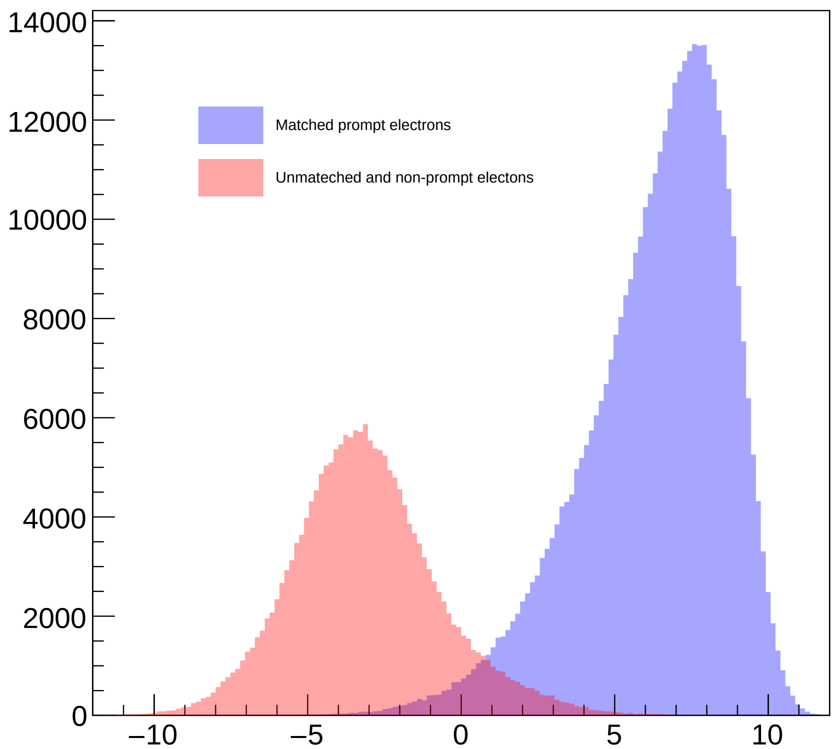



FIGURE 4.2: Output of the multiclassifier discriminant for prompt electrons matched to truth electrons from Z decay (blue) and for fakes (red). Events are all taken from DY events MC sample.

The impact of the transition from the TMVA(v1) to the xgboost(v2) training library is shown in Fig. 4.3. The working points shown are adopted to get the same signal efficiency as the cut on MVA ID and a cut on the PF isolation, in each $p_T$ and $\eta$ bin.

Table 4.2 lists the cut values applied to the BDT score for the chosen working point. For the analysis, loose electrons have to pass this MVA identification and isolation working point.

### Electron Impact Parameter Selection

For lepton to be consistent with some common primary vertex, we require that they have an associated track that corresponds a small impact parameter from the event primary vertex. We use the significance of impact parameter to the vertex of an event, i.e. $|\mathrm{SIP}_{3\mathrm{D}} = \frac{\mathrm{IP}}{\sigma_{\mathrm{IP}}}|$, where IP being three dimensional lepton impact parameter w.r.t the primary interaction vertex, and the $\sigma_{\mathrm{IP}}$ being the uncertainty

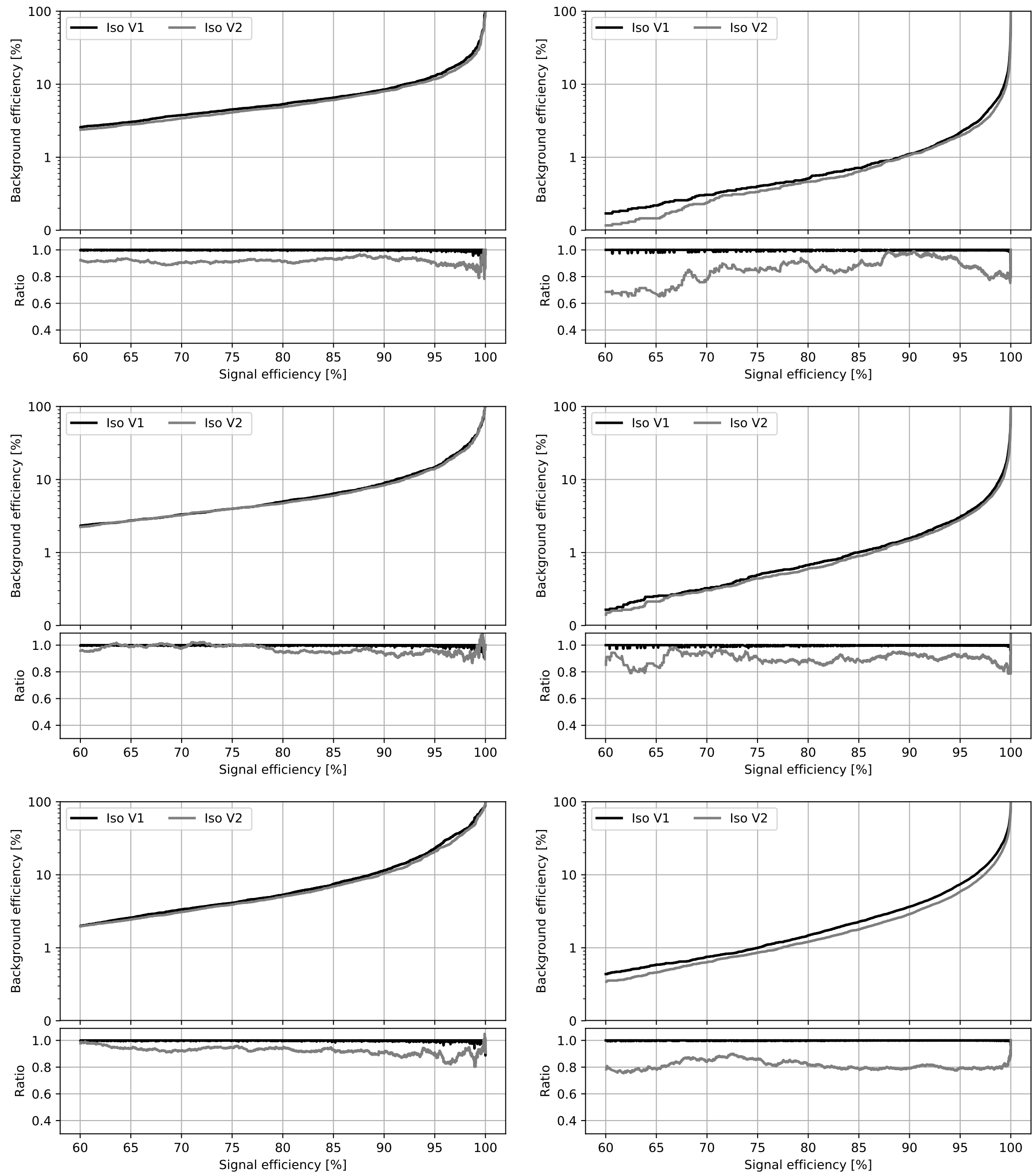



FIGURE 4.3: The receiver operating characteristic curves (background efficiency vs signal efficiency) of the MVA's trained on the 2017 simulation and applied on the 2018 simulation. The training combines identification and isolation variables. Performance are shown for electrons with $5 < p_T < 10$ GeV (left), $p_T > 10GeV$ (right) and $|\eta| < 0.8$ (top), $0.8 < |eta| < 1.479$ (middle) and $|eta| > 1.479$ (bottom). V1 and V2 versions of training are compared, exploiting TMVA and xgboost training libraries respectively.

associated to this. Hence, a lepton satisfying $|\mathrm{SIP}_{3\mathrm{D}}| < 4$ is regarded as "primary lepton".

| | $\|\eta\| < 0.8$ | | |
|---|---|---|---|
| | Cut on BDT score | Signal eff. | Background eff. |
| $5 < p_T < 10$ GeV | 1.264 | 81.04% | 4.4% |
| $p_T > 10$ GeV | 2.364 | 97.1% | 2.9% |
| | $0.8 < \|\eta\| < 1.479$ | | |
| | Cut on BDT score | Signal eff. | Background eff. |
| $5 < p_T < 10$ GeV | 1.178 | 79.3% | 4.6% |
| $p_T > 10$ GeV | 2.078 | 96.3% | 3.6% |
| | $\|\eta\| > 1.479$ | | |
| | Cut on BDT score | Signal eff. | Background eff. |
| $5 < p_T < 10$ GeV | 1.330 | 72.97% | 3.6% |
| $p_T > 10$ GeV | 1.080 | 95.7% | 6.7% |

TABLE 4.2: Minimum value of BDT score for electron to pass identification, together with the corresponding signal and the background efficiencies.

### Electron Energy Calibrations

Electrons in data are corrected for features in ECAL energy scale in bins of $p_T$ and $|\eta|$. Corrections are calculated on a $Z \rightarrow \mathrm{ee}$ sample to align the dielectronʻă mass spectrum in data to that in MC, and to minimize its width.

The resolution of $Z \rightarrow \mathrm{ee}$ mass in the Monte Carlo is made to match with data by applying a pseudorandom Gaussian smearing to electron energies, with varying parameters of Gaussian in $p_T$ and $|\eta|$ bins. This has effect of convoluting the electron energy spectrum with a Gaussian.

The electron energy scale is measured in data by fitting a Crystall-ball function to di-electron spectrum of mass around the Z peak in the $Z \rightarrow \mathrm{ee}$ control region. The energy scale for the full 2018 dataset is shown in Fig. 4.4(a) and agrees decently with the MC with the preliminary corrections released by EGAMMA POG.

### Electron Efficiency Measurements

The Tag-and-Probe study was performed on the EGM dataset using the golden JSON of 58.83, 41.4 and 35.9 fb$^{-1}$ respectively for 2018, 2017 and 2016. More details on this technique can be found in Ref. (**AN-15-277**).

Tag electrons need to satisfy the following quality requirements:

- trigger matched to HLT_Ele32_WPTight_Gsf_L1DoubleEG_v*
- $p_T > 30$ GeV, super cluster (SC) $\eta < 2.17$

For the bin between 7 and 20 GeV, additional criteria are required: the tag has to pass a cut on the MVA ID$> 0.92$, $\sqrt(2 * PFMET * p_T(tag) * (1 - cos(\phi_{MET} - \phi_{tag}))) < 45 GeV$.

Probe electrons only need to be reconstructed as GsfElectron while the FSR recovery algorithm is not applied in efficiency measurement.

The nominal MC efficiencies are evaluated from the LO MadGraph Drell-Yan, while the NLO systematics use the MadGraph_AMCatNLO sample.

For the efficiency measurements a template fit is used. The $m_{ee}$ signal shape of the passing and failing probes is taken from MC and convoluted with a Gaussian. The data is then fitted with the convoluted MC template and a CMSShape (an

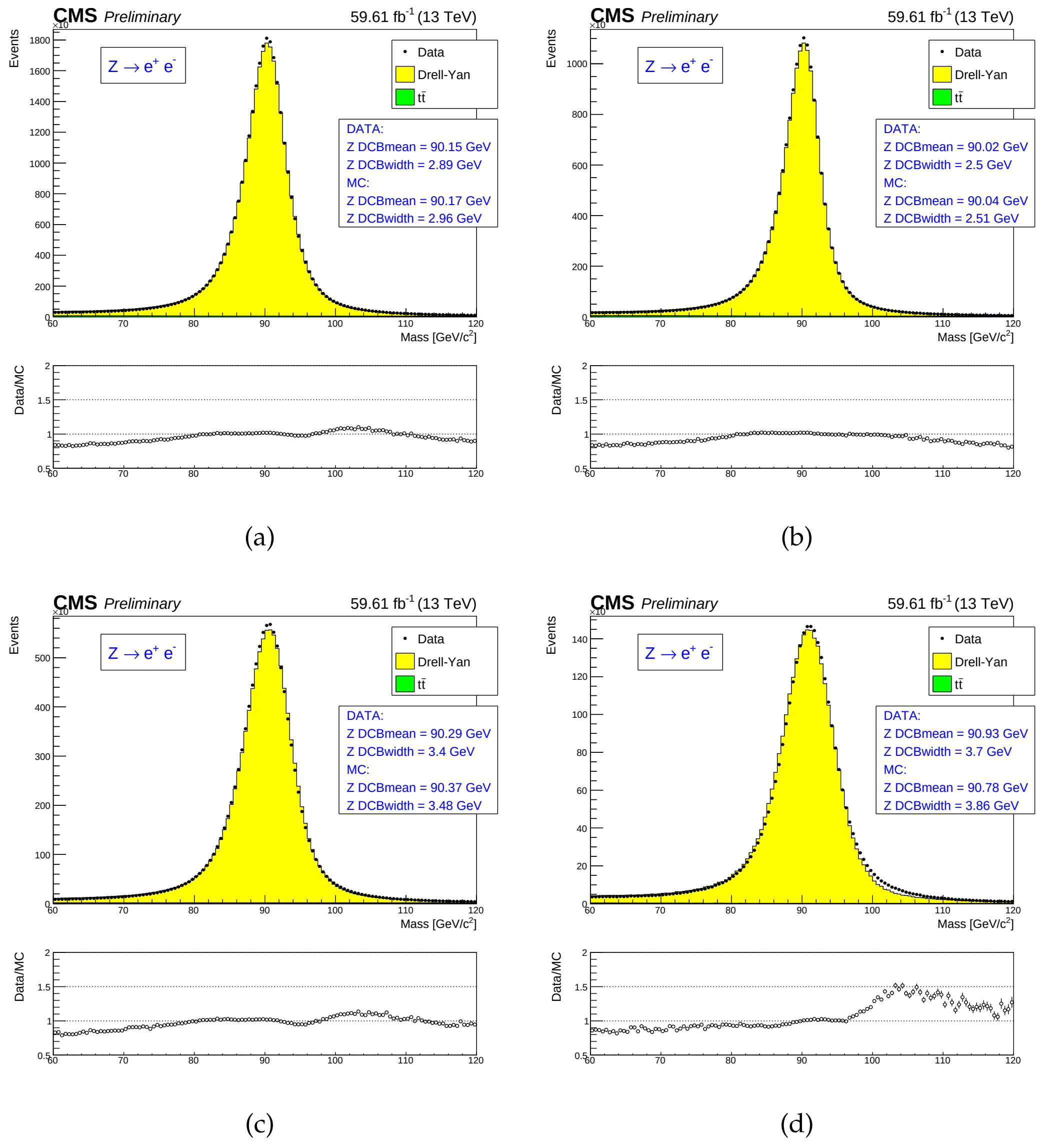
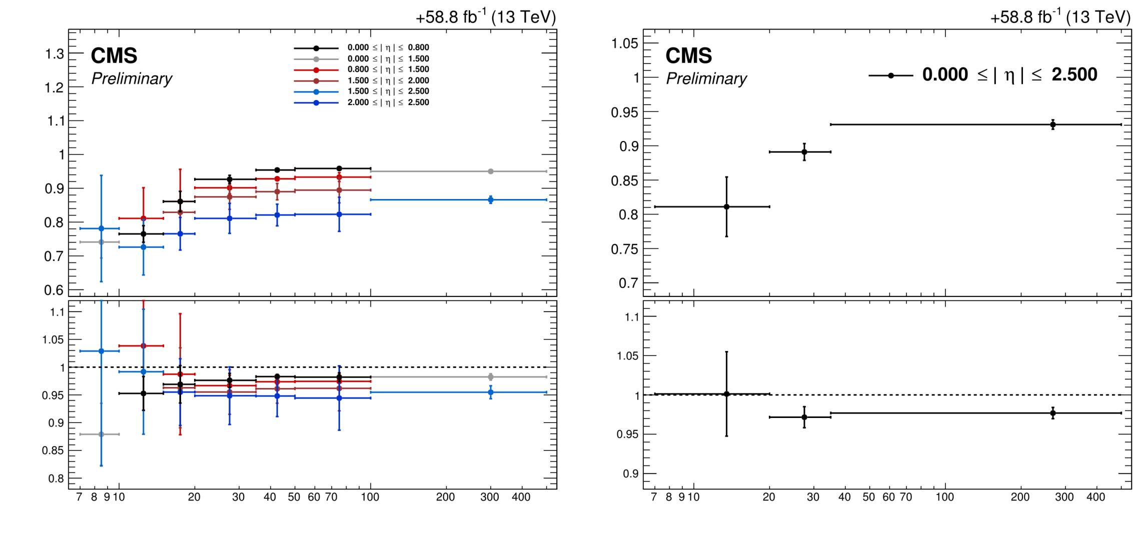



FIGURE 4.4: (a) electron energy scale measured in the $Z \rightarrow ee$ control region for all electrons,(b) for both electrons in barrel, (c) for one electron in barrel and other in endcaps, and for both of the electrons in endcaps (d). In these figures, the results based on fits using Crystall-ball are shown.

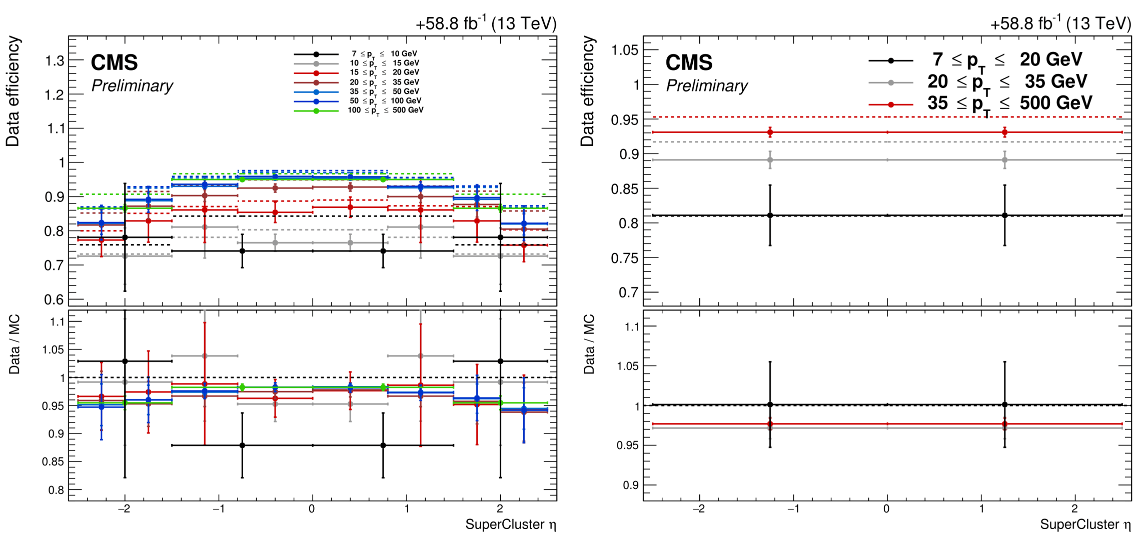
Error-function with a one-sided exponential tail). This change follows from the usage of the T&P tool developed by the EGM POG.

The measurement of electron selection efficiency is made as a function of the probe electron $p_T$ and its SC $\eta$. For electrons falling in the ECAL gaps, it is measured separately. Figure 4.5 and **??** shows the $p_T$ and $\eta$ turn-on curves measured in data.




(a) (b)

FIGURE 4.5: Electron selection efficiencies vs $p_T$ measured using the Tag-and-Probe technique, non-gap electrons (left) and gap electrons (right), together with the corresponding data/MC ratio bottom part of each plot.




(a) (b)

FIGURE 4.6: Electron selection efficiencies vs $\eta$, non-gap electrons (left) and gap electrons (right), together wit the corresponding data/MC ratio at the bottom part of each plot.

The EGM recommendations about evaluation of Tag-and-Probe uncertainties on efficiency measurements are followed. Specifically, we consider

- Variation of the signal shape from a MC shape to an analytic shape (Crystal Ball) fitted to the MC
- Use of analytic function (Crystal Ball) to fit the signal as alternative of MC template
- Using an NLO MC sample for signal templates

Assessment of total uncertainty for measurement of the scale factors is taken as quadratic sum of statistical uncertainties (returned from the fit) and the aforementioned systematic uncertainties.

### 4.1.2 Muons

#### Muon Reconstruction and Identification

We define **loose muons** as the muons that satisfy $p_T > 5$, $|\eta| < 2.4$, $dxy < 0.5$ cm, $dz < 1$ cm, where $dxy$ and $dz$ are defined w.r.t. the PV and using the 'muonBest-Track'. Muons have to be reconstructed by either the Global Muon or Tracker Muon algorithm. Standalone Tracks of muon reconstructed only in the muon system are rejected. Muons with `muonBestTrackType==2` (standalone) are discarded even if they are marked as global or tracker muons. More details on the reconstruction of muons can be seen in Ref. (**AN-15-277**).

Loose muons with $p_T$ below 200 GeVare considered identified muons if they also pass the PF muon ID (note that the naming convention used for these IDs differs from the muon POG naming scheme, in which the "tight ID" used here is called the "loose ID"). Loose muons with $p_T$ above 200 GeVare considered identified muons if they pass the PF ID or the Tracker High-$p_T$ ID, the definition of which is shown in Table 4.3. This relaxed definition is used to increase signal efficiency for the high-mass search. When a very heavy resonance decays to two bosons, both bosons will be very boosted. In the lab frame, the leptons originating from decay of a highly boosted $Z$ will be nearly collinear, and the PF ID loses efficiency for muons separated by approximately $\Delta R < 0.4$, which roughly corresponds to muons originating from bosons with $p_T > 500$ GeV.

TABLE 4.3: The requirements for a muon to pass the Tracker High-$p_T$ ID. Note that these are equivalent to the Muon POG High-$p_T$ ID with the global track requirements removed.

| Plain-text description | Technical description |
|---|---|
| Muon station matching | Matching of muon to the segments in two muon stations at least **NB: It means that muon is an arbitrated tracker muon.** |
| Good $p_T$ measurement | $\frac{p_T}{\sigma_{p_T}} < 0.3$ |
| Vertex compatibility $(x - y)$ | $d_{xy} < 2$ mm |
| Vertex compatibility $(z)$ | $d_z < 5$ mm |
| Pixel hits | One pixel hit at least |
| Tracker hits | Hits in at least six tracker layers |

It may happen that a single muon is reconstruted incorrectly as two muons or even more. To cope with this situation, additional "ghost-cleaning" is performed as:

- Those tracker Muons which are not Global Muons, are needed to be arbitrated.
- If two muons are sharing their segments upto 50% or more, lower quality muon is removed.

### Muon Isolation

Muon isolation used is based on Particle-Flow. So-called $\Delta\beta$ correction is used in order to subtract the pileup contribution for muons, whereby $\Delta\beta = \frac{1}{2}\sum_{\text{PU}}^{\text{charged had.}} p_T$ gives an estimate of the energy deposit of neutral particles (i.e. photons and hadrons) from the pile-up vertices. Relative isolation for muons is then defined as:

$$\text{RelPFiso} = \frac{\sum^{\text{charged had.}} p_T + \max(\sum^{\text{neutral had.}} + \sum^{\text{photon}} - \Delta\beta, 0)}{p_T^{\text{lepton}}} \tag{4.1}$$

The isolation working point for muons was optimized in Ref. (**AN-15-277**) and the working point was chosen to be the same as electrons, namely RelPFiso$(\Delta R = 0.3) < 0.35$.

### Muon Impact Parameter Selection

Selection on muon significance of impact parameter(SIP) is used same as for the electrons which is described in Section 4.1.1:

- $|\text{SIP}_{\text{3D}} = \frac{\text{IP}}{\sigma_{\text{IP}}}| < 4$

### Muon Energy Calibrations

Similar to electrons the momentum scale of muon is measured in data by fitting a Crystall-ball function to the di-muon invariant mass spectrum around the Z peak in the $Z \to \mu\mu$ control region. From Fig. 4.7 shows a very good comparison of data with simulation.

### Efficiency Measurements of muons

Efficiencies of muons are measured using same method used for electron measurements i.e. Tag-and-Probe (T&P) method performed on $Z \to \mu\mu$ and $\to \mu\mu$ events in bins of $p_T$ and $\eta$. More details on the methodology can be found in Ref. (**CMS_AN_2015-277**). Measurements are extracted using 2018 RunA,B,C,D data while the measurements corresponding to 2016 and 2017 datasets have already been reported in Ref. (**CMS_AN_2016-442**) and Ref. (**CMS_AN_2017-342**) respectively. For the measurement of muon reconstruction and identification efficiency at high $p_T$ ($p_T$>20GeV), and impact parameter requirements and isolation efficiencies for all $p_T$, Z sample is used. to measure the muon reconstruction and identification efficiency at high $p_T$, and the efficiency of the isolation and impact parameter requirements at all $p_T$. To measure the reconstruction and identification efficiency at low $p_T$, sample is used as it benefits from a better purity in that kinematic regime. In this case, events are collected using `HLT_Mu7p5_Track2_Jpsi_v*` when probing the reconstruction and identification efficiency in the muon system, and using the `HLT_Mu7p5_L2Mu2_Jpsi_v*` when probing the tracking efficiency.

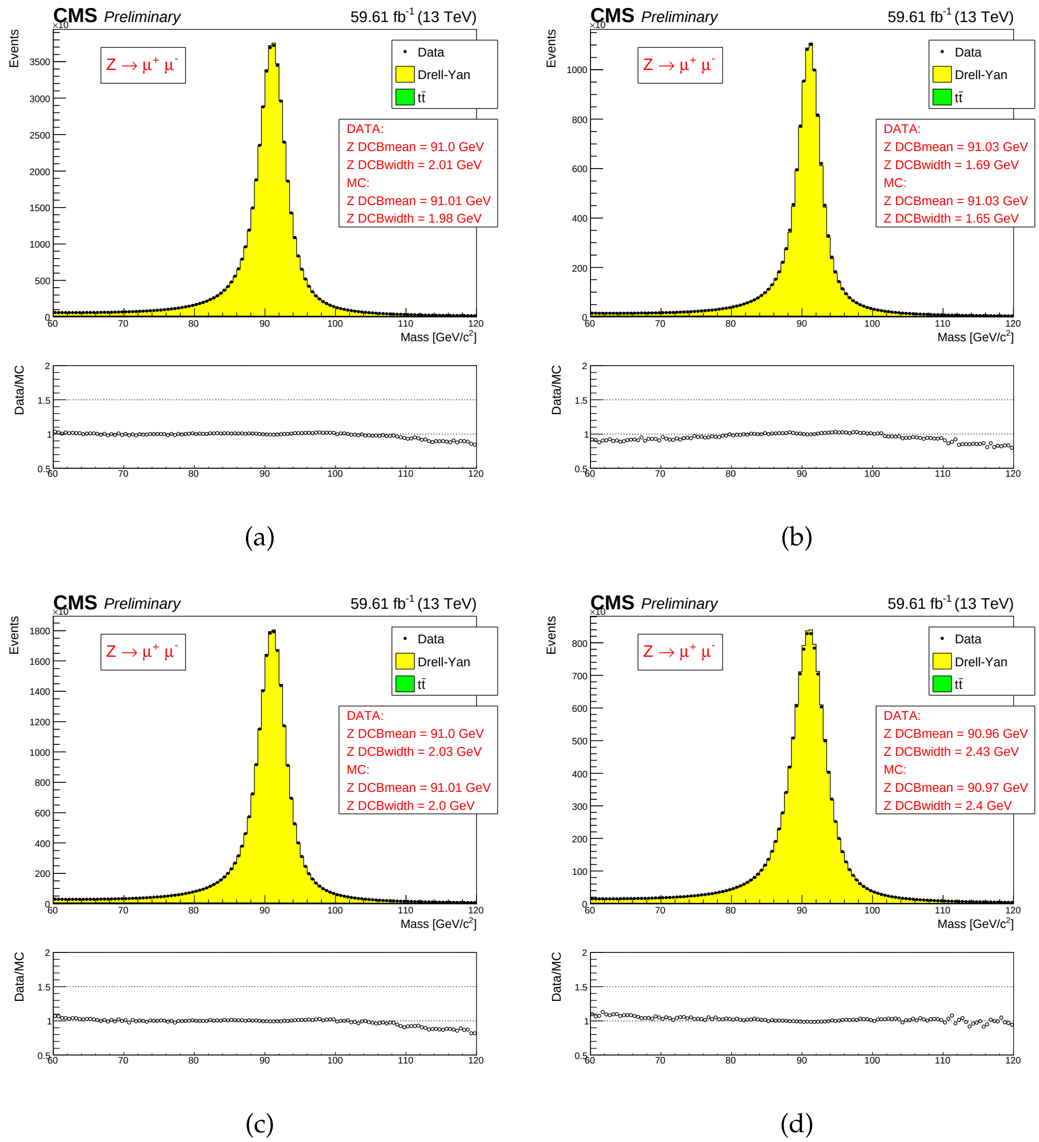
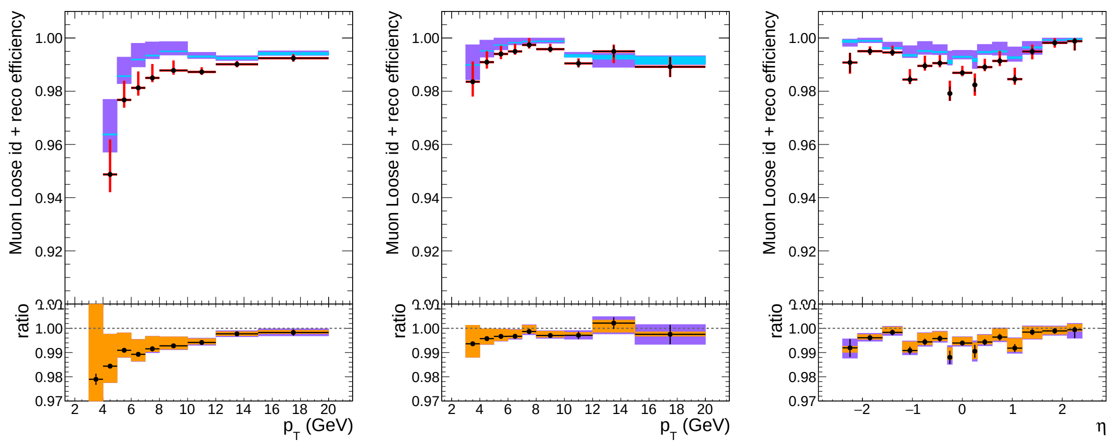



FIGURE 4.7: (a): muon energy scale measured in the $Z \to \mu\mu$ control region for all muons, for both of muons frome barrel region (b), for one muon from barrel, one from endcaps (c) and for both muons in the endcaps (d). The results in the figures are extrated from fits using the Crystall-ball function.

**Reconstruction and identification** Results for muon reconstruction and identification efficiencies for high $p_T$ muons ($p_T > 20$ GeV) have been derived by the Muon POG. The probe in this measurement are tracks of inner tracker, and the passing probes are those which are also regarded as a global or tracker muon and passing the Muon POG Loose muon identification. Results for low $p_T$muons were derived using events, with the same definitions of probe and passing probes. The systematic uncertainties on the measurements are estimated by including several variations e.g. varying the analytical signal and background shape models used to fit distribution of dimuon invariant mass. Furthermore, number of bins in dimuon distribution and mass window are also increased and decreased to include systematic effects. Details on the procedure can be found in Ref. (**AN-15-277**). The efficiency and scale factors used for low $p_T$ muons are the ones derived using single muon prompt-reco dataset.

The data and simulation efficiencies and their ratios in bins are shown in Fig. 4.8.



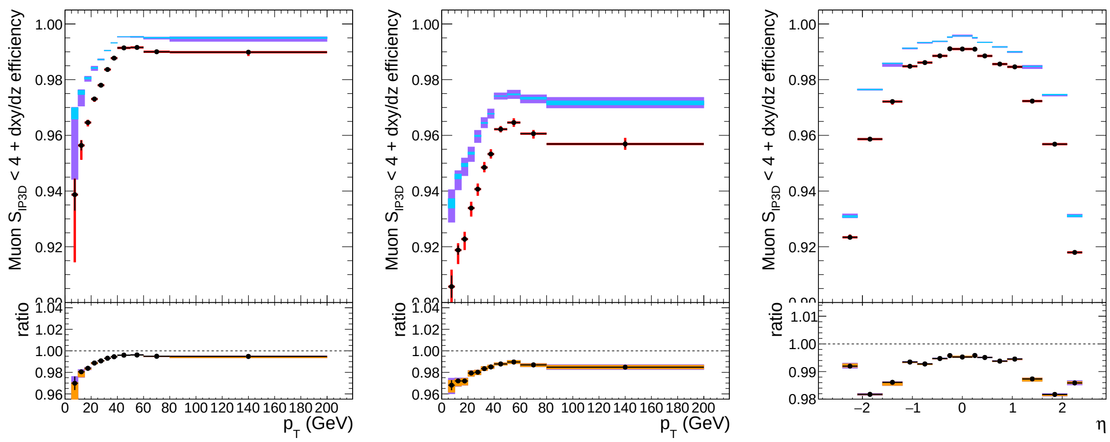

(a) (b) (c)

FIGURE 4.8: Reconstruction and identification efficiencies of muons at low $p_T$, measured with the $T\&P$ method on events, in bins of $p_T$in barrel (left) and endcaps (center), and in terms of $\eta$ for $p_T > 7$ GeV (right). In upper panel of each diagram, the larger uncertainty bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of two efficiencies, black uncertainty bars are for statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

**Impact parameter requirements** The measurement is performed using Z events. Events are selected with `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers. For this measurement, the probe is a muon passing the POG Loose identification criteria, and it is considered a passing probe if satisfies the SIP3D, dxy, dz cuts of this analysis. The results are shown in the Fig. 4.9.

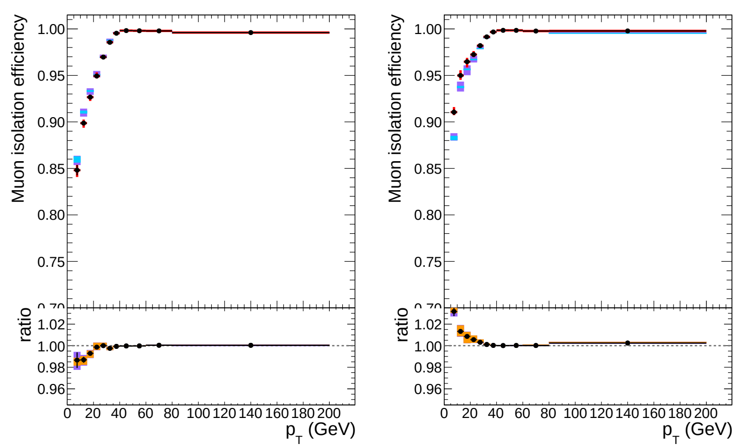
**Isolation requirements** Isolation efficiency is measured using events from the Z decay for any $p_T$. The events are selected with either of `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers. The isolation of the muons are calculated after recovery of the FSR photons and subtracting their contribution to the isolation cone of the muons. There are details described of this method which can be found in Ref. (**AN-16-217**).

The results are shown in Fig. 4.10.




(a) (b) (c)

FIGURE 4.9: Efficiency of the muon impact parameter requirements, measured with the $T\&P$ method on Z events, in bins of $p_T$in barrel (left) and endcaps (center), and in terms of $\eta$ for all $p_T$ (right). In upper panel, the larger uncertainty bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of two efficiencies, black error bars are for statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.




(a) (b)

FIGURE 4.10: Efficiency of muon isolation requirement, measured with the tag&probe method on Zevents, as function of $p_T$in the barrel (left) and endcaps (right). In the upper panel, the larger error bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of the two efficiencies, the black error bars are for the statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

**Tracking** Tracking efficiency (i.e. efficiency to associate a track to some muon in inner detector) is almost 100% and hence muon Physics Object Group no more recommends tracking scale factors in the analyses.

**Overall results of muon Scale Factors** The product of all scale factors for reconstruction, identification, impact parameter and isolation requirements is shown in two dimensional display in Fig. 4.11. The figure also shows overall uncertainty on scale factors.

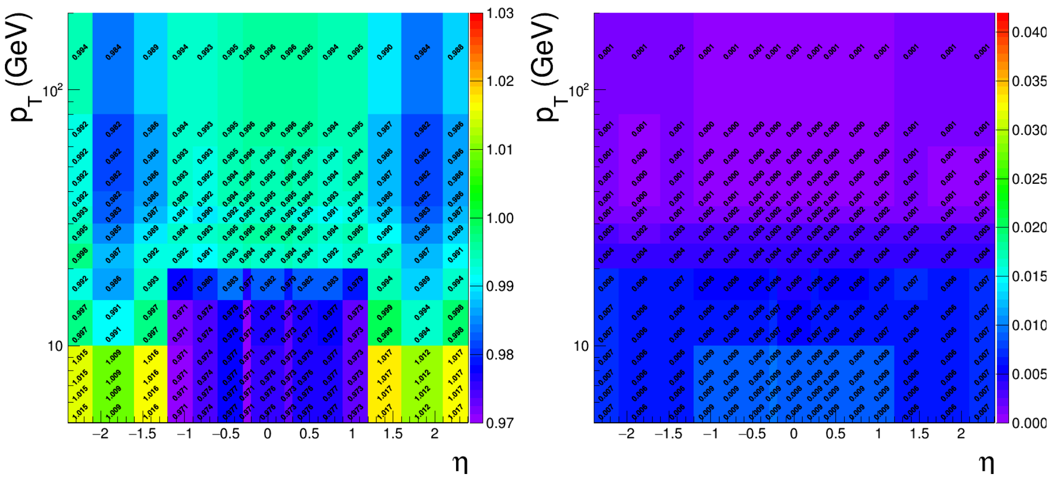



FIGURE 4.11: Left: Overall scale factors for muons, as function of $p_T$and $\eta$. Right: Uncertainties on overall scale factors muons shown in two dimensions of $p_T$and $\eta$.

### 4.1.3 Photons for FSR recovery

The FSR recovery algorithm was considerably simplified with respect to what was done in Run I, while maintaining similar performance. The selection of FSR photons is now only done per-lepton and no longer depends on any Z mass criterion, thus much simplifying the subsequent ZZ candidate building and selection. As regards the association of photons with leptons, the rectangular cuts on $\Delta R(\gamma, l)$ and $E_{T,\gamma}$ have been replaced by a cut on $\Delta R(\gamma, l)/E^2_{T,\gamma}$.

Initially, photons are collected using particle-flow alogorithm. Further selection and their association with a lepton is done as follows:

1. PF photons are preselected by requiring $p_{T,\gamma} > 2$ GeV, $|\eta^\gamma| < 2.4$, and a relative Particle-flow isolation smaller than $1.8$ (computed within a cone of radius $R = 0.3$). On charged hadrons and neutral hadrons or photons $0.2$ GeV and $0.5$ GeV thresholds are applied with a size of veto cone $0.0001$ and $0.01$ respectively, also considering contribution from pileup vertices (with the same radius and threshold as per charged isolation) .

2. Supercluster (SC) veto: All photons matching (direct association with PF candidates) with some electron that passes ID and SIP requirements are removed.

3. Association of photons is made with lepton which is closest and passes ID and SIP requirements.

4. Photons not satisfying the requirements $\Delta R(\gamma, l)/E^2_{T,\gamma} < 0.012$, and $\Delta R(\gamma, l) < 0.5$ are discarded.

5. In case if a photon gets associated with same lepton, the one with less $\Delta R(\gamma, l)/E^2_{T,\gamma}$ is opted.

6. For each selected FSR photon, we exclude that photon from sum of isolation of all leptons in an event which pass both loose ID and SIP requirements. This concerns the photons which are in isolation cone and outside veto of isolation of said leptons ($\Delta R < 0.4$ and $\Delta R > 0.01$ for muons and $\Delta R < 0.4$ and ($\eta^{\text{SC}} < 1.479$ or $\Delta R > 0.08$) for electrons).

More details on the optimization of the FSR photon selection can be found in Ref. (**AN-15-277**; **AN-16-217**).

### 4.1.4 Jets

Higgs production through fusion of vector bosons (VBF) and through other mechanisms normally differ as regards the jet kinematics.

### Jet Identification

Jets are reconstructed through anti-$k_T$ clustering algorithm out of particle-flow candidates, with distance parameter $R = 0.4$, after rejecting the charged hadrons that are associated with a pileup primary vertex.

To reduce instrumental background, the tight working point jet ID suggested by the JetMET Physics Object Group is applied (**JetID2018**). In addition, jets from Pile-Up are rejected using the PileUp jet ID criteria suggested by the JetMET POG (**JetPUID2017**). In this analysis, the jets from within $|\eta| < 4.7$ area are required to have a transverse momentum greater that 30 GeV. In addition, jets are cleaned from any of the tight leptons (passing the SIP and isolation cut computed after FSR correction) and FSR photons by a separation criterion: $\Delta R(\text{jet,lepton/photon}) > 0.4$.

### Jet Energy Corrections

The capability of current calorimeter to measure the energy of jets correctly is not linear. Hence the measured energy cannot be regarded as particle's true energy and we need to estimate corrections to Jet Energy. In the analysis, standard jet energy corrections are implemented to the reconstructed jets, which consist of L1 Pileup, L2 Relative Jet Correction, L3 Absolute Jet Correction for both Monte Carlo samples and data, and also residual calibration for data (**JECMC2018**). Jet Energy resolution corrections are also applied to MC.

### B-tagging

In order to distinguish whether a jet is b-jet or not *DeepCSV* algorithm is used as our b-tagging algorithm. It combines the same information as the previous tagger *CSVv2* i.e. SIP, jet kinematics and secondary vertex but uses information of more tracks. Also, the b-tag output discriminator is computed with a Deep Neural Network. In the analysis, a jet is considered as a b-tagged if it meets requirement of *medium* working point, i.e. if its `|pfDeepCSVJetTags:probb+` pfDeepCSVJetTags:probbb| discriminator is greater than 0.4184 (**BTAG2018**).

Scale factors of data to simulation efficiencies for b-tagging are provided for this working point for the full dataset varied in jet $p_T$, $\eta$ and flavour. They are applied to simulated jets by downgrading (upgrading) the b-tagging status of a fraction of the b-tagged (untagged) jets that have a scale factor smaller (larger) than one.

### Validation on data

Similarly to what was done with electrons and muons, some validation studies are performed to probe agreement of data with simulated samples for jets. Events containing $\mathrm{Z} \to \mathrm{ee}$ or $Z \to \mu^+\mu^-$ were selecting, following the same trigger and lepton selection criteria of the main analysis, but increasing the $p_T$ threshold on leptons to 30 GeV and asking the dilepton invariant mass window of 70-110 GeV. Basic properties of additional jets in an event, requiring $p_T > 30$ GeV and $\eta < 4.7$, were then compared. Figure 4.12 shows comparison of data with the MC in bins of pseudo-rapidity of the leading jet. Simulated samples of Drell-Yan and ¯are used. Uncertainties from the jet energy corrections are also displayed on ratio plots of data to MC.

As it is shown, Jet Energy Resolution corrections are creating the “horns" one can see in the MC for $\eta$ of about 2.8. Although not satisfactory, the disagreement of data from MC we see in this region is covered by the very large uncertainty. After discussion with jet experts in the JetMET POG, it has been decided that this is the configuration we should use for this preliminary result. We also verified that it was not inducing significant migration of events in the signal MC between the categories.

The disagreement of data with MC for the MET distribution prevents us at this stage to use this variable in the analysis, in particular to better target VH events where, for instance, a Z would decay into neutrinos.

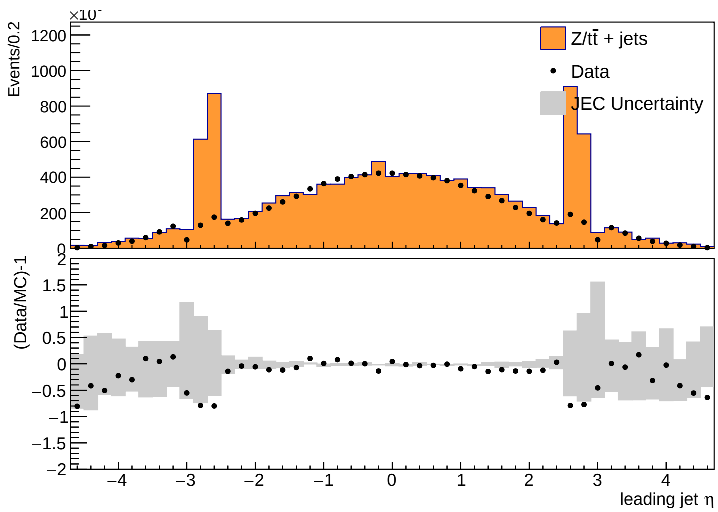



FIGURE 4.12: Comparison of data with MC for leading jet $\eta$ in Z$\rightarrow \ell\ell$ +jets. MC is normalized to data. Jet ID and Jet PUID are applied. MC samples include DY and ttbar. Data/MC ratio plots are shown in the bottom of each plots, together with the uncertainties (shaded histograms) from Jet Energy Corrections.

### 4.1.5 Object Summary

The requirements on all objects used in the analysis (with 2018 data) are summed up in the Table 4.4. In addition, a "ghost-cleaning" procedure is applied to the muons, as described in Sec. 4.1.2.

A lepton is declared **loose** if it passes the reconstruction, kinematics and dxy/dz cuts and declared **tight** if it passes in addition the identification, isolation and SIP3D cut.

TABLE 4.4: Chart of selection of physics objects for the $H \to ZZ \to 4\ell$ analysis used on the 2018 data.

| **Electrons** |
|---|
| $p_T^e > 7$ *and* GeV $\eta^e < 2.5$ |
| $d_{xy} < 0.5\,cm$ and $d_z < 1\,cm$ |
| $\lvert SIP_{3D}\rvert < 4$ |
| BDT ID with isolation (Fall17v2 training) with cuts from Table 4.2 |
| **Muons** |
| Global or Tracker Muon |
| Discard Standalone Muon tracks if reconstructed in muon system only |
| Discard muons with muonBestTrackType==2 even if they are globa or tracker muons |
| $p_T^\mu > 5$ GeV and $\eta^\mu < 2.4$ |
| $d_{xy} < 0.5\,cm$ and $d_z < 1\,cm$ |
| $\lvert SIP_{3D}\rvert < 4$ |
| The PF muon ID if $p_T < 200$ GeV, PF muon ID or High-$p_T$ muon ID (Table 4.3) if $p_T > 200$ GeV |
| $\mathcal{I}_{PF}^{\mu} < 0.35$ |
| **The FSR photons** |
| $p_T^\gamma > 2$ GeV and $\eta^\gamma < 2.4$ |
| $\mathcal{I}_{PF}^{\gamma} < 1.8$ |
| $\Delta R(\ell,\gamma) < 0.5$ and $\frac{\Delta R(\ell,\gamma)}{(p_T^\gamma)^2} < 0.012$ GeV$^{-2}$ |
| **Jets** |
| $p_T^{jet} > 30$ GeV and $\eta^{jet} < 4.7$ |
| $\Delta R(\ell/\gamma, \text{jet}) > 0.4$ |
| Cut-based jet ID (tight WP) |
| Jet pileup ID (tight WP) |
| Deep CSV algorithm for b-tagging (medium WP) |

# Chapter 5

# Probing four lepton final-state for Higgs Cross section measurement

## 5.1 Introduction

Higgs particle has been the first fundamental scalar particle which was observed at LHC, CERN. Some years ago in 2012, both CMS and ATLAS collaborations announced its discovery (**Reference20**; **Reference21**) with its mass around 125 GeV. Later on, these experiments have also studied its comprehensive properties (**Reference22**; **Reference21a**; **Reference21b**; **Reference21c**) which are found to be consistent with the SM Higgs.

In this chapter, measurements of fiducial cross sections (inclusive and differential) of Higgs in the $H \rightarrow 4\ell$ ($\ell = e, \mu$) decay channel using full Run 2 data recorded with CMS from collisions of proton-proton at $\sqrt{s} = 13TeV$ are presented which give a direct test of the perturbative QCD calculations of SM (Higgs sector) at TeV energy scale. Any deviation of the integrated or differential distributions from SM expectations would provide a direct hint to BSM contributions. These measurements with earlier or partial Run2 CMS data has also been performed Ref. [JHEP 04 (2016) 005, CMSPAS HIG-15-004, JHEP 11 (2017) 047, HIG-17-028] . Nearly similar measurements have already been reported by ATLAS experiment in $H \rightarrow \gamma\gamma$ (**ATLAS-CONF-2019-029**) (**ATLAS:Hgg_8TeV**; **ATLAS:Hgg_8TeV_13TeV**), $H \rightarrow 4\ell$ [ATLAS-CONF-2019-025], (**ATLAS:H4l_8TeV**; **ATLAS:H4l_8TeV_13TeV**) decay channels, combination of both channels [ATLAS-CONF-2019-032] (**ATLAS:H4l_Hgg_8TeV**; **ATLAS:H4l_Hgg_8TeV_13TeV**) and by CMS experiment in $H \rightarrow \gamma\gamma$ decay channel (**CMS:Hgg_8TeV**).

The integrated fiducial cross section measurements are performed using collisions data of CMS (LHC) corresponding to 137.1 $fb^{-1}$ integrated luminosity at $\sqrt{s} = 13$ TeV. It is important to minimize results' dependence on underlying models of Higgs production mechanism and its properties. Hence, cross sections are extracted in the fiducial phase space that closely matches with the detector geometry as shown in Figure 5.1. The remaining model dependence is extracted by repeating the measurements for SM and other exotic models and reported as a distinct systematic uncertainty. All measurements includes the corrections for the finite detection efficiency and resolution effects. As $\sigma \times BR$ strongly depends on the mass of Higgs, measurements are extracted assuming $m_H = 125.09$ GeV, as measured from $H \to 2\gamma$ and $H \to 4\ell$ decay modes at CMS experiment of LHC (**Reference39**).

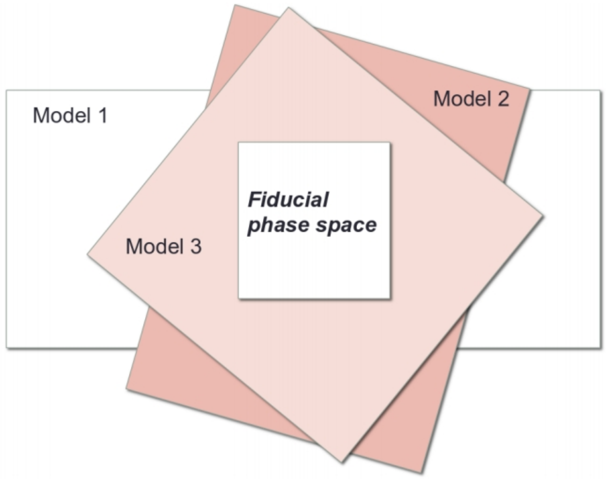



FIGURE 5.1: Fiducial phase definition among different models.

The cross sections are also measured differentially as a function of kinematic observables. In this dissertation we study the differential observables which are sensitive to production mechanism of Higgs: transverse momentum and the rapidity of Higgs, the transverse momentum of leading jet and jet multiplicity.

The structure of this chapter follows as the datasets and MC or simulated samples used are introduced in Section 5.1.1. The analysis strategy is described in Section 5.2, where event selection, fiducial phase space definition and strategy method for extracting inlusive and differential fiducial cross sections. In Section 5.3 details of systematic uncertainties affecting the cross section measurement are explained. In Section 5.4 results of measurements are compared with SM predictions.

### 5.1.1 Datasets, Simulation Samples and Theoretical Predictions

### 5.1.2 Datasets

#### Triggers and Datasets

The analysis uses full Run 2 (2016, 2017 and 2018) CMS data sample corresponding to $137.1\ \mathrm{fb}^{-1}$ intergrated luminosity.

The datasets used are listed and documented in (**HIG-19-001**). The analysis relies on four different primary datasets (PDs), *DoubleMuon*, *MuEG*, *EGamma*, and *SingleMuon*, each of which combines a certain collections of HLT paths. To avoid duplicate events from different primary datasets, events are taken:

- from EGamma if they pass the diEle or triEle or singleElectron triggers,

- from DoubleMuon if they pass the diMuon or triMuon triggers and fail the diEle and triEle triggers,
- from MuEG if they pass the MuEle or MuDiEle or DiMuEle triggers and fail the diEle, triEle, singleElectron, diMuon and triMuon triggers,
- from SingleMuon if they pass the singleMuon trigger and fail all the above triggers.

The HLT paths used for 2018 collision data is listed in (**HIG-19-001**).

#### Trigger Efficiency

Efficiency in data of combination of triggers used for the anlysis is measured by considering $4\ell$ events triggered by single lepton triggers. Among the reconstructed four leptons, is geometrically associated with an object that passes at least one single electron or muon triggers. Rest of three leptons are regarded as “probes". In a $4\ell$ event, the possible tag-probe pairs are 4 and are counted in denominator while computing efficiency. For each of three leptons which are probes, all matching trigger filter objects are collected. Then the matched trigger filter objects of the three probe leptons are combined in attempt to reconstruct any of the triggers used in the anlysis. If any of the analysis triggers can be formed using the probe leptons, probes are call passing probes which are countedas numerator for the efficiency measurement.

This technique is used to measure efficiency on MC and the comparison with data is exploited to estimate the reliability of simulation. Fig. 5.2 shows the measured efficiency both in data and MC (2018) varied with minimum $p_{\mathrm{T}}$ of leptons. MC efficiency seems to describe data well within the uncertainties.

### 5.1.3 Simulation Samples and Theoretical Predictions

Simulations of the interesting events help us build a realistic model of experiment in high energy. It enables us for a detailed study of physics processes (signal and background). Particularly for proton-proton collisions, generation of simulated events can be separated in many stages. For this analysis, in particular, descriptions of the production of SM Higgs through process of gluon-gluon fusion (Fig. 5.3) are acquired using POWHEG V2 (**powheg**; **powhegvv**) generator. At accuracy level of next-to-next-to-leading order (i.e. NNLO) in perturbative QCD (pQCD), $\mathrm{gg} \rightarrow \mathrm{H}$ process description is obtained by partonic level MC generator NNLOPS.

The POWHEG, provides perturbative QCD calculations at next-to-leading (NLO) accuracy with an connection to parton-shower programs. Using the POWHEG, Higgs production via $\mathrm{gg} \rightarrow \mathrm{H}$ with zero associated jets can be described at accuracy level of NLO. Finite masses of top and bottom quarks has been considered for $\mathrm{gg} \rightarrow \mathrm{H}$ production for all generators. The SM Higgs boson production by

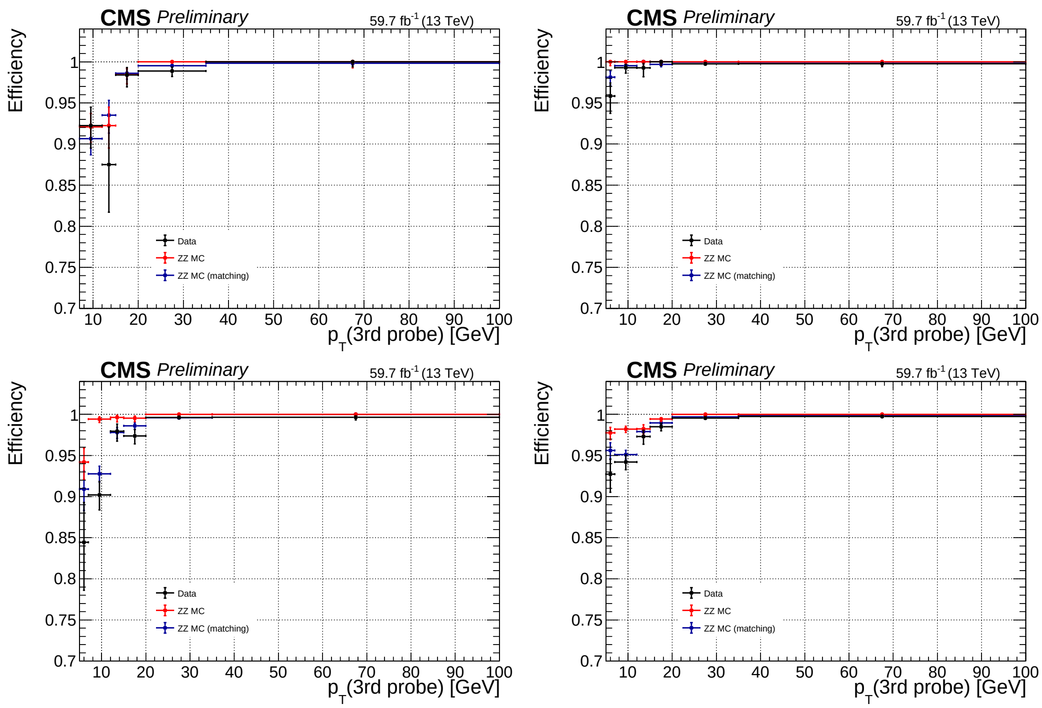



FIGURE 5.2: Measured trigger efficiency in 2018 data where the event collection is made using single electron and muon triggers: $4e$ (top left), $4\mu$ (top right), $2e2\mu$ (bottom left) and $4\ell$ (bottom right) final states.

### Signal Samples

Descriptions of the SM Higgs boson production are obtained using the POWHEG V2 (**Alioli:2008gx**; **Nason:2004rx**; **Frixione:2007vw**) generator for the five main production modes (ggH, VBF, WH, ZH and ttH) as described in section 2.4 (**Bagnaschi:2011tu**), (**Nason:2009ai**), (**Hartanto:2015uka**), (**Luisoni:2013kna**). However, WH and $Z$H production modes are simulated by the MINLO HVJ extension of the POWHEG (**Luisoni:2013kna**). Higgs decay to four lepton is simulated by JHUGEN generator (**Gao:2010qx**). For the WH, ZH and ttH modes, Higgs is allowed to decay to H$\to$ $ZZ$ $\to$ $2\ell2$X such that two leptons from 4-lepton events are produced from the decay of associated W, Z bosons or top quarks are included in simulation. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards according to the POWHEG internal scale. All samples are generated with the NNPDF 3.1 NLO parton distribution functions (PDFs) (**Ball:2014uwa**). The list of signal samples with the corresponding cross sections are shown in (**HIG-19-001**). These samples are also used to asses model dependence of measurement results are indicated in the Table **??**.

The MC samples generated by POWHEG and PYTHIA8.209 are used to extract fiducial phase space acceptance and reconstruction efficiencies for individual models separately as explained in Section 5.2.4. These samples, with the sample generated by using the NNLOPS with alternate representation of $\mathrm{gg} \rightarrow \mathrm{H}$ process, are utilized for the comparison of measurement results with the theoretical predictions based on SM calculation in Section 5.4.

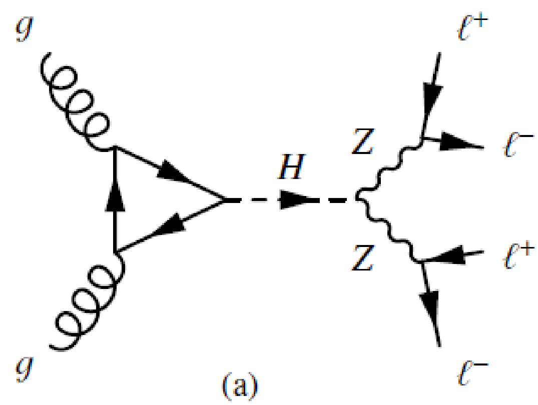



FIGURE 5.3: $\mathrm{gg} \rightarrow \mathrm{H}$ process fynman diagram.

All SM production modes have been produced individually to approximate the dependence of measurements on underlying assumptions of Higgs production mechanism. The details of the method used to asses the model dependence will be described in Section 5.3. All these MC samples are produced using POWHEG for the representation of QCD effects at NLO in Higgs production, and by using JHUGEN (**Gao:2010qx**; **Bolognesi:2012mm**; **Anderson:2013afp**) to represent such exotic resonances decays to four-leptons containing all the spin correlations.

### Background Samples

For the estimations of the backgrounds using MC samples, the dominant process $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ} \rightarrow 4\ell$ was generated at the NLO accuracy level containing s-, t- and u-channel diagrams (Figure 5.4) using POWHEG V2 (**Nason:2013ydw**) and PYTHIA8, with the same configurations as for the Higgs signal. The dynamical renormalization and factorization scales have been chosed equal to $m_{\mathrm{ZZ}}$ since the simulation covers a wide range of invariant masses of ZZ.

The $\mathrm{gg} \rightarrow ZZ$ process (Figure 5.4) is simulated at leading order (LO) with MCFM (**MCFM**; **Campbell:2013una**). In order to have matching in spectra of $\mathrm{gg} \rightarrow \mathrm{H} \rightarrow ZZ$ as predicted at NLO by POWHEG, the showering of these samples has been performed with some different PYTHIA8 settings. This only allows emissions only up to parton-level scale ("wimpy" shower).

The initial-state and final-state radiations are included for all events generators, which provides hints for the existence of extra (hard) photons in the event. Interference between all amplitudes of process $H \rightarrow 4\ell$ are properly taken into consideration both in POWHEG and the JHUGEN. For the generators (at LO, NLO

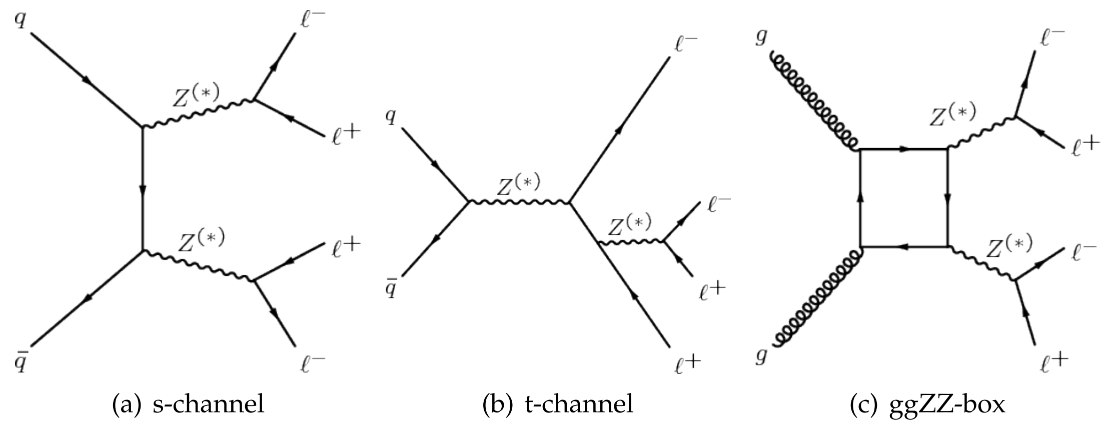



FIGURE 5.4: $q\bar{q}toZZ \to 4\ell$ process s- and t- channel fynman diagrams (a) and (b). $gg \to ZZ \to 4\ell$ process fynman diagram (c).

at NNLO) CTEQ6L (**cteq66**), CT10 (**CT10**) and MSTW2008 (**MSTW08**) PDFs are used.

The generated events were interfaced to PYTHIA 6.4 Tune $Z2^*$ (**Chatrchyan:2011id**) which follows CTEQ6L, to include the fact of the parton shower, underlying events and hadronization.

Processing of all generated events was made via a full simulation for CMS detector using GEANT4 (**GEANT**) and their reconstruction was done by exploiting same algorithms used for the data analysis. Multiple proton-proton interactions are incorporated in the simulations by reweighting the pileup distribution to match the distribution of the observed data. While measuring the energy of jets and isolation of objects (using FASTJET technique) (**Cacciari:2007fd**; **Cacciari:2008gn**; **Cacciari:2011ma**), the energy deposits from pileup interactions and underlying events are subtracted.

All the data-to-MC scale factors are computed by using a "T&P" technique (**CMS:2011lsa**), as described in Sections 4.1.1 and 4.1.2, and are used to correct simulated samples to compensate the differences of leptons' selection efficiencies in the data and MC simulation. There is negligibe dependence of the selection efficiencies used in the analysis over the primary vertices. More details are presented and discussed in Ref. (**Chatrchyan:2013mxa**).

### Pileup Reweighting

For each year, corresponding simulation samples are reweighted to meet pileup distribution in the data. An example of the reweighting procedure for 2018 data is shown in Fig. 5.5.

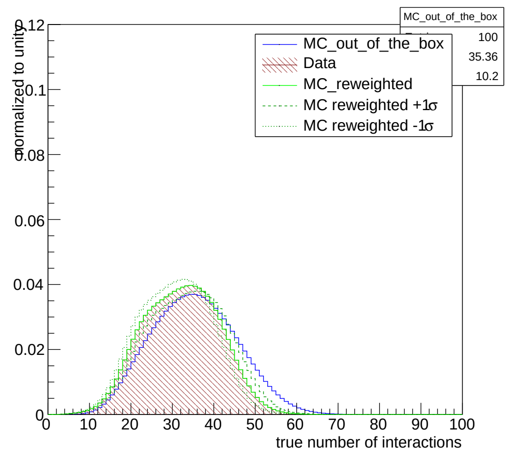



FIGURE 5.5: Pileup distribution in 2018 MC and Data shown before and after the application of PU weights. Up and down variations of 5% in the minimum bias cross section when calculating the weighs are also shown.

## 5.2 Analysis Strategy

In order to minimize the effects of model depedent acceptances, measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region. Fiducial selection mimics the reconstruction level selection, which is optimized to detect a low mass Higgs ($m_H$ 125 GeV) boson decaying to 4 leptons through a pair of Z bosons. Since in SM there are multiple production modes, the fiducial selection and measurement strategy are designed to be independent of how the Higgs boson is produced.

For this reason, the inclusion of isolation in the fiducial phase space definition is necessary to make the reconstruction efficiencies independent of fiducial phase space.

Furthermore, in sub-leading SM production modes there can be associated vector bosons which can decay to leptons, and therefore the reconstructed Higgs boson candidate may not be composed of the correct combination of the leptons. In order to retain the full information from an unbinned simultaneous fit of the signal and the background PDFs to the four-lepton distribution of invariant mass, it is necessary to treat such “wrong combination” events as a background.

Further signal-induced background can be formed due to imperfect correspondence between quantities at particle level and corresponding quantities at the reconstruction level. Signal events which are from within fiducial phase space but fail the reconstruction are accounted in the fiducial efficiency ($\epsilon$) and the expected contribution from events passing the reconstruction selections but are out of the fiducial phase space ($f_{nonfid}$) are referred to as “nonfiducial signal” events and are treated as a background, illustrated in Figure 5.6.

The effects of imperfect detector resolution can in general have a non-negligible

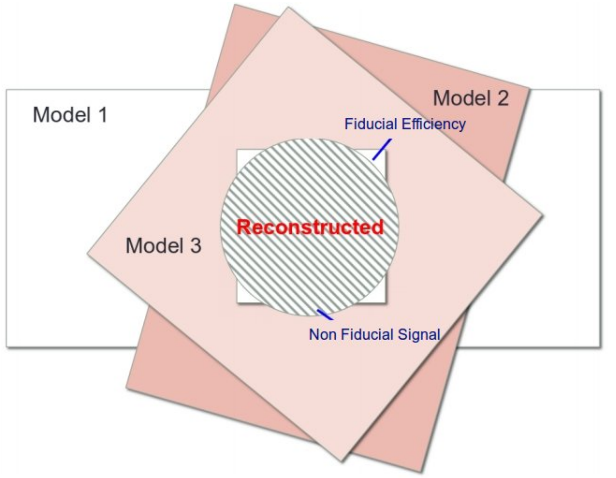



FIGURE 5.6: Fiducial efficiency($\epsilon$) and nonfiducial($f_{nonfid}$) signal definition.

effect on shape of the distribution of measured observables. For this reason, for differential cross section measurements of Higgs, a procedure to correct the detection efficiencies and resolution effects is applied. Throughout this document, this procedure will be referred to as the unfolding procedure, and the unfolded differential distributions will be referred to as distributions at the fiducial level. The reconstruction level selection, fiducial phase space definition, statistical procedure, and their model dependence will be described in the subsequent Sections.

### 5.2.1 Event Selection and Background Modeling

#### Event Selection

This dissertation is based on the analysis for which event selection for measurements are similar to what was used in pervious analyses on same topic and in the same final state (**HIG-18-001**; **HIG-16-041**; **Chatrchyan:2013mxa**; **Khachatryan:2014kca**). The online event selection is based on having fired High Level Trigger paths as described in (**sec:trigpaths**). Unlike in the Run I analysis, the trigger requirement does not depend on the selected final state: it is always the OR of all HLT paths. The reason is in Run II we will be targeting associated production modes that can come with additional leptons, thus improving trigger efficiency further. The events are required to have at least one primary vertex (PV) that fullfills the criteria of having large degree of freedom ($N_{PV} > 4$) with radius of PV being small ($r_{PV} < 2$ cm) and the collisions being restricted along $z-$axis ($z_{PV} < 24$ cm).

For the double lepton triggers, the requirement on minimum $p_T$ for leading and the subleading lepton are imposed to be 17 and 8 GeV respectively; however, for triple electron triggers, it is 15, 8 and 5 GeV in the same order respectively.

### 5.2.2 ZZ Candidate Selection

The four-lepton candidates are built from what we call **selected leptons**, which are the tight leptons (defined in sections 4.1.1 and 4.1.2) that pass the $\mathrm{SIP}_{3D} < 4$ vertex constraint and the isolation cuts (defined in sections 4.1.1 and 4.1.2), where FSR photons are subtracted as briefed in Section 4.1.3. A lepton cross cleaning is applied by discarding electrons which are within $\Delta R < 0.05$ of selected muons.

The construction and selection of four-lepton candidates proceeds according to the following sequence:

1. **Z candidates** are built from oppositely charged and same flavoured leptons ($e^+e^-$, $\mu^+\mu^-$) with additional requirement on dilepton mass as $12 < m_{\ell\ell(\gamma)} < 120 GeV$. Z boson candidate is also required to include FSR photons (if any).

2. **ZZ candidates** are then built from non-overlapping Z boson candidates. The one who is reconstructed with $m_{\ell\ell}$ closer to the SM Z boson mass is regarded as $\mathrm{Z}_1$ and other as $\mathrm{Z}_2$. Additional requierements on ZZ candidates are as under:

    - **Ghost removal** : Selected four leptons are required to have $\Delta R(\eta, \phi) > 0.02$ among each other.
    - **lepton** $p_T$: Among selected four leptons, two are required to fulfill $p_{T,i} > 20$ and $p_{T,j} > 10$ .
    - **QCD suppression**: Dilepton inivariant mass is required to pass (regardless of flavor) $m_{\ell\ell} > 4$ for QCD induced $J/\Psi$ veto.
    - $\mathrm{Z}_1$ **mass**: $40\ GeV < m_{Z1} < 120\ GeV$
    - **four-lepton invariant mass**: $m_{4\ell} > 70$

3. Events containing one selected ZZ candidate at leaset should form the **signal region**.

### 5.2.3 Choice of best ZZ Candidate

Unlike in the Run I analysis, the best ZZ candidate is now chosen after all kinematic cuts, a change that allows to test other selection strategies for this candidate choice. This is especially relevant for events with more than four selected leptons, having in mind search for Higgs boson production modes where associated particles can decay to leptons, such as VH and ttH.

For fiducial cross section measurement, $Z_1$ is chosen with mass closest to the mass of SM Z boson mass while $Z_2$ is chosen from Z candidates whose lepton pair gives higher $p_T$ sum.

Reconstruction of the jets is incumbent to explore the differential cross section measurements with respect to jets related observables. Jets are reconstructed with $p_T \geq 30$ GeV within $|\eta| \leq 4.7$, discussed in Section **??**. To remove overlapping of jets overlap from leptons or selected photons, it is required to have at least $\Delta R > 0.5$ where $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ (**Chatrchyan:2013mx**) with respect to any of the selected leptons and FSR photons candidates.

### Background Modeling

The dominant and sub-dominant irreducible backgrounds for $\mathrm{H} \rightarrow 4\ell$ process are the production of ZZ via annihilation of a $\mathrm{q\bar{q}}$ pair and the gg fusion respectively. Other background is reducible which is of the instrumental nature and arises from the process where partial jets are misidentified as leptons. The main processes which contributes in this background are: production of jets in association with Z boson, production of the jets in association with WZ boson pair and the $\mathrm{t\bar{t}}$ pair production. This background is as Z+X.

TABLE 5.1: Number of observed candidates, expected background and the signal events passing all selections in mass range $m_{4\ell} > 70$ GeV corresponding to three different years. Z+X is estimated by control regions in the data while ZZ and signal events are estimated using MC simulation.

| **Channel** | 4e | $4\mu$ | $2e2\mu$ | $4\ell$ |
|---|---|---|---|---|
| 2016 (13 TeV) | | | | |
| $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ | $192.7^{+18.6}_{-20.1}$ | $360.2^{+24.9}_{-27.3}$ | $471.0^{+32.6}_{-35.7}$ | $1023.9^{+68.9}_{-76.0}$ |
| $\mathrm{gg} \rightarrow \mathrm{ZZ}$ | $41.2^{+6.3}_{-6.1}$ | $69.0^{+9.5}_{-9.0}$ | $101.7^{+14.0}_{-13.3}$ | $211.8^{+28.9}_{-27.5}$ |
| Z + X | $21.1^{+8.5}_{-10.4}$ | $34.4^{+14.5}_{-13.2}$ | $59.9^{+27.1}_{-25.0}$ | $115.4^{+31.9}_{-30.1}$ |
| Sum of backgrounds | $255.0^{+23.9}_{-25.1}$ | $463.5^{+31.9}_{-33.7}$ | $632.6^{+44.2}_{-46.1}$ | $1351.1^{+85.8}_{-91.2}$ |
| Signal ($m_{\mathrm{H}} = 125$ GeV) | $12.1^{+1.3}_{-1.4}$ | $23.8 \pm 2.1$ | $30.3 \pm 2.6$ | $66.3 \pm 5.6$ |
| Total expected | $267.1^{+24.9}_{-26.1}$ | $487.4^{+33.1}_{-34.9}$ | $662.9^{+45.7}_{-47.5}$ | $1417.4^{+89.1}_{-94.3}$ |
| Observed | 293 | 505 | 681 | 1479 |
| 2017 (13 TeV) | | | | |
| $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ | $233.86^{+y.y}_{-z.z}$ | $441.08^{+y.y}_{-z.z}$ | $569.51^{+y.y}_{-z.z}$ | $1244.45^{+y.y}_{-z.z}$ |
| $\mathrm{gg} \rightarrow \mathrm{ZZ}$ | $49.11^{+y.y}_{-z.z}$ | $81.50^{+y.y}_{-z.z}$ | $121.05^{+y.y}_{-z.z}$ | $251.66^{+y.y}_{-z.z}$ |
| Z + X | $12.39^{+y.y}_{-z.z}$ | $36.39^{+y.y}_{-z.z}$ | $44.45^{+y.y}_{-z.z}$ | $93.24^{+y.y}_{-z.z}$ |
| Sum of backgrounds | $295.37^{+y.y}_{-z.z}$ | $558.97^{+y.y}_{-z.z}$ | $735.02^{+y.y}_{-z.z}$ | $1589.35^{+y.y}_{-z.z}$ |
| Signal ($m_{\mathrm{H}} = 125$ GeV) | $13.85^{+y.y}_{-z.z}$ | $28.64^{+y.y}_{-z.z}$ | $35.55^{+y.y}_{-z.z}$ | $78.04^{+y.y}_{-z.z}$ |
| Total expected | $309.22^{+y.y}_{-z.z}$ | $587.60^{+y.y}_{-z.z}$ | $770.57^{+y.y}_{-z.z}$ | $1667.39^{+y.y}_{-z.z}$ |
| Observed | 307 | 602 | 797 | 1706 |
| 2018 (13 TeV) | | | | |
| $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ | $333^{+57}_{-53}$ | $622^{+31}_{-44}$ | $815 \pm 73$ | $1770^{+98}_{-101}$ |
| $\mathrm{gg} \rightarrow \mathrm{ZZ}$ | $75.1^{+14.3}_{-13.5}$ | $116.6^{+11.7}_{-12.8}$ | $176.9 \pm 23.0$ | $368.5^{+29.5}_{-29.6}$ |
| Z + X | $19.3 \pm 7.2$ | $50.8 \pm 15.2$ | $64.6 \pm 15.6$ | $134.7 \pm 22.9$ |
| Sum of backgrounds | $428^{+59.2}_{55.2}$ | $790^{+36.4}_{-48.3}$ | $1057 \pm 78.1$ | $2274^{+104.9}_{-107.7}$ |
| Signal ($m_{\mathrm{H}} = 125$ GeV) | $19.6^{+3.3}_{-3.1}$ | $40.8^{+2.5}_{-2.9}$ | $50.7 \pm 5.6$ | $111.1^{+6.9}_{-7.0}$ |
| Total expected | $447^{+59.3}_{55.2}$ | $830^{+36.5}_{-48.4}$ | $1108 \pm 78.3$ | $2385^{+105.1}_{-107.9}$ |
| Observed | 462 | 850 | 1130 | 2442 |

For fiducial cross section measurement for $\mathrm{H} \rightarrow 4\ell$, both of irreducible backgrounds ($\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ and $\mathrm{gg} \rightarrow \mathrm{ZZ}$) are estimated from Monte Carlo (MC) simulation as done earlier in Ref. (**Chatrchyan:2013mxa**). In addition, following the study in

Ref. (**Bonvini:2013**), in these measurements the expected yield for the gg $\rightarrow$ ZZ background has been increased by assigning the same ($m_{4\ell}$ dependent) k-factor as signal $\mathrm{H} \rightarrow 4\ell$ process to leading order (LO) cross section of background. In the region ($105 < m_{4\ell} < 140$ GeV) this factor leads to an increase of expected gg $\rightarrow$ $ZZ$ background by factor of $\sim 2.66$. After inclusion of this k-factor, the gg $\rightarrow ZZ$ process is still a sub-dominant background in all measurements.

Reducible background (Z+X) is estimated using data. A tight-to-loose "fake rate" technique is used in the control regions of data by following Ref. (**Chatrchyan:2013mxa**). Z+X background yields are determined using two methods:

- Opposite Sign (OS) method
- Same Sign (SS) method

Both methods are found to be consistent with each other within the statistical uncertainties. First control region is attained with the requirement that two leptons do pass loose selections which does fulfill the criteria of $\mathrm{Z}_1$ candidate (failing ID and isolation requirements), while the other two leptons in the event passes the criteria of $\mathrm{Z}_1$ candidate. Thus, the sample is represented as "2 Prompt + 2 Fail" (2P2F) which is populated with the events which contain two prompt leptons (which are mostly Drell-Yan with small fraction of $\mathrm{Z}/\gamma^*$ and $\mathrm{t\bar{t}}$ events) only. Second control region is attained by requiring that three leptons pass the final requirements. It is represented as "3 Prompt + 1 Fail" (3P1F) sample.

The reducible background's $m_{4\ell}$ shape is attained from the OS method by performing a fit of $m_{4\ell}$ distributions of 3P1F and 2P2F events independently by empirical analytical functional constructed by Landau (**landau:1944**) in Ref. (**Chatrchyan:2013mxa**). Reducible background prediction in all three channels fitted together using this with total uncertainty indicated is shown in Figure **??**. For differentiall measurements in $\mathrm{H} \rightarrow 4\ell$, shape of reducible background is attained from templates built from the control regions in the data for each differential bin of the observable considered for measurement. This procedure is similar to what was used for spin-parity studies as described in the Ref. (**Chatrchyan:2013mxa**). More details on this procedure of differential measurements of $\mathrm{H} \rightarrow 4\ell$ will be given in Section 5.2.7.

The number of estimated signal and the background events, along with observed events in data passing all selections in mass range $m_{4\ell} > 70$ GeV are listed in Table 5.1 for full Run 2 datasets.

Reconstructed four-leptons invariant mass distribution for whole Run 2 datasets and MC samples in full and high mass range is shownn in the Figure 5.7.

### 5.2.4 Definition of the Fiducial Volume

Fraction of all $\mathrm{H} \rightarrow 4\ell$ events that get selected following the event selection procedure can change significantly for various production mechanisms of Higgs and for its distinct exotic models. In order to minimize the effects of model dependent acceptances, measurements on Higgs cross sections (Inclusive and differential) in $\mathrm{H} \rightarrow 4\ell$ are made in a fiducial region. This fiducial phase space is defined so as to

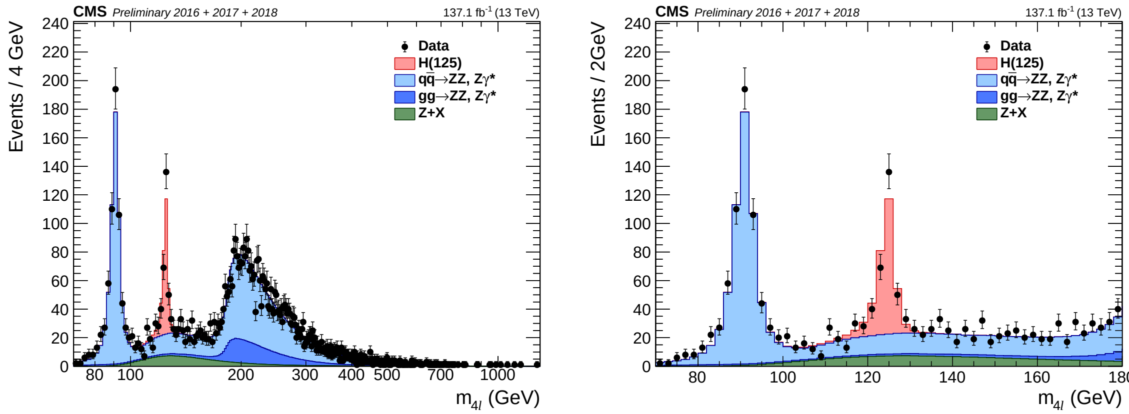



FIGURE 5.7: Invariant mass distributions of invariant mass of four-leptons using full Run 2 datasets in the full (left) and the low (right) mass range, along with predictions for Standard Model Higgs ($m_H = 125.0$ GeV), including $Z \to 4\ell$ resonance.

closely match with the experimental acceptance as defined from selections made at reconstruction level.

Requirement of fiducial phase space is minimum four leptons having $p_T >$ 7 GeV and $|\eta| < 2.5$ in case of electron, and $p_T > 5$ GeV and $|\eta| < 2.4$ in case of muon. Besides, the leading and sub-leaing leptons are required to have $p_T >$ 20 GeV and $p_T > 10$ GeV respectively. The four-momenta of each lepton is calculated at the hard scattering level (PYTHIA status code $= 3$), before any final state radiation occurs. Each lepton isolation is calculated by adding $p_T$ of all stable particles (PYTHIA status code $= 1$) within a cone size $\Delta R < 0.3$ around the lepton. All neutrinos and FSR photons are excluded from above summation. Then the lepton's relative isolation which is defined by ratio of this sum and $p_T$ of considered lepton and is required to be less than $0.35$, consistent with reconstruction level isolation requirement (**Chatrchyan:2013mxa**). The requirement of isolation in definition of fiducial phase space is crucial to control the difference of signal selection efficiency among different models. The studies based on simulation samples show that signal efficiency can change up to 45% among different models without requiring lepton isolation at fiducial level as shown in Table **??**, the maximum difference comes from $t\bar{t}$H and gg $\to$ H production modes. This behavior is expected as $t\bar{t}$H events has more hadronic activity around leptons as compared to gg $\to$ H events, so the cut at relative isolation kills more events in $t\bar{t}$H than gg $\to$ H.

Furthermore, fiducial phase space is defined by several other topological and kinematics selections. There must be two SFOS lepton pairs where $Z_1$ being the one with invariant mass closest to the SM Z boson and reconstructed with $40 \leq m(Z_1) \leq 120$ GeV. $Z_2$ is reconstructed by remaining SFOS pair of leptons within inv. mass range of $12 \leq m(Z_2) \leq 120$ GeV. In case of more than one $Z_2$ passing the criteria, leptons pair with highest $\Sigma|p_T|$ is selected. To remove the overlapping between leptons, we ask for $\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$. To remove contamination from QCD low masses resonance, any opposite-charge pair of leptons is required to meet the condition $m(\ell^+\ell'^-) \geq 4$ GeV. Finally, We require that all selected leptons must emerge from Higgs boson decay and invariant mass of four-leptons must fulfill 105 GeV $< \mathrm{m}(4\ell) <$ 140 GeV. Demand for $\mathrm{m}(4\ell)$ helps to control the contamination from off-shell gg fusion production which is of considerable size, capable to increase the total cross sextion by a few percent (**Kauer:2013cga**; **Kauer:2012hd**). The requirements that defines fiducial phase space are also gathered in Table 5.2.

TABLE 5.2: Overview of all requirements utilized to define the fiducial phase space.

| **Requirements on $\mathrm{H} \to 4\ell$ fiducial phase space** | |
|---|---|
| Lepton kinematics and the isolation | |
| $p_T$ of the leading lepton | $p_T > 20$ GeV |
| $p_T$ of sub-leading lepton | $p_T > 10$ GeV |
| Additional electrons (muons) $p_T$ | $p_T > 7\ (5)$ GeV |
| Pseudorapidity of electrons (muons) | $\lvert\eta\rvert < 2.5\ (2.4)$ |
| Sum of the scalar $p_T$ of all the stable objects within $\Delta R < 0.3$ from the lepton | $< 0.35 p_T$ |
| Event topology | |
| From leptons satisfying criteria above, minimum two SFOS lepton pairs | |
| Invarivant mass of $Z_1$ candidate | $40 < m(Z_1) < 120$ GeV |
| Invariant mass of $Z_2$ candidate | $12 < m(Z_2) < 120$ GeV |
| Separation between selected four leptons | $\Delta R(\ell_i \ell_j) > 0.02$ |
| Invariant mass of any OS lepton pair | $m(\ell_i^+ \ell_j^-) > 4$ GeV |
| Invariant mass of selected four leptons | $105 < m_{4\ell} < 140$ GeV |

It has been established based on simulation samples that with including isolation for the events passing fiducial phase space definition reconstruction efficiencies faintly depends on Higgs properties and its production modes. The remaining model dependence will be quoted as a separate uncertainty as explained in Section 5.3. Part of the signal events passing fiducial phase space definition ($\mathcal{A}_{\text{fid}}$), and reconstruction efficiencies ($\epsilon$) for SM production modes are registered in Table 5.3.

### 5.2.5 Signal Model and Statistical Procedure

The purpose is to extract integrated and differential fiducial cross sections after corrections of detector effects, resolution and systematic biases. To achieve this target, the corrections are extracted by using simulated samples and then encoded in the parametrization of $m_{4\ell}$ spectra at the reconstruction level. Then we directly extract the parameter of interest ($\sigma_{fid}$) by performing the maximum likelihood fit of background and signal parametrization to the observed $m_{4\ell}$ distribution. All the systematic uncertainties in the measurements are encoded as nuisance parameters, which are profited properly in fit procedure. In order to obtain results, we used asymptotic approach (**Cowan:2010js**) with test statistics that is based on profile likelihood ratio (**LHC-HCG**). Maximization of likelihood is performed by fitting

TABLE 5.3: Summary of different SM signal models. The MC samples are from full Run 2 yearly productions (2016, 2017 and 2018) assuming $m_{\mathrm{H}} = 125.0$ GeV. Luminosity averaged fiducial acceptance ($\mathcal{A}_{\text{fid}}$) defined the fraction of events passing fiducial phase space selections, reconstruction efficiency ($\epsilon$) represented by the ratio of reconstructed to accepted events. Also, $f_{\text{nonfid}}$ which are reconstructed events but does not pass fiducial phase space definition at generator level. Values are obtained using 13 TeV signal models and the uncertainties shown are only statistical uncertainties.

| **Signal process** | $\mathcal{A}_{\text{fid}}$ | $\epsilon$ | $f_{\text{nonfid}}$ | $(1+f_{\text{nonfid}})\epsilon$ |
|---|---|---|---|---|
| Individual production modes of Higgs boson | | | | |
| gg→H (POWHEG) | 0.402 ± 0.001 | 0.592 ± 0.002 | 0.053 ± 0.001 | 0.624 ± 0.002 |
| VBF (POWHEG) | 0.444 ± 0.002 | 0.605 ± 0.003 | 0.043 ± 0.001 | 0.631 ± 0.003 |
| WH (POWHEG+MINLO) | 0.325 ± 0.002 | 0.588 ± 0.003 | 0.075 ± 0.002 | 0.632 ± 0.004 |
| ZH (POWHEG+MINLO) | 0.340 ± 0.003 | 0.594 ± 0.005 | 0.081 ± 0.004 | 0.643 ± 0.006 |
| ttH (POWHEG) | 0.314 ± 0.003 | 0.585 ± 0.006 | 0.169 ± 0.006 | 0.684 ± 0.007 |

simultaneously, in each bin, the signal cross section at the fiducial level with all reconstruction bins and all three final states. The procedure applied for $\mathrm{H} \to 4\ell$ fiducial (integrated and differential) cross section measurements will be given in Sections 5.2.6 and 5.2.7. Finally, the unfolding procedure to extract fiducial cross section kinematic observables' bins at the fiducial level, corrected for the detection efficiency and resolution effects, is performed as discussed in Ref. (**Prosper:2011zz**) and similar to what is adopted for $H \to \gamma\gamma$ decay channel (**CMS:HIG-14-016**).

Double Crystal Ball function is used for shape description of signal resonant contribution ($\mathcal{P}_{\text{res}}(m_{4\ell})$) (**Chatrchyan:2013mxa**) as is given below, with the yield proportional to the parameter of interest ($\sigma_{\text{fid}}$).

$$f_{dCB}(x;\alpha,n,s,\sigma) = N \left\{ \begin{array}{ll} e^{-\frac{(x-s)^2}{2\sigma^2}} & \text{if } -\alpha_L < \frac{x-s}{\sigma} < \alpha_R \\ A_L(B_L - \frac{x-s}{\sigma}^{-n_L}) & \text{if } \frac{x-s}{\sigma} \leq -\alpha_L \\ A_R(B_R + \frac{x-s}{\sigma}^{-n_R}) & \text{if } \frac{x-s}{\sigma} \geq \alpha_R \end{array} \right\} \tag{5.1}$$

$$A_L = (\tfrac{n_L}{\alpha})^{n_L} \times e^{-\frac{|\alpha_L|^2}{2}}$$

$$A_R = (\tfrac{n_R}{\alpha})^{n_R} \times e^{-\frac{|\alpha_R|^2}{2}}$$

$$B_L = \frac{n}{\alpha_L} - |\alpha_L|$$

$$B_R = \frac{n}{\alpha_R} - |\alpha_R|$$

$N$ is the normalization constant.

In case of $\mathrm{W}(Z)H$, and $t\bar{t}\mathrm{H}$ modes of Higgs production, the event selection procedure will occasionally select $4\ell$ candidates where minimum one lepton is produced from associated W or Z bosons or jets, and not from $\mathrm{H} \to 4\ell$ process. Fraction of such signal events in $105$ GeV $< \mathrm{m}(4\ell) < 140$ GeV mass range mounts to about 4%, 18% and 22% for WH, ZH and $t\bar{t}$H modes of production, respectively. These events have a broad invariant $m_{4\ell}$ mass distribution that depends on the exact production mechanism. The shape of the contribution by this background , $\mathcal{P}_{\text{comb}}(m_{4\ell})$, is described a Landau distribution with parameters extracted from simulations.

$f_{\text{nonfid}}$ represents the ratio of the events reconstructed from out of fiducial region to those reconstructed from within fiducial volume. For a particular bin j at reconstruction level, it is given as:

$$f^{j}_{\text{nonfid}} = \frac{N(\text{j}_{\text{reco}}\ \&\&\ \text{not fid.})}{N(\text{j}_{\text{reco}}\ \&\&\ \text{fid.})}. \tag{5.2}$$

Such events arise from imperfect detector resolution which causes the differences for quantities calculated at generated and reconstructed levels. Fraction of these events are also treated as background and will be referred "non-fiducial signal“ throughout this disseration. It has been demonstrated with MC samples that these events can be described using same shape as resonant component. The values of such fraction are extracted from simulation repeatedly for SM. This fraction varies among different production modes of Higgs e.g. it is 4% for the $\text{gg} \to \text{H}$ and 3% for $ttH$ production mode. The values for each model are listed in the Tables **??**, **??** and **??** for 2016, 2017 and 2018 respectively.

To make comparison with the theoretical predictions, measurements need corrections for resolution of detector and effects of efficiency. To unfold reconstruction-level distribution to the distribution before all detector effects, the efficiency matrix ($\epsilon$) is constructed using the signal MC samples. The efficiency matrix ($\epsilon$) provides the probability of an event that was generated in $i$th bin at the fiducial level and is reconstructed in j-th bin as:

$$\epsilon_{i,j} = \frac{N(\text{j}_{\text{reco}}\ \&\&\ \text{i}_{\text{gen}})}{N(\text{i}_{\text{gen}})}\ , \tag{5.3}$$

where $N(i_{gen})$ is total number of the events in $i$th bin at fiducial level and $N(\text{j}_{\text{reco}}\ \&\&\ \text{i}_{\text{gen}})$ is total number events which $i$th bin at the fiducial level but reconstructed in $j$th bin. For integrated measurements, where we have only one bin, fiducial efficiency is about 65% which can change upto 7% for other signal models. For differential measurements, it is described by detector response matrix for each observable. Figure **??** shows the efficiency matrix for $p_T(H)$ measurement using different SM production MC samples. The efficiency matrix is dominated by diagonal elements, but in general it can have non-negligible off-diagonal elements specially for jet related observables due to jet energy scale uncertainty. In case of differential variable, jet multiplicity, neighbour elements of diagonal varies from 3% to 21% as shown in Figures **??**-**??** in the Appendix **??**.

The estimated number events for a studied differential observable in a final state $f$ and in the defined bin $i$ are described in terms of $m_{4\ell}$ given by:

$$\begin{aligned} N^{\mathrm{f},i}_{\mathrm{obs}}(m_{4\ell}) &= N^{\mathrm{f},i}_{\mathrm{fid}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{nonfid}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{comb}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{bkg}}(m_{4\ell}) \\ &= \sum_j \epsilon^{\mathrm{f}}_{i,j} \cdot \left(1 + f^{\mathrm{f},i}_{\mathrm{nonfid}}\right) \cdot \sigma^{\mathrm{f},j}_{\mathrm{fid}} \cdot \mathcal{L} \cdot \mathcal{P}_{\mathrm{res}}(m_{4\ell}) \\ &\quad + N^{\mathrm{f},i}_{\mathrm{comb}} \cdot \mathcal{P}_{\mathrm{comb}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{bkg}} \cdot \mathcal{P}_{\mathrm{bkg}}(m_{4\ell}). \end{aligned} \tag{5.4}$$

Where $N^{\mathrm{f},i}_{\mathrm{fid}}(m_{4\ell})$, $N^{\mathrm{f},i}_{\mathrm{nonfid}}(m_{4\ell})$, $N^{\mathrm{f},i}_{\mathrm{comb}}(m_{4\ell})$ and $N^{\mathrm{f},i}_{\mathrm{bkg}}(m_{4\ell})$ denote the resonant fiducial signal, the nonfiducial resonant, the combinatorial contribution of fiducial signal, and background contributions in a defined bin $i$ as functions of $m_{4\ell}$ (four lepton invariant mass), respectively. Then, probability density functions corresponding to the resonant (fiducial and the nonfiducial) signal, the combinatorial contributions of fiducial signal, and contribution from the background are $\mathcal{P}_{\mathrm{res}}(m_{4\ell})$, $\mathcal{P}_{\mathrm{comb}}(m_{4\ell})$ and $\mathcal{P}_{\mathrm{bkg}}(m_{4\ell})$ respectively. $\epsilon^{\mathrm{f}}_{i,j}$ is the response matrix that connects reconstructed level events in a defined bin $i$ with fiducial events in the bin $j$ for studied differential observable. The $f^{i}_{\mathrm{nonfid}}$ fraction represents the ratio of nonfiducial signal to the fiducial signal contributions in a defined bin $i$ at reconstruction level. $\sigma^{\mathrm{f},j}_{\mathrm{fid}}$ is the parameter of interest that represents signal cross section for a final state f in a defined bin $j$ of fiducial region.

### 5.2.6 Extraction of Integrated $\mathrm{H} \to 4\ell$ Fiducial Cross Section

#### Signal Model and Statistical Procedure

For the integrated or inclusive cross section measurement within the defined region fiducial phase space, we follow a general procedure of differential cross sections measurement as outlined in Section 5.2.5, treating it as a special case where the number of considered bins is one.

Table 5.4 shows the reconstruction efficiency ($\epsilon$) for fiducial events, while Table 5.5 shows the ratio ($f_{nonfid}$) of events reconstructed from out of fiducial region and the events reconstructed within fiducial volume, per final state for different Standard Model signal models. The primary source of the “nonfiducial signal” contributions (described with the factor $f_{nonfid}$) is the imperfect correspondence of isolation at particle and reconstruction levels, as discussed at the beginning of Section 5.2. In a similar way, fraction of the signal events in mass range 105.0–140.0 GeV where minimum one lepton selected does not originate from the Higgs boson decay are shown in Table 5.6.

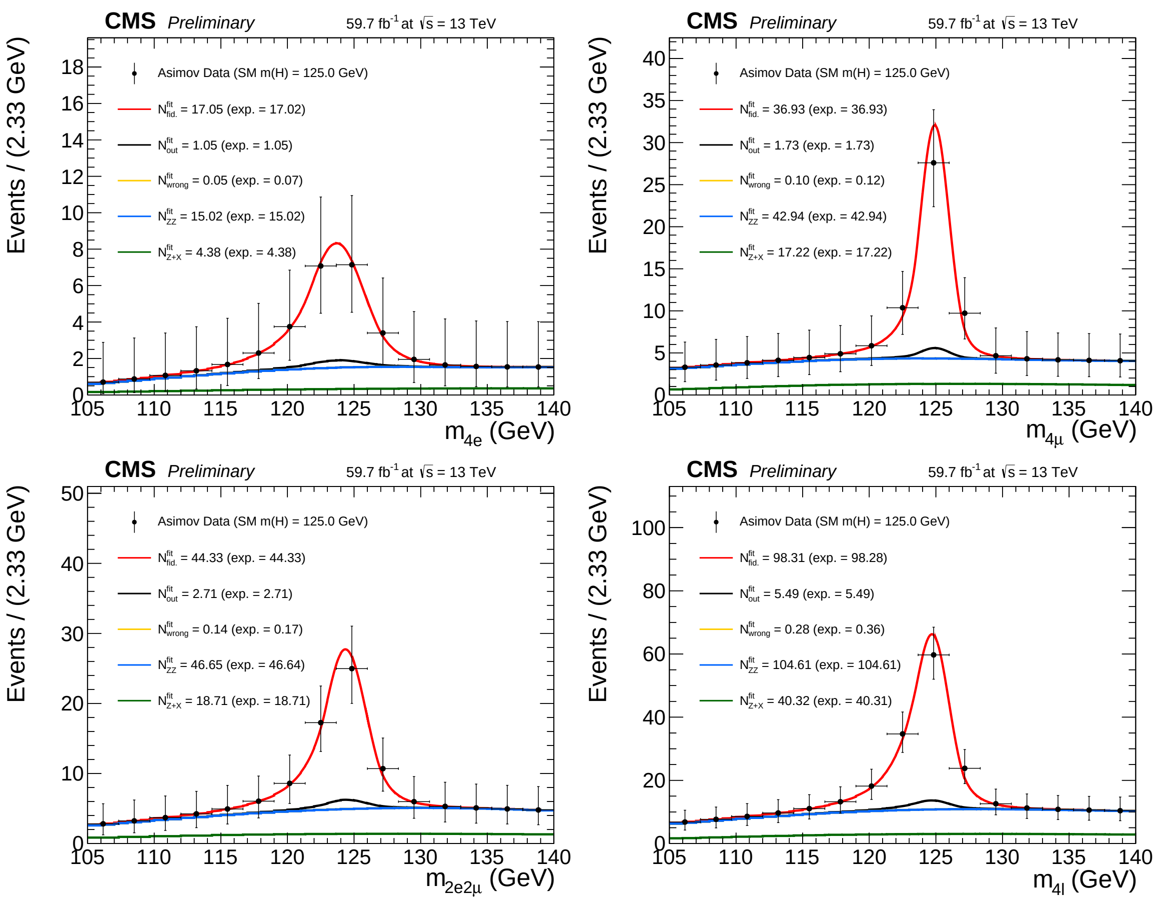



FIGURE 5.8: Inclusive four lepton mass distributions for an Asimov Dataset generated using SM cross section values and SM efficiencies where the gg→H prediction is from POWHEG+JHUGEN and resulting fitted values of the signal and background pdfs in different final states at $\sqrt{s} = 13$ TeV with 2018 data.

TABLE 5.4: Reconstruction efficiency ($\epsilon$) for fiducial events per final state for different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 data.

| Sample | $4e$ | $4\mu$ | $2e2\mu$ | $4\ell$ |
|---|---|---|---|---|
| gg→H (POWHEG) | 0.404 ± 0.002 | 0.799 ± 0.002 | 0.572 ± 0.002 | 0.593 ± 0.001 |
| VBF (POWHEG) | 0.431 ± 0.003 | 0.805 ± 0.002 | 0.586 ± 0.002 | 0.608 ± 0.002 |
| WH (POWHEG+MINLO) | 0.415 ± 0.004 | 0.780 ± 0.003 | 0.580 ± 0.003 | 0.593 ± 0.002 |
| ZH (POWHEG+MINLO) | 0.428 ± 0.006 | 0.780 ± 0.005 | 0.581 ± 0.005 | 0.597 ± 0.003 |
| ttH (POWHEG) | 0.429 ± 0.006 | 0.745 ± 0.005 | 0.573 ± 0.005 | 0.585 ± 0.003 |

TABLE 5.5: Ratio of events reconstructed from out of fiducial volume and the events reconstructed within the fiducial region ($f_{nonfid}$) per final state in different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 data.

| Sample | $4e$ | $4\mu$ | $2e2\mu$ | $4\ell$ |
|---|---|---|---|---|
| gg→H (POWHEG) | 0.060 ± 0.002 | 0.046 ± 0.001 | 0.060 ± 0.001 | 0.055 ± 0.001 |
| VBF (POWHEG) | 0.050 ± 0.002 | 0.037 ± 0.001 | 0.046 ± 0.001 | 0.043 ± 0.001 |
| WH (POWHEG+MINLO) | 0.098 ± 0.004 | 0.058 ± 0.002 | 0.080 ± 0.002 | 0.075 ± 0.001 |
| ZH (POWHEG+MINLO) | 0.105 ± 0.007 | 0.071 ± 0.004 | 0.091 ± 0.004 | 0.087 ± 0.002 |
| ttH (POWHEG) | 0.278 ± 0.012 | 0.127 ± 0.005 | 0.193 ± 0.006 | 0.185 ± 0.004 |

TABLE 5.6: Signal event fraction in the mass range 105.0–140.0 GeV where minimum one lepton selected is originating from Higgs decay at $\sqrt{s} = 13$ TeV with 2018 data.

| Sample | $4e$ | $4\mu$ | $2e2\mu$ | $4\ell$ |
|---|---|---|---|---|
| gg→H (POWHEG) | 0.002 | 0.002 | 0.002 | 0.002 |
| VBF (POWHEG) | 0.002 | 0.003 | 0.003 | 0.003 |
| WH (POWHEG+MINLO) | 0.036 | 0.031 | 0.042 | 0.037 |
| ZH (POWHEG+MINLO) | 0.199 | 0.207 | 0.212 | 0.208 |
| ttH (POWHEG) | 0.175 | 0.156 | 0.164 | 0.163 |

### 5.2.7 Extraction of Differential $\mathrm{H} \to 4\ell$ Fiducial Cross Sections

#### Kinematic Observables

Measurements on differential cross section which are performed as function of the kinematic properties of four-lepton system and as well as jet multiplicity are unfolded to the particle level. In this dissertation, mainly the kinematic properties of four-lepton system are exploited and the results are reported for transverse momentum and rapidity of Higgs or the four lepton system($p_T(\mathrm{H})$ or $p_T(4\ell)$, $|y(\mathrm{H})|$ or $|y(4\ell)|$), N(jets), and $p_T(\mathrm{jet})$ which probe the production mechanism. For each kinematic or differential observable, selection procedure of the binning boundaries has been optimized in order to have a similar expected measured cross-section for every bin.

### Background Model

In order to perform the differential measurements, the background is to be subtracted at the reconstruction level for each bin of the considered observable. In general, the background mass spectrum shape is not the same in all bins of the considered observable because the background does not have a resonance structure. The mass spectrum also can have non-negligible correlation with the observables.

In case of the irreducible backgrounds, the four-lepton mass spectra are modeled as the $m_{4\ell}$ template shapes using simulated events for each bin of the observable considered for the measurement. Similarly, in case of the reducible backgrounds the templates are built from the control regions in the data. This procedure is similar to the one described and used in the Ref. (**Chatrchyan:2013mxa**).

In each differential bin of $p_T(4\ell)$ fractions of background events are presented in the Table 5.7.

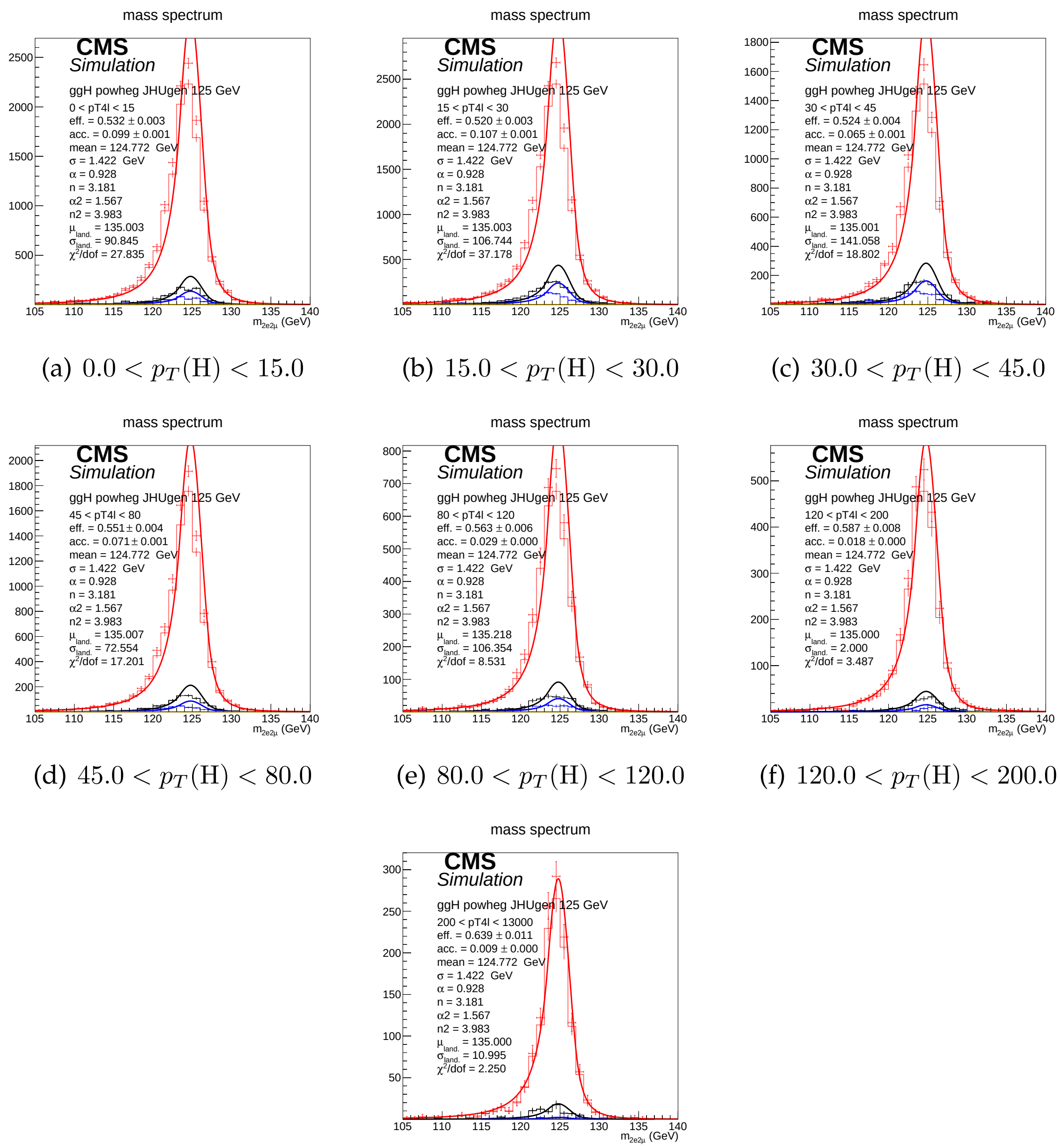
TABLE 5.7: Estimated fraction of background events in each bin of $p_T(4\ell)$, for each final state using 2018 samples.

| Background | final state | $0-15$ | $15-30$ | $30-45$ | $45-80$ | $80-120$ | $120-200$ | $200-13000$ |
|---|---|---|---|---|---|---|---|---|
| qqZZ | $2e2\mu$ | 0.483668 | 0.239196 | 0.130151 | 0.105276 | 0.0266332 | 0.011809 | 0.00326633 |
| qqZZ | $4e$ | 0.459665 | 0.234399 | 0.136986 | 0.114916 | 0.0388128 | 0.00989346 | 0.00532725 |
| qqZZ | $4\mu$ | 0.492632 | 0.241203 | 0.110977 | 0.109173 | 0.0297744 | 0.0129323 | 0.00330827 |
| ggZZ | $2e2\mu$ | 0.310446 | 0.309859 | 0.191901 | 0.168134 | 0.0187793 | 0.000880282 | 0.0 |
| ggZZ | $4e$ | 0.3085 | 0.306332 | 0.19247 | 0.17433 | 0.0169994 | 0.00136908 | 0.0 |
| ggZZ | $4\mu$ | 0.321332 | 0.319311 | 0.194863 | 0.150987 | 0.0128703 | 0.000638196 | 0.0 |
| Z+X (CR) | 4l | 0.0992521 | 0.211153 | 0.209307 | 0.305327 | 0.120303 | 0.0483797 | 0.00627828 |

### Signal Model and Statistical Procedure

The fiducial differential cross section is extracted in the bins of kinematic observables at the fiducial level following the procedure outlined in Section 5.2.5. For signal yields, mass spectrum of signal is built in each cell of the response matrix using a Double Crytal Ball (DCB) function, similar to what was done in Ref. (**Chatrchyan:2013mxa**). In practice, the same parameters of the CB function are used in each cell and the systematic uncertainties assigned on the energy scale and resolution generally cover discrepancy of signal mass spectrum among each cell.

Examples of the fitted signal line shapes for differential bins of $p_T(\mathrm{H})$ from gg→H (POWHEG+JHUGEN 125 GeV 2018 sample) in the individual final states are shown in Figure 5.9, and examples of the efficiency matrices for $p_T(\mathrm{H})$ for the $2e2\mu$ final state are shown in Fig 5.10. Many more examples can be seen in Appendix **??**.




(a) $0.0 < p_T(\mathrm{H}) < 15.0$

(b) $15.0 < p_T(\mathrm{H}) < 30.0$

(c) $30.0 < p_T(\mathrm{H}) < 45.0$

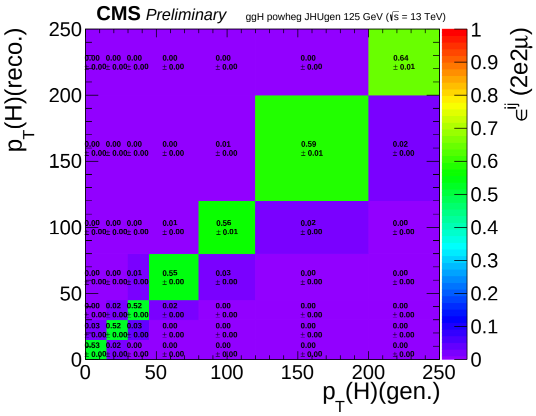
(d) $45.0 < p_T(\mathrm{H}) < 80.0$

(e) $80.0 < p_T(\mathrm{H}) < 120.0$

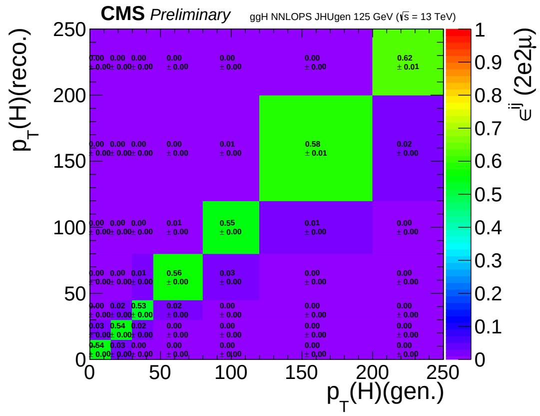
(f) $120.0 < p_T(\mathrm{H}) < 200.0$

(g) $200.0 < p_T(\mathrm{H}) < 13000.0$

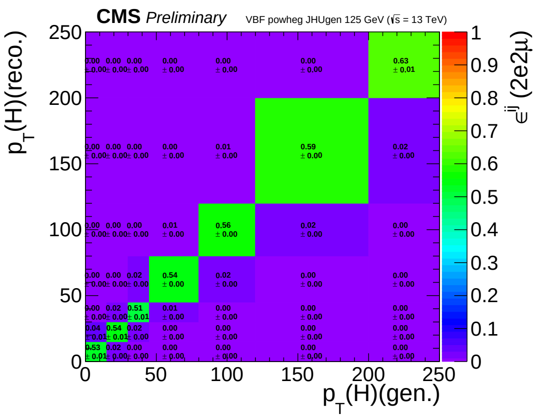
FIGURE 5.9: Distributions of m($4\ell$) in $2e2\mu$ final state for the $gg \to$ H production mode from POWHEG+JHUGEN in different bins of $p_T(\mathrm{H})$.



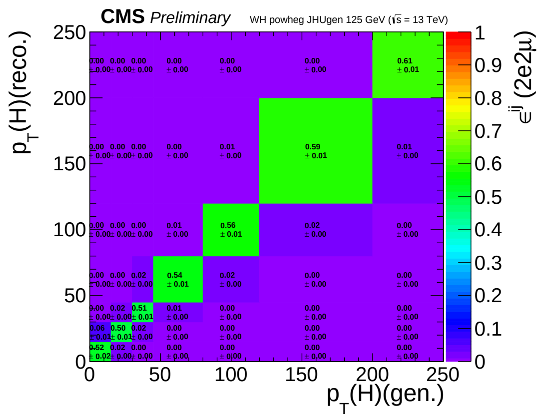

(a) $gg \to$H (POWHEG+JHUGEN)



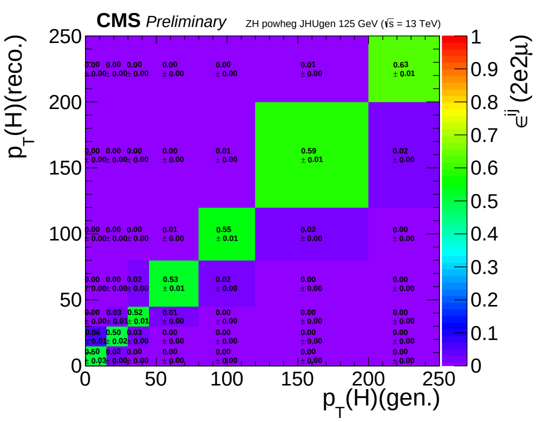

(b) $gg \to$H (MINLOHJJ)



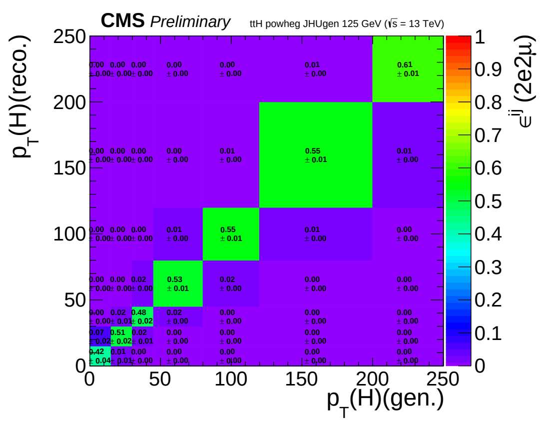

(c) VBF (POWHEG)




(d) WH (PYTHIA)



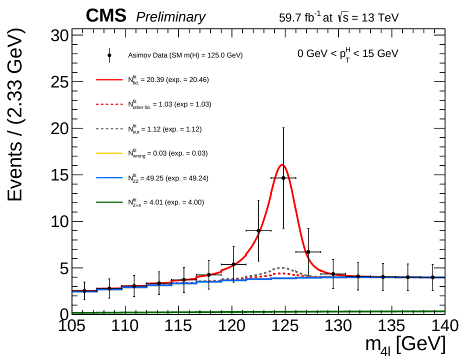

(e) ZH (PYTHIA)



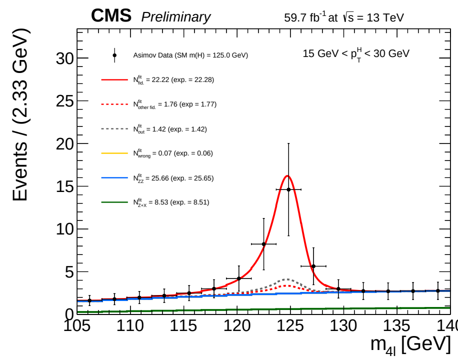

(f) ttH (PYTHIA)

FIGURE 5.10: Efficiency matrices for $p_T$(H) for different SM production modes in the $2e2\mu$ final state using 2018 samples.

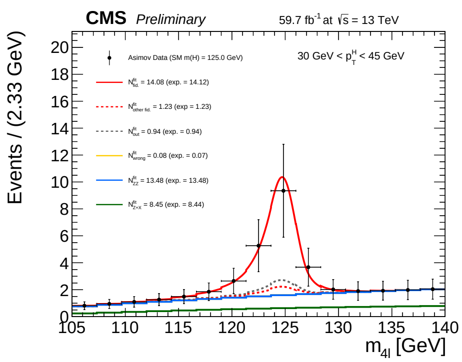
Figure 5.11 shows the four lepton mass distributions for an Asimov dataset generated using SM cross section values and efficiencies from the gg→H production mode from POWHEG+JHUGEN, and resulting fitted values of PDFs of signal and background for different bins of $p_T(\mathrm{H})$ (all final states combined).



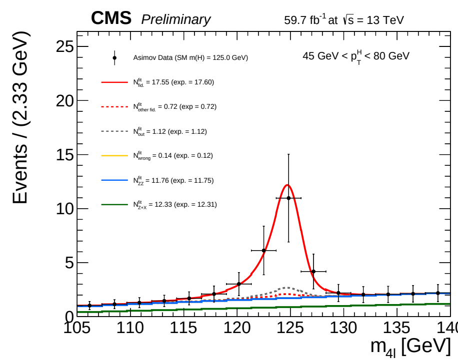

(a) 0.0 GeV $< p_T(\mathrm{H}) <$ 15.0 GeV



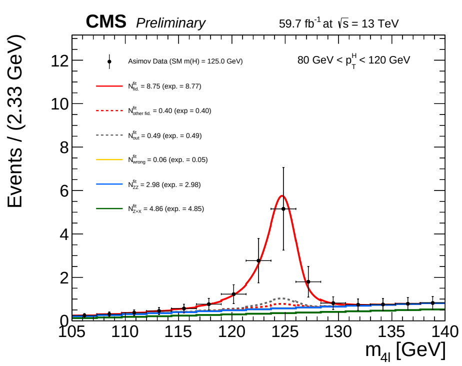

(b) 15.0 GeV $< p_T(\mathrm{H}) <$ 30.0 GeV



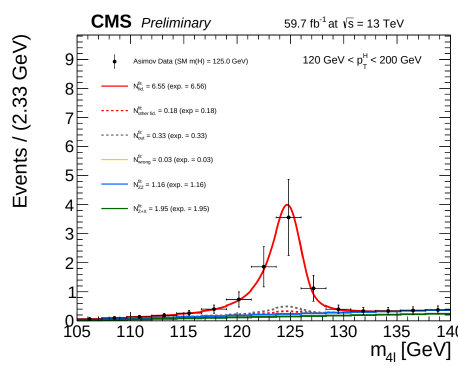

(c) 30.0 GeV $< p_T(\mathrm{H}) <$ 45.0 GeV



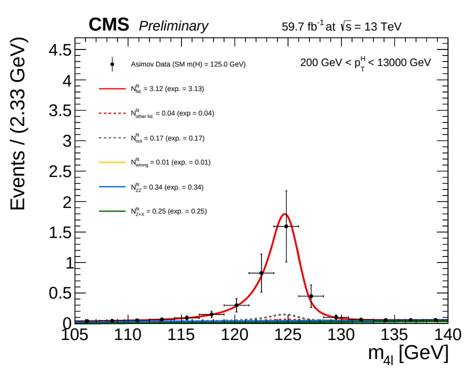

(d) 45.0 GeV $< p_T(\mathrm{H}) <$ 80.0 GeV




(e) 80.0 GeV $< p_T(\mathrm{H}) <$ 120.0 GeV




(f) 120.0 GeV $< p_T(\mathrm{H}) <$ 200.0 GeV




(g) 200.0 GeV $< p_T(\mathrm{H}) <$ 13000.0 GeV

FIGURE 5.11: Four lepton mass distributions for an Asimov dataset generated using SM cross section values and SM efficiencies where the gg→H prediction is from POWHEG+JHUGEN and resulting fitted values of PDFs of signal and background for different bins of $p_T(\mathrm{H})$ (all final states combined).

## 5.3 Systematic Uncertainties

### 5.3.1 Experimental uncertainties souces

Main experimental uncertainties affecting signal and background are uncertainty on integrated luminosity (from 2.3% to 2.5%, depending on the year of data taking) and the uncertainty on efficiencies of lepton reconstruction and identification (ranging from 2.5–16.1% in $4\mu$ and $4e$ channels, respectively). Uncertainties on estimation of reducible backround, as discussed in described in Section 5.2.3, originating from the background composition and misidentification rate uncertainty vary for different final states. Mismatch in composition of backgrounds between different samples where the fake rate is derived and where it is applied accounts between 24 and 43%.

Lepton energy scale uncertainty is assessed by propagating down to the four-leptons invariant mass the uncertainty associated the lepton momentum corrections on each indiviudal lepton. It is estimated by exploiting $Z \to \ell\ell$ invariant mass distributions both in data and the simulation. Events are categorized on the basis $p_T$ and $\eta$ of either(randomly determined) of two leptons and integrating on the other lepton. The dilepton mass distributions are then fit by parameterization of a Breit-Wigner convoluted with a DCB function. The uncertainty is treated as uncorrelated (all up or all down). The new four-leptons invariant mass obtained this way is then fitted with only the mean value floating. Difference between new mean value and default one gives an estimation of impact of lepton energy scale. Values of the differences are reported in the Table 5.8, together with estimation of the impact of lepton momentum resolution, following a similar procedure.

TABLE 5.8: Difference (in %) between the nominal mean and the width of four-leptons or Higgs invariant mass distribution and those where the lepton momentum scale and resolution uncertainties have been propagated.

| | 4e | $4\mu$ | $2e2\mu$ |
|---|---|---|---|
| Scale Difference (%) | 0.087 | 0.049 | 0.069 |
| Resolution Difference (%) | 17 | 12 | 17 |

The results obtained this way suggest that the uncertainties from 2016 (20% on the resolution, 0.3% for the $4e$ channel for scale) are still valid.

### 5.3.2 Theoretical uncertainties

Estimation of the theoretical uncertainties affecting signal and the background include uncertainties from factorization and renormalization scale is done by varying the two scales 0.5-2 times the nominal values with their ratio kept in between 0.5-2. In the case of ggF production mode, the QCD scale uncertainty is broke down to 9 different sources. Other theoretical uncertainty source is PDF set choice which is determined RMS of the variation when different replias of default NNPDF set is used. Additional uncertainty on the background which is 10% on K factor applied on $ggZZ$ prediction is considered as described in Section 5.2.3. Systematic uncertainty of about 2% on the branching ratio of $\mathrm{H} \to 4\ell$ is only applicable to signal yield. Additional sources are the imprecise jet energy scale and resolution knowledge (from 2% to 22% ), the efficiency on b-tagging and the light quarks (*i.e.* u, d, s, c) and gluon jet mistag rate. While cross section measurement, signal cross section uncertainties arising from theoretical sources are removed.

TABLE 5.9: A chart of experimental systematic uncertainties in the $\mathrm{H} \to 4\ell$ cross section measurements of 2016 data.

| **Relative systematic uncertainties** | |
|---|---|
| Uncertainties from common experimental sources | |
| Luminosity | 2.6 % |
| Lepton selection efficiencies | 2.5 – 9 % |
| Uncertainties related to background | |
| Reducible background (Z+X) | 36 – 43 % |
| Uncertainties related to Signal | |
| Lepton energy scale | 0.04 – 0.3 % |
| Lepton energy resolution | 20 % |

TABLE 5.10: A chart of experimental systematic uncertainties in the $\mathrm{H} \to 4\ell$ cross section measurements of 2017 data

| **Relative systematic uncertainties** | |
|---|---|
| Uncertainties from common experimental sources | |
| Luminosity | 2.3 % |
| Lepton selection efficiencies | 3. – 12.5 % |
| Uncertainties related to background | |
| Reducible background fake rate variation (Z+X) | 6 – 32 % |
| Uncertainties related to Signal | |
| Lepton energy scale | 0.04 – 0.3 % |
| Lepton energy resolution | 20 % |

### 5.3.3 Systematic uncertainties treatment when combining yearly measurements on data

In combination of three data taking periods, theoretical uncertainties as well as the experimental ones related to leptons or jets are treated as correlated while all other ones from experimental sources are taken as uncorrelated.

TABLE 5.11: TO BE UPDATED. COPY PASTE NUMBERS OF 2017 FOR NOW. A chart of experimental systematic uncertainties in the $\mathrm{H} \to 4\ell$ cross section measurements of 2018 data

| **Relative systematic uncertainties** | |
|---|---|
| Uncertainties from common experimental sources | |
| Luminosity | 2.3 % |
| Lepton selection efficiencies | 3. – 12.5 % |
| Uncertainties related to background | |
| Reducible background fake rate variation (Z+X) | 6 – 32 % |
| Uncertainties related to Signal | |
| Lepton energy scale | 0.04 – 0.3 % |
| Lepton energy resolution | 20 % |

TABLE 5.12: Table theory uncertainties in the $\mathrm{H} \to 4\ell$ Inclusive cross section measurements

| **Summary of inclusive theory uncertainties** | |
|---|---|
| BR($H \to \mathrm{ZZ} \to 4\ell$) | 2 % |
| QCD scale ($\mathrm{qq} \to \mathrm{ZZ}$) | +3.2/-4.2 % % |
| PDF set ($\mathrm{qq} \to \mathrm{ZZ}$) | +3.1/-3.4 % |
| Electroweak corrections ($\mathrm{qq} \to \mathrm{ZZ}$) | $\pm$ 0.1 % |

## 5.4 Results

### 5.4.1 Measurement of Integrated Fiducial Cross Section

Resulting fits to the inclusive $\mathrm{m}(4\ell)$ distributions (extracted in $m_{4\ell}$ range $viz$ 105 GeV to 140 GeV) for measurement of integrated $\mathrm{H} \to 4\ell$ fiducial cross section in different final states, are shown in Figure 5.13. Extracted inclusive cross sections as function of colliding center of mass energy has been shown in the Figure 5.14(a) while the extracted integrated cross section results for different final states at 13 TeV corresponding to different years of data taking are shown in Figure 5.14(b). Systematic on the measurement is represented by red bar, whereas black error bar represents total uncertainties, summed from the individuals in quadrature. Contributions of the sub-dominant component of the signal i.e. $\mathrm{XH} = \mathrm{VBF} + \mathrm{VH} + t\bar{t}\mathrm{H}$ is indicated in green separately. These results are also given in Tables 5.13 and 5.14.

The inclusive $\mathrm{H} \to 4\ell$ measurements are compared with SM NNLO theoretical estimations where acceptance of dominant $\mathrm{gg} \to \mathrm{H}$ contribution is estimated using POWHEG at $\sqrt{s}$=13 TeVand HRES (**Grazzini:2013mca**; **deFlorian:2012mx**) at $\sqrt{s}$=7 and 8 TeV. Total gluon fusion cross section and associated uncertainty has

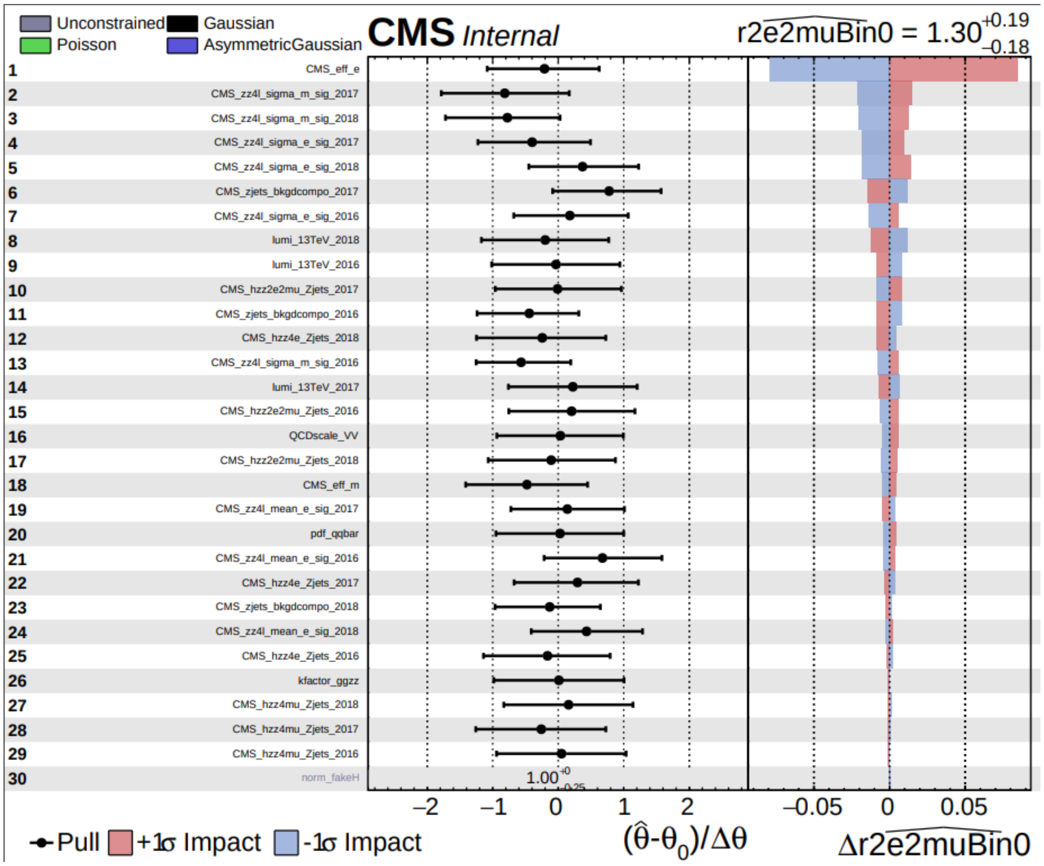


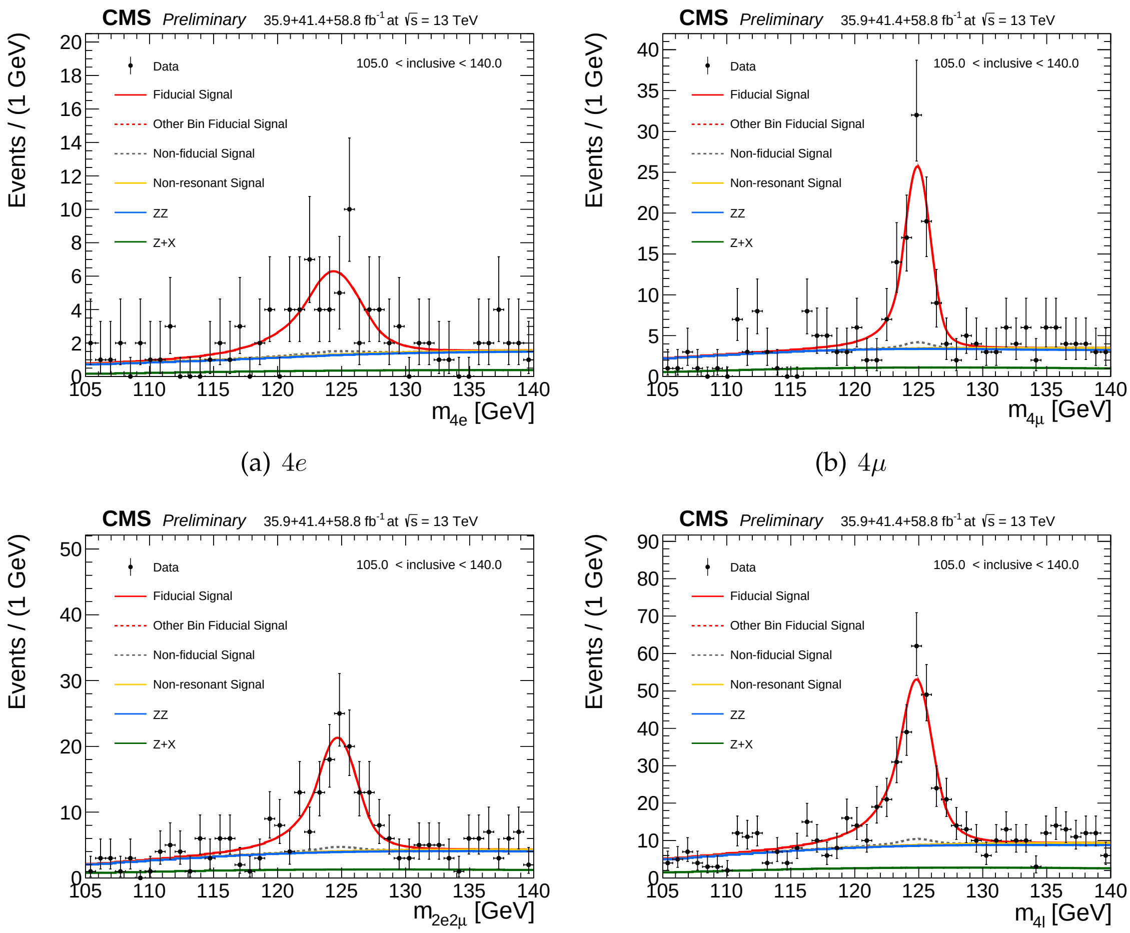

FIGURE 5.12: Pulls. Overall impact of different uncertainties sources on the inclusive measurement ($2e2\mu$ final state)

been taken from Ref. (**Anastasiou2016**). Definition of fiducial volume for $\sqrt{s}$=6–9 TeVuses the lepton isolation definition from Ref. (**CMSH4lFiducial8TeV**), while for $\sqrt{s}$=12–14 TeVthe definition described in the text is used.

When extracting the integrated H$\to 4\ell$ fiducial cross section, the parameter of interest is total fiducial cross section and the fractions of $4e$, $4\mu$, and $2e2\mu$ is floated in the fit. When extracting the fiducial cross section in the final states, three parameters of interest are the cross sections of three individual final states.

TABLE 5.13: Measured H $\to 4\ell$ Inclusive fiducial cross section in $m_{4\ell}$ ranging from 105 GeV to 140 GeV in proton collisions at 13 TeV.

| **Fiducial cross section** H $\to 4\ell$ **at 13 TeV** | |
|---|---|
| Measured | $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.})$ fb (model) |
| gg$\to$H(POWHEG) + XH | $2.72^{+0.14}_{-0.14}$fb |
| gg$\to$H(NNLOPS) + XH | $2.72^{+0.14}_{-0.14}$fb |




(a) $4e$ (b) $4\mu$

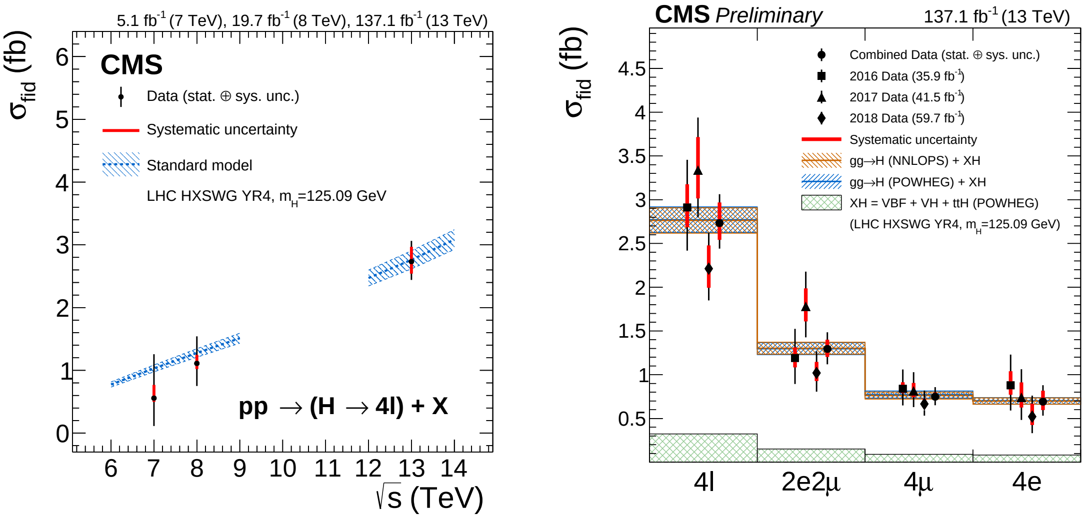
(c) $2e2\mu$ (d) $4\ell$

FIGURE 5.13: Inclusive four lepton invariant mass distributions in data and resulting fitted values of signal and the background PDFs in different final states.

Measured integrated $\mathrm{H} \to 4\ell$ fiducial cross section using 13 TeV data is in nice agreement with theoretical predictions within corresponding uncertainties. Dominant and sub-dominant uncertainties on the measurement are systematic and statistical uncertainties which are about 24% and 23% respectively. The theoretical uncertainty is about 11%.

### 5.4.2 Measurement of Differential Fiducial Cross Sections

Observed data along with prediction of SM Higgs boson ($m_H = 125.09$ GeV) and expected background in each differential bin for chosen observables are shown in Figures 5.15 for production related observables. The differential results are only produced using 13 TeV data. The results of Higgs differential cross section measurements are produced for four-leptons transverse momentum and rapidity,




(a) $\mathrm{H} \to 4\ell$ (7, 8 and 13 TeV)

(b) $\mathrm{H} \to 4\ell$ (13 TeV)

FIGURE 5.14: The measured fiducial cross section as a function of $\sqrt{s}$ (left). The measured inclusive fiducial cross section in different final states (right). The acceptance is calculated using at $\sqrt{s}$=13 TeVand HRES (**Grazzini:2013mca**; **deFlorian:2012mx**) at $\sqrt{s}$=7 and 8 TeVand the total gluon fusion cross section and uncertainty are taken from Ref. (**Anastasiou2016**). The fiducial volume for $\sqrt{s}$=6–9 TeVuses the lepton isolation definition from Ref. (**CMSH4lFiducial8TeV**), while for $\sqrt{s}$=12–14 TeVthe definition described in the text is used.

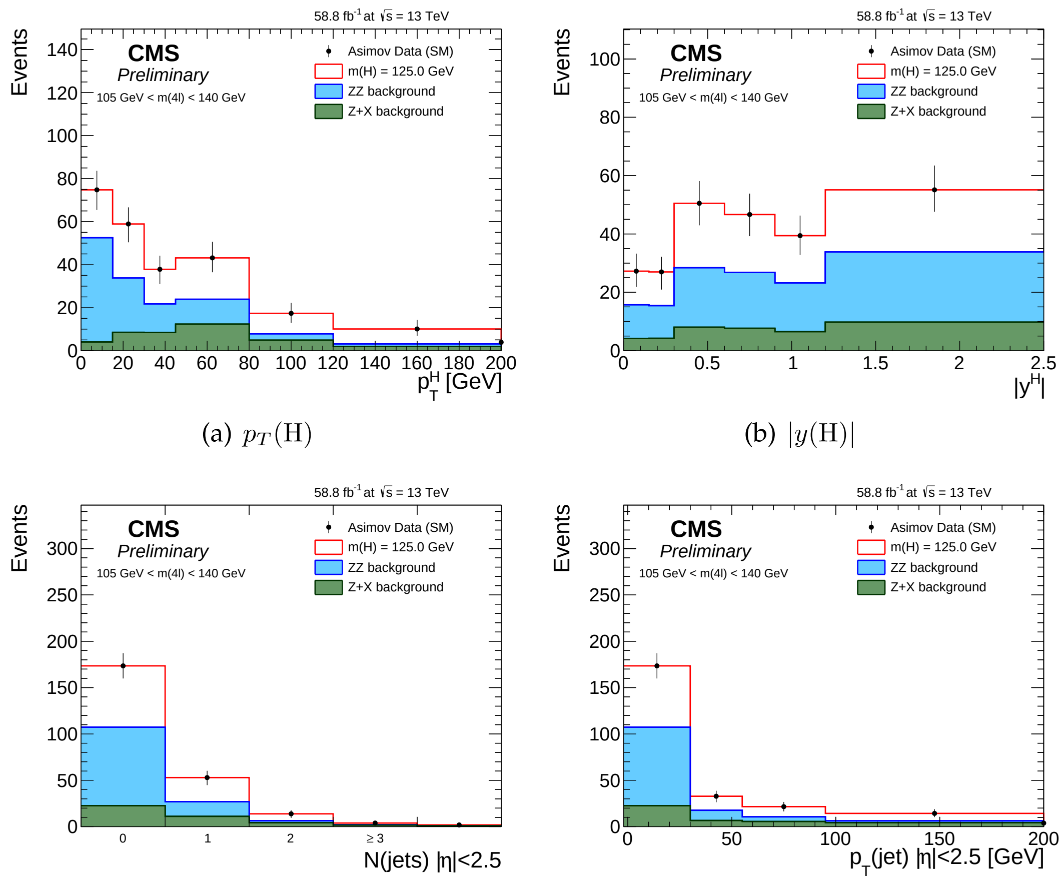
TABLE 5.14: Measured H→ $4l$ integrated or inclusive fiducial cross section performed in $m_{4\ell}$ ranging from 105 GeV to 140 GeV.

| Fiducial cross section (Inclusive) of H→ $4l$ (fb) at 13 TeV. | | | | |
|---|---|---|---|---|
| | 4l | $2e2\mu$ | $4\mu$ | $4e$ |
| Measured (stat. ⊕ sys.) | $1.11^{+0.43+0.08}_{-0.36-0.02}$ | $0.48^{+0.27+0.03}_{-0.22-0.03}$ | $0.26^{+0.17+0.01}_{-0.13-0.01}$ | $0.37^{+0.30+0.04}_{-0.22-0.02}$ |
| gg→H(POHWHEG) + XH | $1.16^{+0.11}_{-0.11}$ | $0.54^{+0.05}_{-0.05}$ | $0.33^{+0.03}_{-0.03}$ | $0.30^{+0.03}_{-0.03}$ |
| gg→H(NNLOPS) + XH | $1.17^{+0.11}_{-0.11}$ | $0.54^{+0.05}_{-0.05}$ | $0.32^{+0.03}_{-0.03}$ | $0.29^{+0.03}_{-0.03}$ |

transverse momentum of leading jet, jet multiplicity and are shown in the Figures 5.16. The uncertainty in the theoretical prediction for dominant process i.e. $\mathrm{gg} \to \mathrm{H}$ is estimated in the each differential bin of studied observable with POWHEG+JHUGEN which carries specific signal description. The uncertainty in the theoretical prediction for associated production mechanisms are assumed to be constant among differential bins and are obtained from Ref. (**LHCHiggsCrossSectionWorkingGrou**
The differential cross sections as a function of Higgs transverse momentum and rapidity are sensitive to pQCD calculations of $\mathrm{gg} \to \mathrm{H}$ production mode which is mediated by a loop, in which Higgs $p_T(H)$ is anticipated to be balanced with emission

of the soft quarks and gluons. The differential distribution as a function rapidity of Higgs also probes PDFs of the colliding protons and representation of the gluon fusion production modes. The differential observables, transverse momentum $p_T$) of leading jet and jet multiplicity N(jets) probe hard quark and gluon radiations theoretical modelling in the process. Comparison of measurements with theoretical predictions from POWHEG and NNLOPS for the gg$\rightarrow$H process (displayed in the blue and brown colors, respectively) together with predictions for XH = VBF (POWHEG) + VH + ttH(PYTHIA) (shown in green) is also shown. The error bars represents statistical and systematic uncertainty. The results are found to be compatible with SM based theoretical predictions within the uncertainties.

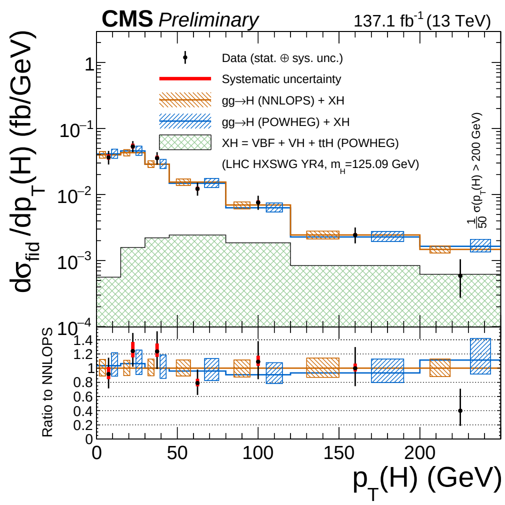
While extracting the differential measurements, the fractions of the $4e, 4\mu$ and $2e2\mu$ contributions to the fiducial cross sections in each bin is allowed to float. Results where the fractions are set to their SM predictions are included in Appendix **??**. Similarly, results where the model dependence uncertainty is assessed by taking into account the experimental constraints from the existing CMS measurements are included in Appendix **??**.



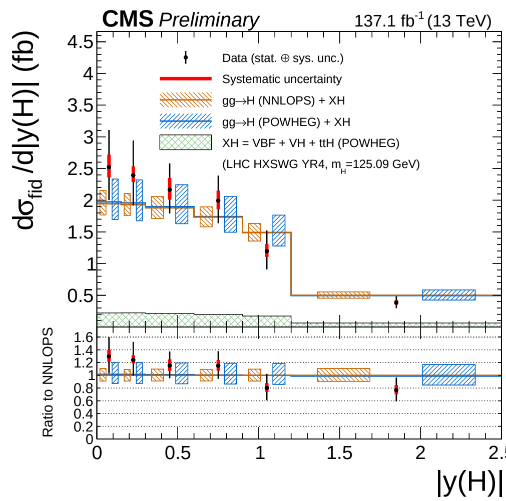

(a) $p_T(\mathrm{H})$ (b) $|y(\mathrm{H})|$

(c) N(jets), $|\eta| < 2.5$ (d) $p_T(\mathrm{jet}), |\eta| < 2.5$

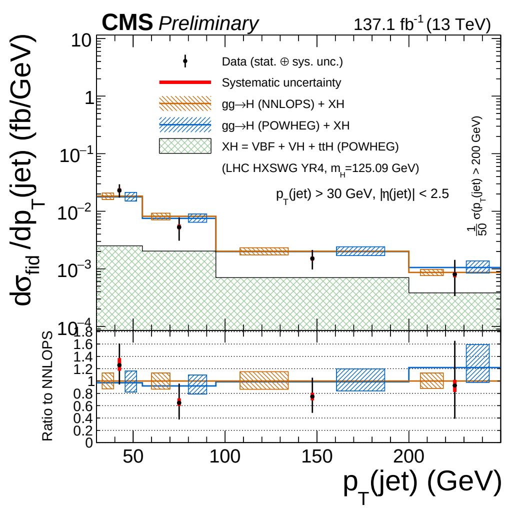
FIGURE 5.15: Expected yield using 2018 Asimov data along with SM signal ($m_H$ = 125.09 GeV) and background expectations for different observables and for all final states combined, in mass window $105 < m_{4\ell} < 140$ GeV which is used forH $\to 4\ell$ measurements. The extraction of the signal is performed by exploiting the differences in the expected $m_{4\ell}$ spectra of the signal and the background contributions in each differential bin of an observable.



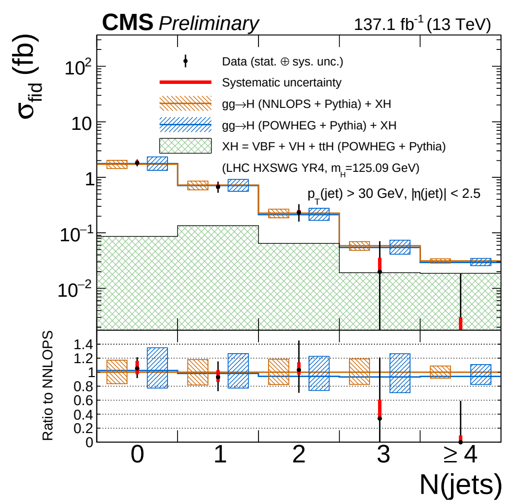

(a) $p_T(\mathrm{H})$




(b) $|y(\mathrm{H})|$




(c) $p_T(\text{leading jet}), |\eta| < 2.5$




(d) N(jets), $|\eta| < 2.5$

FIGURE 5.16: Differential fiducial cross section results for $p_T(\mathrm{H})$ (top left), $|y(\mathrm{H})|$ (top right), $p_T$ of leading jet (bottom left) and N(jets) (bottom right) in $\mathrm{H} \rightarrow 4\ell$ using full Run 2 CMS data and comparison with the theoretical predictions. Systematic uncertainty is indicated by red bars and black bars show total statistical and the systematic uncertainties, combined in quadrature. The blue and brown colors show the theoretical predictions. The acceptances and the theoretical uncertainties in differential bins are computed using POWHEG. Sub-dominant production contributions $\mathrm{XH} = \mathrm{VBF} + \mathrm{VH} + t\bar{t}\mathrm{H}$ are shown in green separately. The inclusive cross section is normalized to SM estimates calculated at NNLL+NNLO level of accuracy for all theoretical predictions. The systematic uncertainties correspond to the generators accuracy are also taken in to account for differential predictions. The fraction of $4\mu$, $4e$ and $2e2\mu$ in each differential bin is allowed to float in fit.

# Chapter 6

# Conclusions

The work that covers the fiducial cross section measurements of the Higgs boson production in the $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) in proton-proton collisions at $\sqrt{s} = 13$ TeV is presented in this dissertation. These measurements help to probe the structure, nature of interaction of Higgs boson with decaying particles. These measurements provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale. Any deviation of integrated or differential distributions from SM expectations would provide a direct hint to BSM contributions. The differential distributions has been extracted for 13  TeVdataset.

These measurements are made using data from pp collisions corresponding to integrated luminosity of 137.1 $\mathrm{fb}^{-1}$. To minimize the dependence of the results on the underlying model of the Higgs boson production mechanism and its properties, the cross sections are extracted in a fiducial phase space closely matching with the experimental acceptance. The remaining model dependence is extracted by repeating the measurement for SM and other exotic model and reported as a distinct systematic uncertainty. All measurements include the correction for the finite detection efficiency and resolution effects. As $\sigma \times BR$ strongly depend upon Higgs mass, the measurement has been extracted assuming $m_H = 125.09$ GeV, as measured using the $H \rightarrow 2\gamma$ and $H \rightarrow 4\ell$ decay channels by the CMS experiment (**Reference39**).

The differential fiducial cross sections are presented using the full Run 2 CMS data collected at 13  TeV  as a function several kinematic observables which are sensitive to the Higgs-boson production mechanisms: rapidity and transverse momentum of the four-lepton system, associated jet multiplicity and transverse momentum of the leading jet. Full Run 2 data enabled us to explore the differential observables in more fine bins than before. The integrated fiducial cross sections are measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\mathrm{stat.})^{+0.24}_{-0.19}(\mathrm{syst.})$ fb at 13 TeV.

All results have been corrected for the detection efficiency and resolution effects to allow for direct comparison with the SM predictions. The agreement between SM predictions and measured $H \rightarrow 4\ell$ cross section measurement at $s$ =13 TeV is found be good. These model independent cross section measurements can be compared with any theoretical predictions in future with better accuracy.

# Chapter 7

# Outlook

After the discovery of the Higgs boson, it became very natural and important to measure Higgs boson properties as precise as possible. Precision measurements of the Higgs boson properties can provide us direct or indirect hint of any new physics for which we are looking desperately. This dissertation presented the measurements of integrated and differential Higgs boson via $H \to 4\ell$ decay channel which is an important Higgs boson property. These measurements provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale. Any deviation of integrated or differential distributions from SM expectations would provide a direct hint to BSM contributions. The differential distributions has been extracted only for 13 TeV dataset. These differential distributions are sensitive gluon fusion production mechanism, PDFs of colliding proton, theoretical modeling of hard quark radiation and relative contribution of different Higgs boson production mechanism.

The current measurements are emerged with significant statistical uncertainties. The studies based on simulations showed that the uncertainties on Higgs cross section measurements with 300 $\text{fb}^{-1}$pp collisions data would be significantly less than the current theoretical uncertainties. LHC has aim to deliver 3000 $\text{fb}^{-1}$of pp collisions data. The Higgs cross section measurements would be extremely important along with other properties with more data.